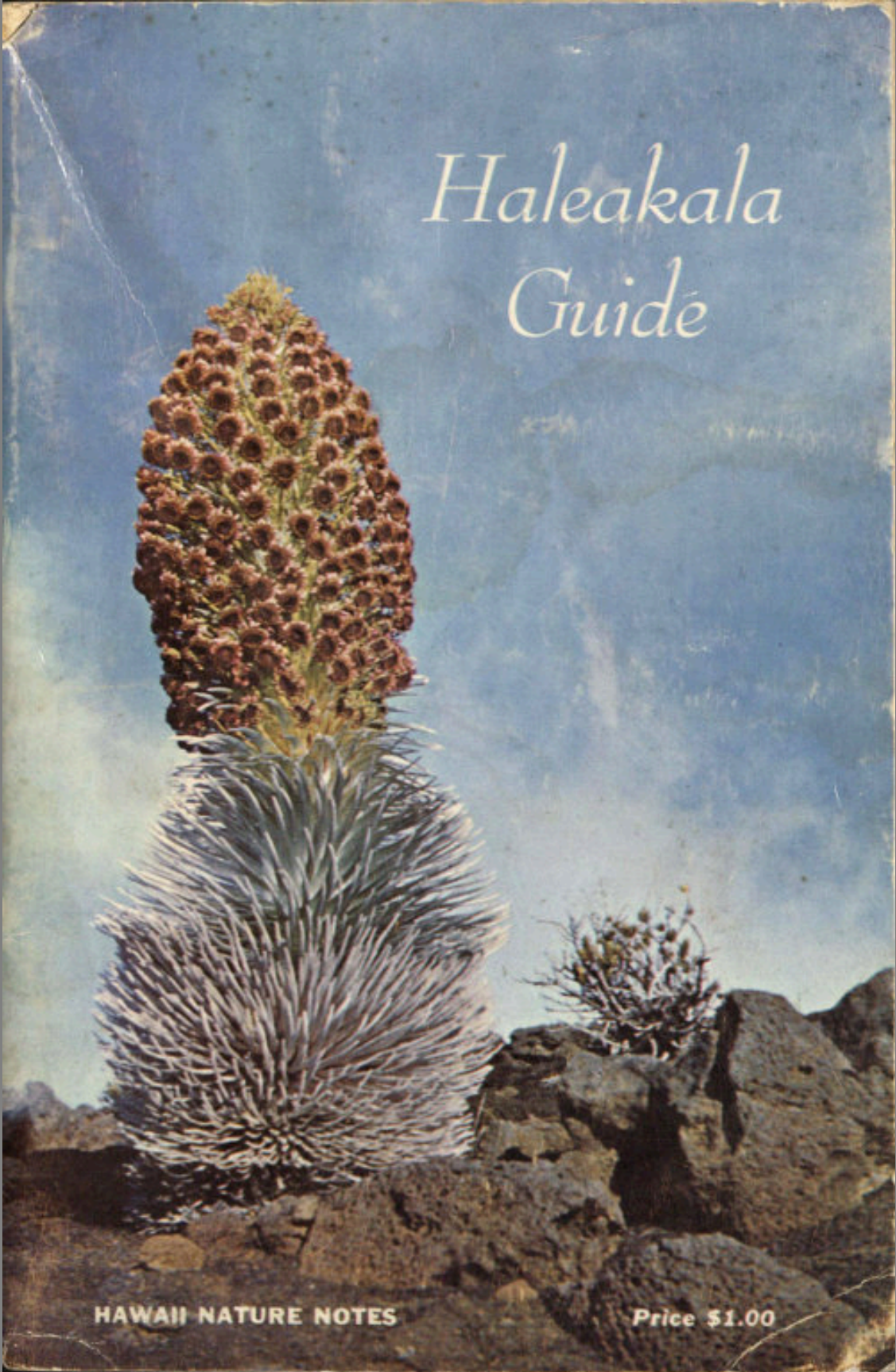


Haleakala Guide



HAWAII NATURE NOTES

Price \$1.00



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Illustrator: Donald M. Black

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NATIONAL PARK: A GUIDE FOR THE HALEAKALA SECTION,
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UNITED STATES DEPARTMENT OF THE INTERIOR
NATIONAL PARK SERVICE



PUBLISHED IN COOPERATION
WITH THE NATIONAL PARK SERVICE

Covers: Silversword in bloom



HAWAII NATIONAL PARK

A Guide for the Haleakala Section Island of Maui, Hawaii

by

GEORGE C. RUHLE

illustrated by Donald M. Black



PUBLICATION OF THE
HAWAII NATURAL HISTORY ASSOCIATION
JUNE, 1959



On the Sliding Sands Trail

Preface

Most of us yearn to travel, and the preliminary to travel is to choose a place that others, people or books, say is interesting, then find out more about it.

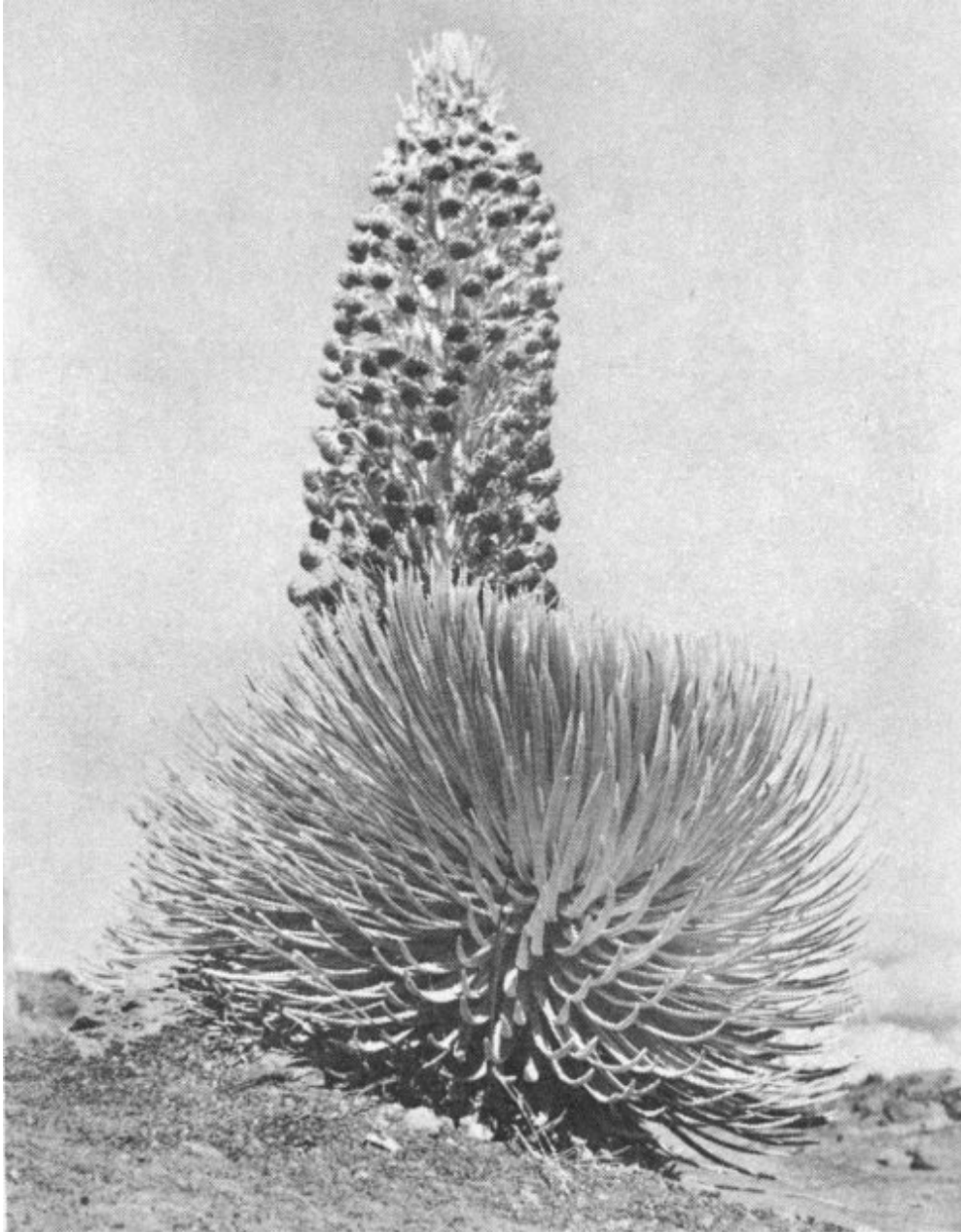
This guide is to help you find out more about Haleakala. It is neither a reference book nor a treatise. It sums up what many have studied and observed. It skims over the myths that the mountain itself created in the imagination of old Hawaiians. It reflects also the labor and thought of the compiler. Its aim is to satisfy your interest while you are here on the brim, or at some other point. For some of you it may be the start of a deeper curiosity, to be satisfied by further reading elsewhere.

Think of this booklet as a chatty companion along the way, and a ready reminder after you have left, of your pleasant experience at Haleakala.

The system of 29 National Parks contains areas of superlative scenic and scientific grandeur essentially in the primitive state. The National Park Service of the Department of the Interior administers these, as well as 152 other areas of outstanding national significance. The law of the land enjoins us to use them in such manner that they may be passed unimpaired for the enjoyment of future generations.

The story of **HAWAII NATIONAL PARK** is the story of active volcanism singularly marked by eruptions of very fluent lava. The park is in two sections; that on the island of Maui, discussed in this guide, includes the great eroded crater of Haleakala Volcano; that on the island of Hawaii embraces the summits of Mauna Loa and Kilauea Volcanoes.

HALEAKALA



The silversword is the pride and distinction of Haleakala.

HALEAKALA is a great volcano, 33 miles long and 10,025 feet high. During a long period of inactivity, stream erosion cut two deep valleys, Keanae and Kaupo, into its sides. These joined near the summit. When volcanic activity once again resumed, flows of aa and blankets of cinders were spread on the valley floors. A multicolored cover, emphasized by symmetrical cones, formed the new floor of the depression, now loosely called Haleakala Crater.

The well chosen name, Hale-a-ka-la, means House of the Sun. Old Hawaiians associated Maui, a trickster demi-God, with the mountain. He was a legendary figure throughout Polynesia long before a few of its inhabitants discovered and settled in Hawaii, bringing their gods with them.

How Maui brought down or ensnared the sun has several versions. Maui's mother, Hina, had trouble drying bark cloth, kapa, because the day was too short, its warmth insufficient. The sun just sped too fast across the sky. So Maui fashioned a strong net to snare it in its course. A slight variant, possibly less used, appeals more strongly. In early dawn, one can watch strong streamers of light from the rising sun break through the clouds and stalk across the crater. With these spidery legs the sun progresses through the heavens. As one by one they were placed over Koolau Gap, Maui seized them and bound them with strong thongs to an ohia tree. Thus captured, the sun pleaded for release. This Maui granted on promise of a slower gait, for which Hina as well as the rest of us can be eternally thankful.

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YOUR VACATION

Anticipate a restful, invigorating interlude. Islanders consider vacation on the cool mountain an inexpensive, pleasant variant from a mainland trip.

Silversword Inn at an elevation of 6,800 feet is popular with luncheon guests and with those staying overnight to view sunset or sunrise from the summit of the great mountain. Attractive, friendly, comfortable, it is the loftiest hostelry in the islands. There is no formal atmosphere: warm, casual clothing is worn; it is strictly “come in as you are.”

Hiking and riding in the vicinity of the inn are favorite pastimes. Adjacent groves of trees of the Temperate Zone impart an aspect novel to the islands. A visit is highlighted by trips into the crater and to the summit, less than thirty minutes distant by car. The cup runneth over for photographers and nature enthusiasts. You can enjoy cool, restful nights between daytime drives to the many points of interest on Maui. For further details, reservations, and rates, consult the Manager, Silversword Inn, R.R. 53, Waiakoa, Maui, Hawaii, or Mr. William S. Ellis, Jr., 900 Nuuanu Ave., Honolulu 17, Hawaii.

ACCESS

The Haleakala road climbs through plantations and ranchland from Kahului Harbor and Kahului Airport to the park entrance at an elevation of 6,740 feet. The distance by the shortest route is thirty miles. The highway continues eleven miles further to the Park Observatory on the western rim of Haleakala Crater and to the 10,025-foot summit. No bus service exists, but taxis and U-drive cars are hired at the airport and in the towns of Kahului and Wailuku. The sole access into the crater is over good trails for travel on foot or by horse.

WHAT TO DO AND SEE

The start of a drive to the park is made by one of three paved routes. The shortest is Pukalani Road. The other two turn inland at Paia or Haiku and traverse more interesting country. The three routes converge at Pukalani Junction ten miles up the mountain. **PUKALANI** means a hole in heaven, which picturesquely describes the fact that the sun breaks through at this place despite a general overcast elsewhere. 2

As the road rises up and ever up, it unfolds distant views of fields of sugar cane and pineapple, of West Maui Range, 6,000 feet high, and of Molokai, Lanai, Kahoolawe, and Molokini Islands beyond channels of blue sea. 100 miles southward, the tops of the snowy volcanoes on the Island of Hawaii float on billowy clouds. 10,000 feet below, the aquamarine Pacific fringes Maui with white surf.

Three viewpoints along the road overlook the great crater: Leleiwi, at the 9,000-foot switchback; Kalahaku, two miles below the summit; and the Park Observatory, near the top. The roadway extends along the crest one mile southwestward to a scenic point beyond the park boundary and a communication station of the Civil Aeronautics Administration.

Just above the park entrance, Silversword Inn, a National Park concession, offers meals, rooms, souvenirs, horseback riding and guided horseback trips into the crater. Across from the inn, a paved spur road leads a half mile to Hosmer Grove Campground and Picnic Area.

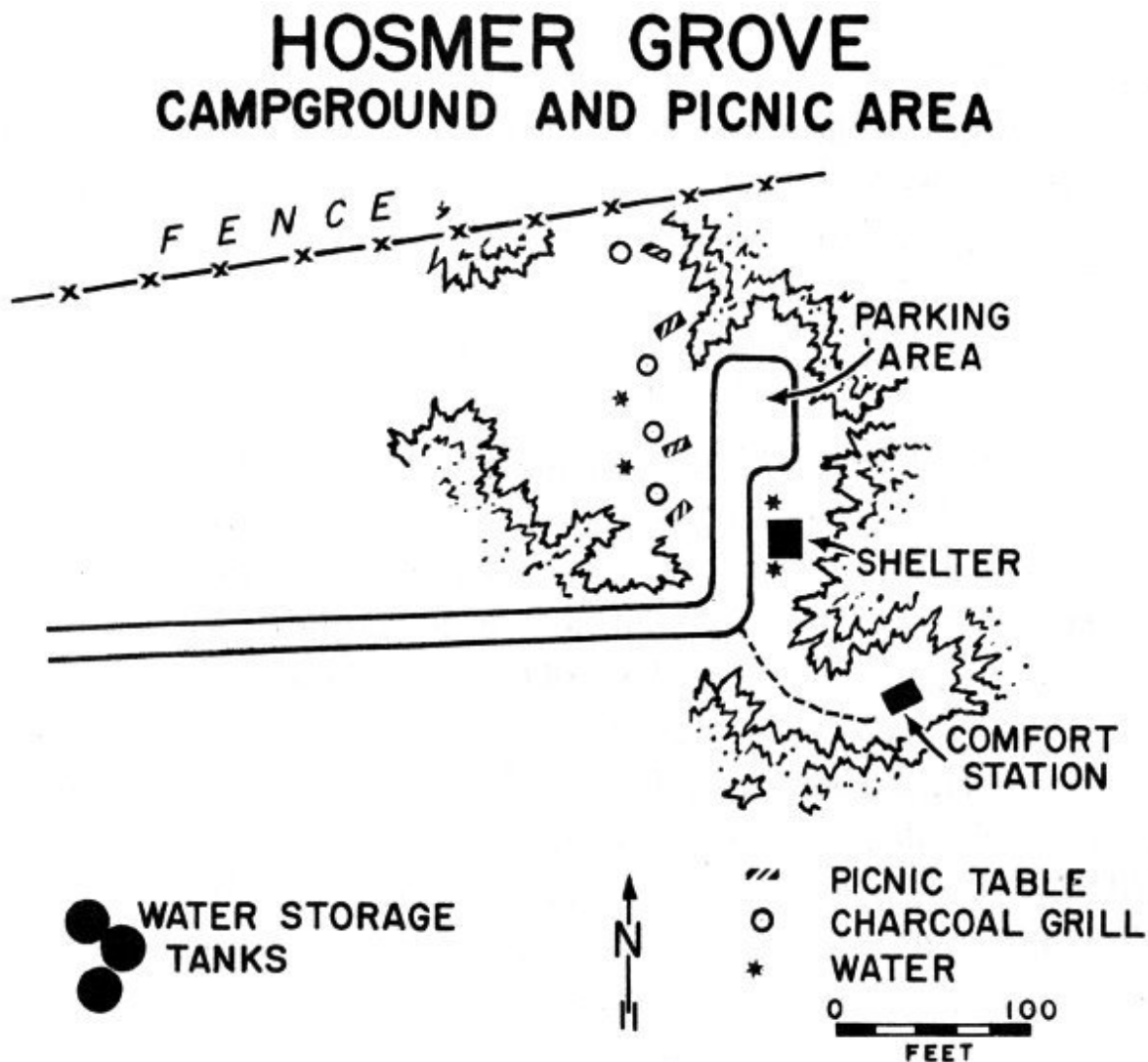
Haleakala Crater is a favorite with those who like the back-country; its inspiring scenery and restful solitude are great reward for time and effort. The National Park Service maintains three cabins on the crater floor and 30 miles of well-marked trails for hikers and horseback parties.

HOSMER GROVE CAMPGROUND AND PICNIC AREA

A quarter mile above the park entrance, opposite the driveway to the inn, a paved lane, one-half mile long, leads to the Hosmer Grove Campground and Picnic Area. It has a shelter for rainy weather that contains two tables and two charcoal burners. Four additional tables with adjacent charcoal burners are in an open site below the road. Running water, parking space for eight

cars, and sites for pitching tents are provided. Charcoal may be purchased at the inn. A self-guiding nature trail leads through the grove.

3



HOSMER GROVE CAMPGROUND AND PICNIC AREA

The grove was named for the first Territorial forester, Dr. Ralph S. Hosmer, who experimented with planting temperate trees at high altitudes on Haleakala and Mauna Kea. Trees, planted here in 1910, include the deodar, *Cedrus deodara* from the Himalayas; the tsugi, *Cryptomeria japonica* from Japan; eucalypti from Australia; and from the mainland a cypress, *Cupressus*

arizonica; a juniper, *Juniperus virginiana*; Douglas fir, *Pseudotsuga taxifolia*; incense cedar, *Libocedrus decurrens*; two spruces, *Picea canadensis*, *P. excelsa*; and seven pines: lodgepole, *Pinus contorta*, Coulter or big-cone, *P. coulteri*, Jeffrey, *P. jeffreyi*, longleaf, *P. palustris*, ponderosa or western yellow, *P. ponderosa*, white, *P. strobus*, and Scotch, *P. sylvestris*. Many of these have survived and have borne fruits. The huge keeled cones of Coulter pines are cherished as ornaments in some homes.

Native plants associated with the area are the shrubs: Haleakala sandalwood, mamane, pukiawe, aalii, mountain pilo, ohelo, silver geranium, kupaoa; 4 two or three ferns; two sedges; and three native grasses.

Two thirds of the distance to the grove, the crater trail from the inn starts up the mountain to the left. This is a connecting link, 1¾ miles long, to the Halemauu Trail which it joins a half mile below its start on the highway. See *Numbered Points of Interest*, [No. 9](#).

THE TRAIL SYSTEM

The Sliding Sands Trail, the popular route into the crater, starts from the parking area at the Observatory. It is constructed along the south side of the crater to Kapalaoa Cabin six miles away. Connecting trails go to Paliku Cabin, four miles farther. The Halemauu Trail has two upper ends, at the 8,000-foot elevation on the highway and on the Hosmer Grove Campground Spur near Silversword Inn. Halemauu Trail goes down Leleiwi Pali, the west wall, to Holua Cabin, four miles from the road or six miles from the lodge. The trail continues easterly from Holua for six miles along the north side of the crater floor to Paliku. Branch trails are built to points of interest. The Kaupo Trail through Kaupo Gap leaves Paliku Cabin and the crater to make a rapid descent of the southern, sun-drenched slope.

PARK CABINS

Each of the three visitor cabins within the crater, Kapalaoa, Paliku, and Holua, is equipped with running water, a wood-burning cookstove, firewood, kerosene lamps, cooking and eating utensils, twelve bunks, mattresses, and blankets. Use of these cabins by hikers on a priority reservation basis is

granted free of charge by the Park. In consideration for their use *cabins should be left clean and in order* by each party. The following arrangements are necessary: write the Park, giving an outline of your proposed trip, number in the party, exact calendar dates, and names of specific cabins which you wish to use each night. The address is: "Hawaii National Park, Haleakala Section, P. O. Box 456, Kahului, Maui, Hawaii." Cabin reservations can also be made by telephoning the Park. When you arrive in the Park, stop at the Administration Building for your permit, cabin key, and orientation.

5

SUGGESTED HIKING TRIPS

For safety reasons, all visitors are required to obtain permits from the rangers for all trips into the crater, other than those with Silversword Inn guides.

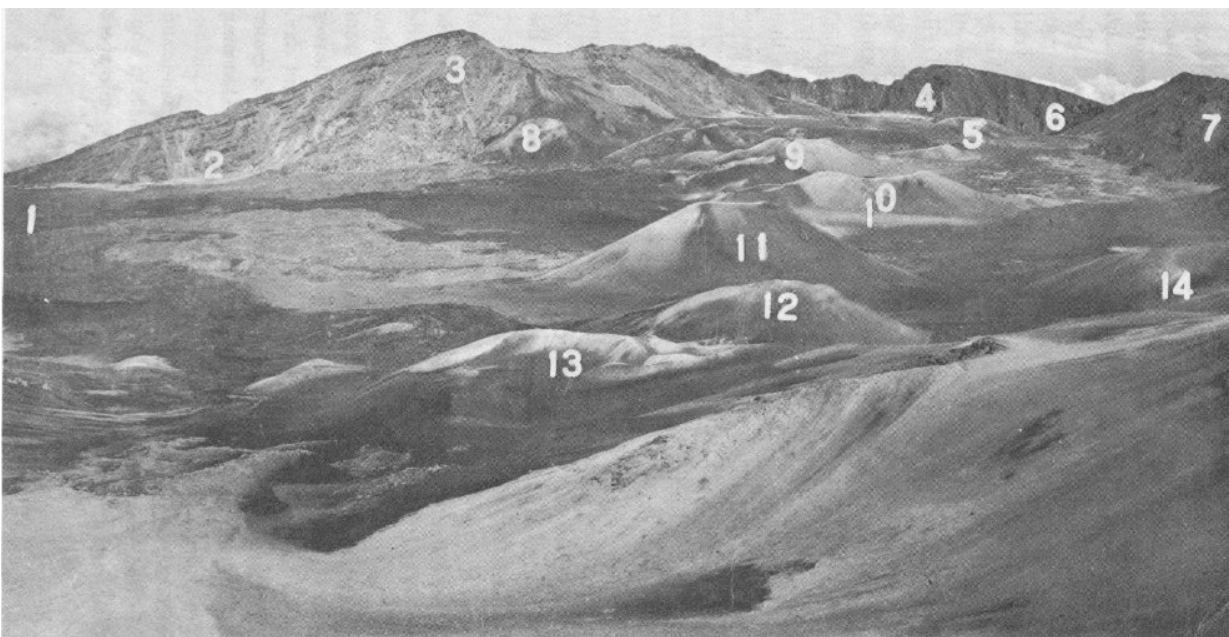
Short Walks for the Day Visitor: (1) Along Halemauu Trail from the highway to the Crater Rim, three-quarters of a mile. Views down Keanae Valley, across Koolau Gap to Hanakauhi, and of Halemauu Trail. (2) A short distance down the Sliding Sands Trail. *Be careful not to travel too far down.* The return climb is exhausting at this high altitude. (3) To the top of White Hill. The trail winds among ancient Hawaiian stonewalled encampment sites.

One-day hikes into crater: (1) Down Halemauu Trail to Holua Cabin and return to the highway, a scenic trip of eight miles that can be taken by any reasonably good hiker. It is not recommended when clouds blanket the pali and Koolau Gap. Allow a half-day for hiking. (2) Down Sliding Sands Trail to the crater floor, across Ka Moa o Pele Trail to Halemauu Trail at Pele's Paint Pot or Bottomless Pit, and return to the highway via Holua and the Halemauu Trail. This colorful, spectacular twelve-mile trip is recommended only for good hikers. Allow eight hours of hiking time. *At your risk*, rangers can often arrange to move your car for you to the place at which the Halemauu Trail emerges on the highway.

Overnight crater hikes: Hike down Sliding Sands Trail; spend the night at either Paliku, Kapalaoa, or Holua Cabin; return via Halemauu Trail. The choice overnight hike to Paliku, a 20-mile round-trip, is recommended for good hikers only. The trip with an overnight stop at Kapalaoa is 14 miles long. The shortest route from the foot of Sliding Sands Trail to Holua via Ka Moa o Pele leaves only the four-mile climb via Halemauu Trail for the second day. *The Sliding Sands Trail is not recommended for return from a crater trip.* The long climb to an elevation of 10,000 feet is too exhausting for most people.

Two and three day crater hikes: Entry via the Sliding Sands and return via Halemauu Trail is recommended. A three-night trip stopping at Kapalaoa, Paliku, and Holua allows leisurely enjoyment of the crater. Good health and fair walking ability are all that are required for these longer trips.

6



1. Koolau Gap; 2. Waikau; 3. Hanakauhi Peak; 4. Paliku; 5. Puu Maile; 6. Kaupo Gap; 7. Haleakala Peak; 8. Puu Kumu; 9. Puu Naue; 10. Ka Moa o Pele; 11. Puu o Maui; 12. Kamohoalii; 13. Ka Lua o ka Oo; 14. Puu o Pele.

No guide service is available or necessary for parties hiking or riding their own stock into the crater. However, a permit is required before you

7

start your trip within the crater. For details, consult the section labelled “[Park Cabins](#).”

AN OUTFIT FOR HIKING IN HALEAKALA CRATER

Clothing should consist of hiking shoes, slacks, shirt, jacket, hat, and preferably a light raincoat. Basketball shoes or keds are preferred by some. Because of the chill climate at elevations of seven to ten thousand feet, warm clothing is advisable. In climbing, temperature goes down as you go up. The top of Haleakala averages thirty degrees cooler than sea level. You should bring your food for the trip, a knapsack, sunburn lotion, soap, hand towel, dish towel, matches, and simple first aid. As cooking facilities in the cabins are adequate, food need not be precooked.

HORSEBACK CRATER TRIPS

Silversword Inn provides horseback trips on good stock with a competent guide. Food is provided, cooking is done by the guide, and sleeping accommodations are arranged by the management. A guest need only concern himself about personal effects and clothing suitable for riding in a cool climate. For rates and trip reservations, telephone or write to the Manager, Silversword Inn, R.R. 53, Waiakoa, Maui, Hawaii, or to Mr. William S. Ellis, Jr., 900 Nuuanu Ave., Honolulu 17, Hawaii.

If you have your own horses make the same arrangements as hikers for entry into the crater and for cabin use. You may travel on any of the crater trails. Fenced horse pastures are adjacent to each of the cabins.

NUMBERED POINTS OF INTEREST ON THE MAP

Basic data for this section was compiled by the park staff and submitted by Eugene J. Barton, Assistant Superintendent in charge of Haleakala from 1949-1955. The map is in the center of the booklet.

1. *Park entrance* (elev. 6,740') *and inn*: The park entrance, marked by a rustic sign, is on the slope of Puu Nianiau, an ancient cinder cone. *Nianiau* is Hawaiian for swordfern. One quarter mile above the entrance turn right to the Silversword Inn for meals, lunches, overnight accommodations, color pictures, slides, and souvenirs of the park. The inn arranges guided trips through Haleakala Crater.

2. *Ralph S. Hosmer Grove*: Across from the lodge, a paved spur road leads to the Hosmer Grove Campground and Picnic Area.

3. *Headquarters of the Haleakala Section of Hawaii National Park*: Stop at the Administration Building beside the road for information, permits, cabin keys needed on crater trips, and for assistance in case of trouble or accident. The park maintenance area is located behind the station. As you drive toward the summit, note the small native trees and shrubs growing along the road. This elfin forest was all but destroyed by goats and cattle; it has recovered under National Park protection. The book, "Plants of Hawaii National Park," and the section on [plants](#) in this guide may help you identify the different species. These and other publications of the Hawaii Natural History Association may be purchased at the inn or at Park Headquarters.

4. *Leleiwi Overlook; Kalahaku Overlook*: At the switchback near the 9,000-foot contour, a parking space has been constructed that is labelled LELEIWI OVERLOOK. This is above Holua and the Halemauu Trail, so that parties can be watched as they go down Leleiwi Pali. On clear days, the whole length of Keanae Valley can be seen through Koolau Gap. The lateral view

extends from Hana Airport across the big isthmus to Kihei on Maalaea Bay. On afternoons, clouds roll into the gap, making this a good place to see the Brocken Specter. One's shadowy image appears within a circular 9 rainbow, projected against the bank of cloud. **KALAHAKU OVERLOOK** (elev. 9,325') is a remarkable view point on the crater rim that is reached by a short spur off the main park road. It is the site of the Old Rest House, in which travellers on foot or horseback stayed overnight before the road was built. Silversword plants may be seen just below the main parking area. **KALAHAKU** is the name of the rugged pali forming the crater wall at this point. The name means "meeting place of the headmen or chiefs."

5. *The Observatory* commands a spectacular view of the crater near the summit of Haleakala. It was erected in 1937 by the National Park Service for your comfort and convenience. The observatory is the main objective of most visitors to Haleakala. Besides being a welcome shelter during inclement weather, it has a large relief map and interpretive orientation devices. Modern rest rooms are located to the rear of the building.

6. *White Hill, Pakaoao* (elev. 9,865'); *start of Sliding Sands Trail*: The rocky hill, just above the observatory, is capped with andesite, a volcanic rock lighter in color and different in composition from the common basaltic lava of Hawaii. The southwest slope, up which the trail leads, is covered with Hawaiian sleeping shelters consisting of oval stone-walled enclosures. These gave protection against the wind, fog, and cold that is common here. They may have been used long ago by wayfarers or by sentries and groups of warriors stationed at this commanding site. They may also have served as ambush for professional robbers, *aihue*, who waylaid travellers in lonely places. Refer to the topic, "[The Tradition of Kaoao](#)" in "Haleakala Hawaiiana" for more of the story.

7. *Red Hill, Pakaoao, Summit of Haleakala* (original English name, Pendulum Peak; elev. 10,025'): To drive to the summit take the paved road, turning sharply left below the observatory parking area. Red Hill is the highest of the three prominent recent cinder cones. Early morning is a good time to view extended panoramas and distant seascapes, before streaks of clouds form shelves along the sides of the mountain. But afternoon is better for viewing the crater, as its features appear in excellent light and color at that time. A pointer table by the road indicates the names of islands and

peaks seen. An Army radar and radio communication station was located on Red Hill during World War II.

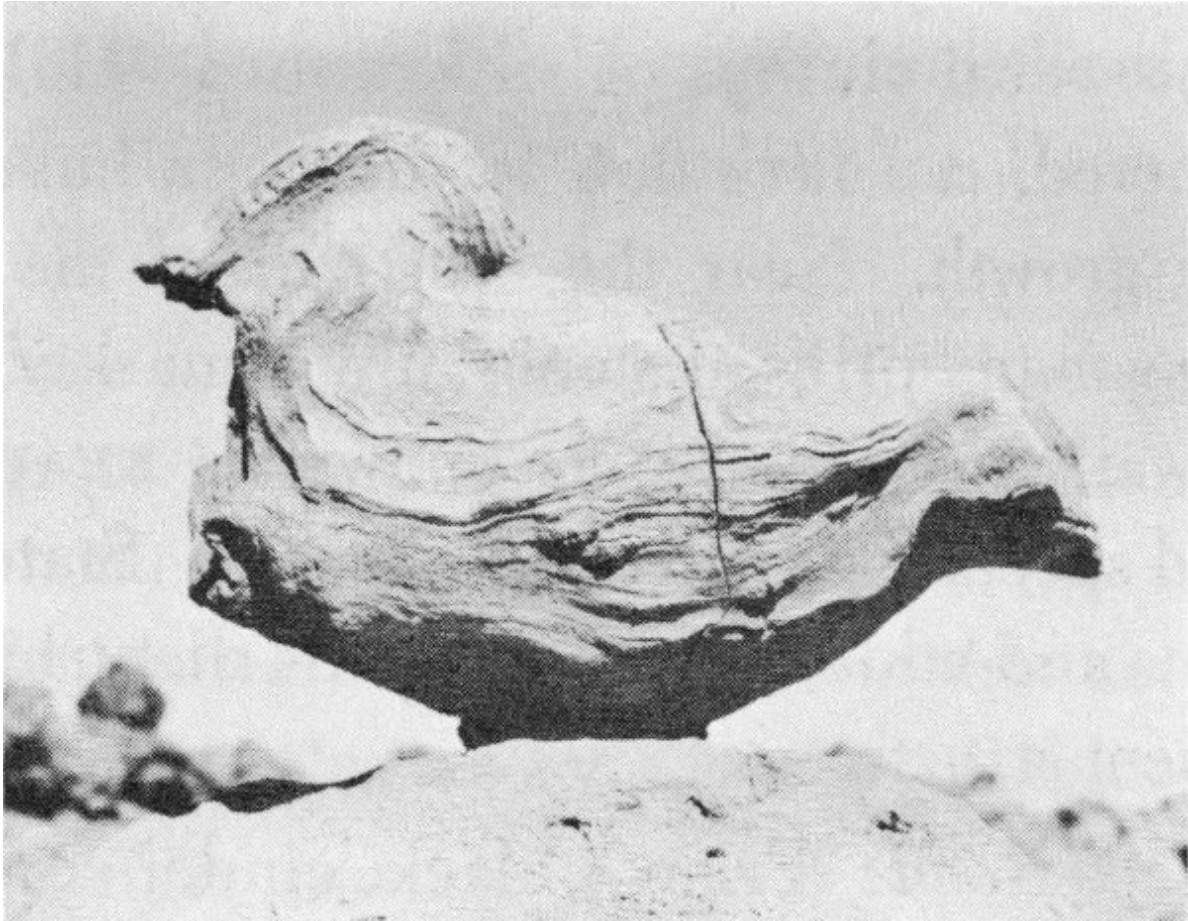
10



Halemauu Trail on Leleiwi Pali.

8. *Skyline Drive*: An improved driveway to the Civil Aeronautics Repeater Station leaves the park at the pass between Red Hill and Kolekole. Views stretch down the vast, precipitous southwest outer slope of Haleakala to the Lualailua Hills and a desert seashore, ten thousand feet below. Morning will most likely yield the cloud-free views. The Aeronautics Station is open to visitors. A parking area is just outside the gate to the station. Kolekole is the site of television relays and a satellite tracking station. It is a mile from the Observatory to the summit of Kolekole. It is 1.7 miles from the Observatory at the FAA Station. Magnetic Peak is across the road from Red Hill. A huge curiously shaped bomb on its skyline has a silhouette that looks much like a sitting duck.

11



Volcanic bomb on Magnetic Peak.

9. *The Halemauu Trail* starts from a curve on the park road at the 8,000 foot elevation. It leads down a gentle slope for eight-tenths of a mile to the crater rim. An alternate start is on the driveway into Hosmer Grove Campground, near Silversword Inn. Built by the National Park Service in 1937, the trail drops with a gentle grade down the 1500-foot Leleiwi Pali to the crater floor. It suggests thrilling trails in Glacier National Park and other rugged mainland areas as it swings and clings to the vertical cliffs. Hike to the crater rim for spectacular views of Windward Maui and Keanae Valley in fair weather; if clouds are rolling up Koolau Gap in the evening, you may possibly be greeted by the Specter of the Brocken. Silver geraniums, Hinahina, small spherical shrubs peculiar to Haleakala, grow along the upper trail and road. **HALEMAUU**, grass house, is said to be derived from one formerly located near the head of the trail.

10. *Holua Cabin* rests in a grassy plot at the foot of the towering, 3,000-foot Leleiwi Pali. Introduced grasses grow in the meadow near it. The pale “moss” on the rough lava flow is *Stereocaulon*, actually a lichen, that strange combination of an alga and a fungus growing together to appear as one plant. This pioneer on barren new lavas is sometimes called Hawaiian Snow. The comparatively recent flow in front of the cabin contains lava tubes or caves. These are rough and dangerous, and should not be entered except by especially equipped parties. Holua Cave, above the cabin, was used as a night shelter before the cabin was built.

11. *Silversword Loop*, a quarter-mile long, deviates from Halemau Trail past several clumps of silversword. Hollows and slopes of the old, weathered, red lava flow in this area have favorable conditions for their growth. Near the east end of the loop, the main trail passes among many piles of stones, ahus, markers, and platforms built by the Hawaiians. This place, known as **KEAHUOKAHOLO**, appears to have had special or sacred significance. State law, as well as Hawaiian customs and ethics, strictly forbids disturbing or damaging any of these ancient structures.

12. *Kihapiilani Road*: From a black, cinder-covered flat, an old Hawaiian pathway crosses the rough shoulder of the small cone on Halalii to smoother surface between Mamane Hill and Puu Kumu. It is six to eight feet wide and is paved with flat stones. The start, difficult to locate, is to the right of a small, horned spattercone seen from the trail as you look toward Hanakauhi Peak. A chief, Kihapiilani, built the trail over Mauna Hina along the North Rim to a pond, Wai Ale, probably the present Wai Anapanapa on the exterior slope. Refer to “[The Legend of Kihapiilani](#)” in “Haleakala Hawaiiana” for the story.

13. *Bottomless Pit* is a yawning black well, ten feet in diameter, rimmed with colored lava spatter a few feet high. *Use caution approaching its edge.* Baseless legend claims that it sinks to sea level. Although no bottom can be seen, debris chokes the opening sixty or seventy feet below. It is a vent, through which superheated gases were emitted in an eruption of long ago. The rim around the throat indicates that a little lava sputtered out with the gas. During the flank eruptions of Mauna Loa and Kilauea similar vents exhale columns of blue, incandescent gases at intervals. Old Hawaiians at Kaupo say that the pit was used for disposition of umbilical cords of babies.

Various reasons are assigned to the custom, such as to make the child strong, or to prevent its becoming a thief.

14. *Ka Moa o Pele Trail* branches from the foot of the Sliding Sands Trail to join the Halemau Trail across the crater. It is a scenic route between silverswords on Ka Moa o Pele, a red cinder cone. Flowering plants can usually be seen from June to September. Pa Puaa o Pele, Pele's Pig Pen, is the rim of a spatter cone, now buried, in the low pass between Halalii and Ka Moa o Pele. Hikers from Sliding Sands to Holua should go around the right or east side of Halalii to see the Bottomless Pit and Pele's Paint Pot, a colorful pass just a short distance beyond.

15. *Bubble Cave* is a large, collapsed bubble with heavy walls. It was blown by gases in ancient molten lava. Only a small segment collapsed in the center of the roof, which serves as entrance and smoke-hole; old-timers were wont to camp in this natural shelter. It was the rest stop now supplanted by Kapalaoa Cabin.

16. *Wailuku Cabin*, outside the park, was built by the State Board of Agriculture and Forestry for the use of hunters. Arrangements for its use must be made at the Maui Office of the Board in Kahului. Trails to it are not of park standards and are not recommended to sightseers.

17. *Old and New Volcanics*: Dikes that protrude as slabs of light-colored rock from slopes of the ridge toward Hanakauhi mark part of the ancient divide between Koolau and Kaupo Valleys, the great erosional canyons worn into the mountain during an age of quiescence. This divide is deeply buried inside the crater under the cinders and flows of renewed eruptions that form the present floor. Adjacent Mauna Hina and Namana o ke Akua, green with shrubs, grass, and scrub mamane, are cinder cones from ancient eruptions. Puu Nole, a garish, black youngster in the community, has only 14 silverswords on its barren slopes. Towards Paliku the trail is flanked by some of the most recent lava flows, only a few hundred years old. Their source vents are visible on the slopes of Hanakauhi and the north wall of the crater.

18. *Lauulu Trail* is plainly visible as it zigzags up the north wall. Although not maintained at present, it is passable and allows rugged enthusiasts to

climb the rim for views of the crater. On the outer slopes, moist grassy plots, often fog-bound, blend into jungle that drops to the Hana Coast, 8,000 feet below. Kalapawili Ridge extends from the summit of Hanakauhi Mountain eastward around the head of Kipahulu Valley to the tiny lake, Wai Anapanapa. It is readily traversable for the good hiker.

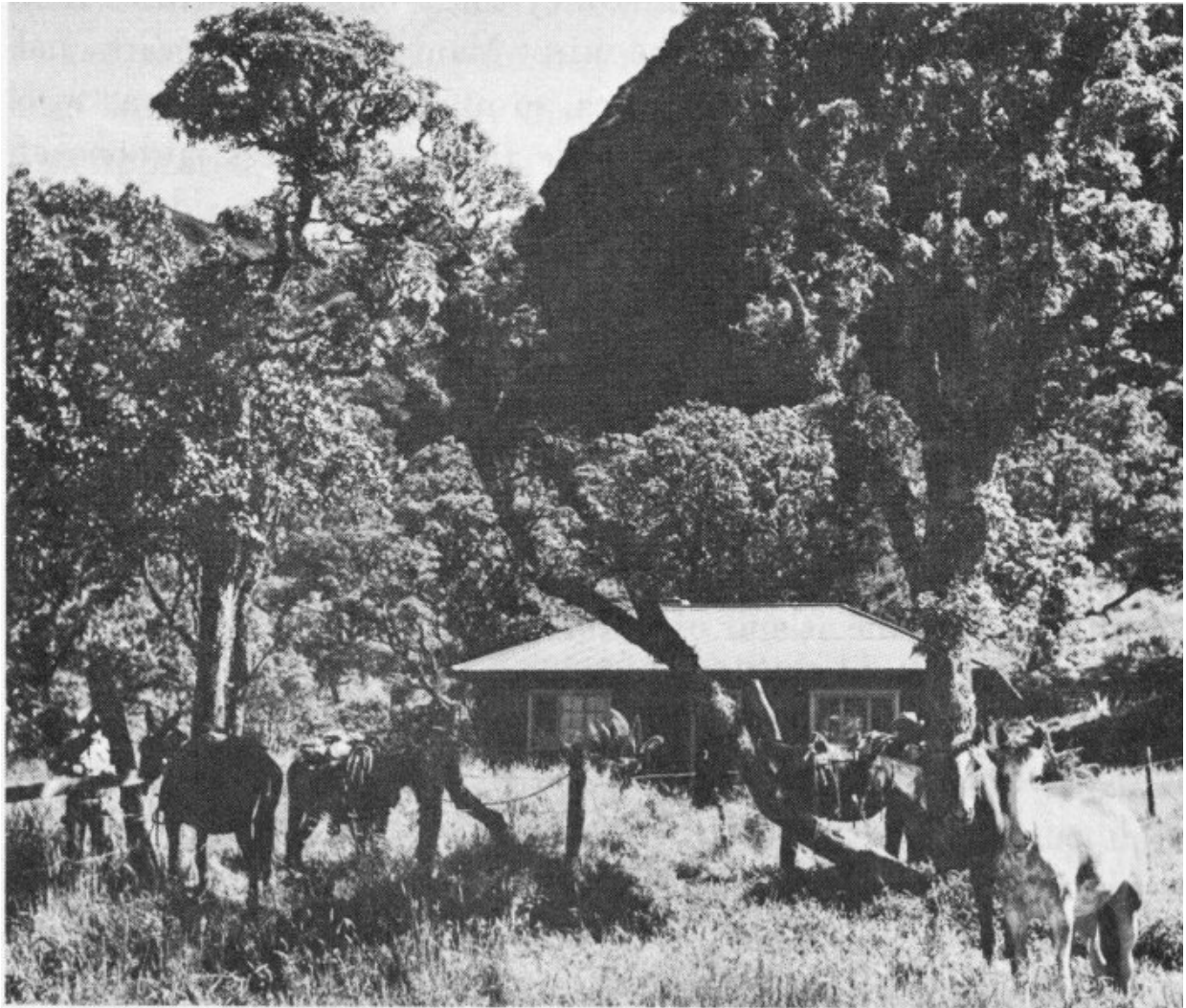
19. *Paliku* is very different from the desert wastes in the other parts of the crater. Lush grass and ferns, overhanging forest trees, and a verdant cover on a towering pali result from abundant clouds pushed by the trade winds over the east wall. **AKALA**, the giant Hawaiian raspberry, ripens abundantly back of the cabins in July. It is excellent for pie, preserves, or dessert. Native Hawaiian trees include: **MAMANE**, far larger than the scrubby growth in the crater; **OHIA**, with gray-green leaves and red lehua flowers; **KOLEA**, with thick magnolia-like leaves four inches long; **OLAPA**, with leaves that tremble in the slightest breeze, like those of quaking aspen. The conifer above the Ranger Cabin is a cryptomeria, a Japanese evergreen planted early in this century.

20. *Kipahulu Valley*, beyond the eastern rim, is a remote, jungle wilderness walled in by loftiest cliffs; a no man's land, it has barely been explored, let alone touched or altered by civilization. Look from the cabin at Paliku across the pasture to the lowest notch in the sheer eastern pali. For a view into Kipahulu, a good climber can follow an old goat trail, steep and tortuous, that leads up the left side of the draw to this notch. A chorus of bird songs rises from the primeval forest in Kipahulu. With them you may hear complacent squeals of wild pigs that have been undisturbed for generations.

21. *Kaupo Trail* is a good trail that winds down Kaupo Gap, across little meadows, through groves of small **AALII** trees, and under spreading **KOA** trees to the park boundary which is below the 4,000-foot contour. 15

Wild goats may be heard on the pali or on the lavas in the gap. Kaupo Ranch extends below the park. From the redwood water tanks, a jeep trail descends through pastures to ranch headquarters. It is 8 miles from Paliku to ranch headquarters, which can be hiked in a half day. Kaupo Village on the belt road around Haleakala is 1½ miles further. The route is interesting, but arrangements must be made in advance to be met by car either at the ranch or the village. It is fifty miles from Central Maui by the shortest road. Although Kaupo Trail crosses private property, advance permission is not required to use it; but, in due consideration, please use care to close gates

and do not molest or damage any property. *The reverse trip from Kaupo Village into the crater is not recommended* because of the long, arduous climb up the unsheltered, south-facing mountain slope.



Paliku Cabin.

HALEAKALA HAWAIIANA

MAUI LEGENDS

Thus starts the story of Maui, beloved demi-god of all Polynesia in the never-never land of long ago. Not wanted as a babe, for he was scrawny and deformed, his mother, Hina, wrapped him in a lock of her hair and cast him into the sea. But jellyfish rescued and mothered him, and the god, Kanaloa, gave him protection. For all that, the growing youngster yearned for his own, so that one day he crept back and stealthily mingled with his four brothers. He was accepted in the family circle only after much pleading by the eldest boy.

Maui-of-a-thousand-tricks is the favorite nickname given this delightful, oft thoroughly human scamp. Throughout his escapades he was ever faithful to man, so frequently fickle and unworthy. Maui made the birds visible, for at first they could only be heard as they sang and fluttered through the air. Maui invented spears and barbed fishhooks. His greatest catch, so runs his fish story, was with a huge hook of powerful magic made from the lifeless jawbone of his grandmother, who was remarkable, for only one side of her body was alive. He tricked his brothers into the task of manning the paddles while he, equipped with his potent tackle, fished up the Hawaiian Islands from the depths of the ocean. Maui first had strictly admonished the brothers not to watch him, but only to look straight ahead. When curiosity overcame the belabored paddlers, they disobeyed. The line parted as they looked back, leaving the land only partly emerged, a chain of islands instead of a continuous whole.

The story of the island of Maui thus begins and—so runs one version—Maui lived in plain fashion on simple fare in a humble grass hut at Kauiki, the famous foothill of Haleakala in the district of Hana. Hawaii at the time was covered by darkness and fog, so Maui pushed the heavens to their present position far above the highest mountains, Haleakala, Mauna Kea, Mauna Loa, and Hualalai. Before this feat, man had to stoop and crawl, pressed

closely to the ground, for the skies were low, and were held up by the plants whose leaves became flattened by their burden, even as they remain 17
today. Yet even now, when Maui is asleep, the heavens rush back as somber clouds and darken the country with storms. From Kauiki, Maui made his important journey to the crater to capture the sun and force it to move slowly through the heavens that tapa may be dried and fruits can mature. Also there are more daylight hours for fishing and ceremonial preparations in the *heiau*. This legend, the most pertinent to this guide, is sketched in the [Introduction to Part I](#).

Most Hawaiian legends place Maui's home in the black lava bank of the Wailuku River above Hilo, while his mother chose the dark cave behind Rainbow Falls. His exploits, so often capricious, more frequently reflect a benefactor's concern for mankind. Maui disclosed the art of making fire by rubbing sticks together. Previously, man was dependent on Pele's volcanic furnace or on embers carefully nurtured to kindle his fires. Maui wrung the secret from Alae, the reluctant mud-hen, who alone knew it and guarded it with greatest care. Exasperated by repeated frustration and deception, Maui, once he had gained her secret, punished the stubborn bird by rubbing her head with a stick so roughly that all the feathers came off and raw flesh appeared, which is how it remains today.

Maui in his noblest moment gave his all for man, whose frailty brought about the downfall of both. For Maui, too, was doomed to die some day, since his father, like Achilles' nurse Cynosura, had neglected a part of the proper ceremony to make him immortal. Maui abhorred the fact that man must die, for he regarded death as degrading and an insult to the dignity of man. The secret of life was hidden within the heart of Hina-nui-kepo, the dread ogress of death. To win immortality one had to steal through her jaws, which had sharp basalt teeth, enter into the inky blackness of the stomach, and tear out the heart. This could only be done while Hina slept. First, Maui turned man into a little bird, and advised him to keep very quiet, lest he awaken the fearful goddess from her slumber. Then he went stealthily about his fearsome task, speedily and alone. All went well until the return journey; alas! poor, weak man could not restrain himself, but burst into uncontrollable laughter as he watched the plight of the demigod within the ugly, 18
gaping fish-mouth of Hina. The great jaws closed with a snap that crushed Maui and left Death ever after the victor. Greater love hath no man!

It would seem that the crater so high in the sky, so remote in location, so difficult in access, so desolate in appearance, so dread in origin, should have been shunned by the early native. Quite the contrary, many marks of frequent and varied use may be discovered in the crater. A trail through it connected the busy sites on West Maui and the isthmus with Kaupo and Hana near the eastern shore. This direct route avoided the wet heavily forested northern slopes of the mountain as well as the precipitous, arid, rough terrain on the southwest. It was easily traversed, in spite of the climb involved.

THE LEGEND OF KIHAPIILANI

“Kihapiilani was one time King of Maui. It was he who caused the road from Kawaipapa to Kahalaoaka to be paved with smooth rocks, even to the forests of Oopuloa in Koolau, Maui. He also was the one who built the road of shells on Molokai.” “And,” the great Hawaiian antiquarian, Abraham Fornander, might have continued, “he caused the trail across the crater of Haleakala to be paved with water-worn stones, to the foot of Hanakauhi of the mists.” Kihapiilani was the great public works king of the islands.

Because of Kihapiilani a most remarkable event of ancient Hawaiian history came to pass, the expedition of numberless canoes.

Three and a half centuries ago, Piilani, a king of Maui, had four children: two sons, Lonoapii who was the eldest child, and Kihapiilani, the youngest; and two daughters, Piikea, who became the wife of King Umi of Hawaii, and Kihawahine, now regarded as the lizard god. When the old king was dying, he adjured his successor, Lonoapii, to take his place as father and to be kind to the younger brother. But alas, the young prince was neglected and treated with contempt. One day at Waihee, two calabashes of small fish, *nehu*, still wet with sea water, were brought to Lonoapii. These he gave to 19 everybody except the younger brother, who, therefore, reached out and helped himself. This angered the king so that he hurled the calabash and its contents into Kihapiilani's face. Without a word, Kihapiilani arose and travelled to Kula. After some time, he told his story to the *kahuna* Apuna and asked what he should do. Apuna replied that he should seek advice at Keanae from a *kahuna* named Kahoko. Kahoko sent him to Hana, and from

there to Hawaii, following a certain dark object as guide. Kihapiilani and his retinue travelled the windward side of Kohala on foot, swimming through shark-infested waters around the bold headlands. Everywhere people gathered to him, for he was handsome and brave. When he reached Laupahoehoe, in Hamakua, he found his sister, Piikea, living with her husband, Umi. Umi had already become a great chief for he had overthrown his tyrant brother, Hakau, but his greatest deeds were yet to follow. These include the union of all of the island of Hawaii under his rule.

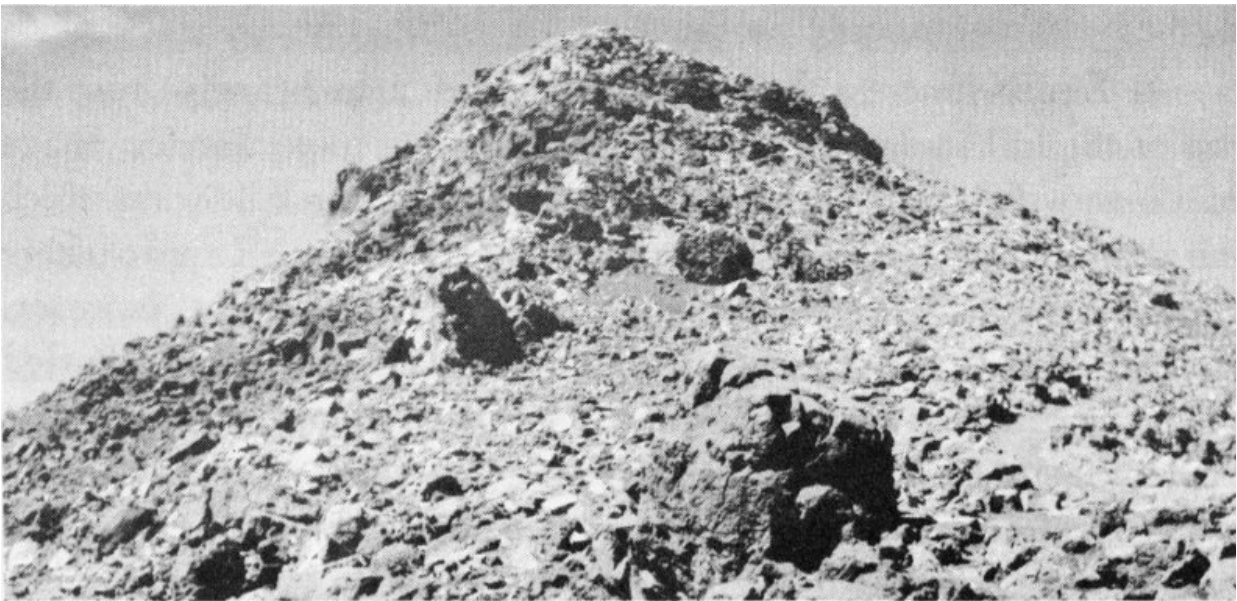
When Kihapiilani explained that he was seeking someone to avenge him, for Lonoapii had thrown salt water into his face, Piikea goaded Umi to help him, for he had crossed the seas. So Umi sent messengers with orders that *koa* be felled and many canoes made ready for the crossing to Maui. These were so numerous that when the first canoe reached Kauiki, the last were still in Waipio. The sea was covered with canoes. Umi ordered the canoes to be fastened bow to stern by twos, and in this way the men walked across instead of sailing, the canoes being a dependable road. This is known in legend as the sailing of numberless canoes.

At Kauiki fortress the leader, Hoolae, fought bravely from the top of the hill in daytime, but at night he set a huge wooden image of a man with a bristling war club at the head of the ladder up which the attackers had to climb. The trick frightened away all approaching enemies, while the defenders slept in peace. One night, however, Piimaiwaa took his war club and approached the giant. He hurled insults at him from a safe distance. As his taunts drew no response or movement, Piimaiwaa twirled his war club, threatened with gestures, and gradually crept closer to discover the clever ruse that had fooled the forces of Umi. With that obstacle gone, Umi's men surprised and slew the enemy. Hoolae was captured and killed on the eastern side of Haleakala. War spread over all of Maui, until Lonapii was slain at Waihee and Kihapiilani became king of the island. It could well be that during this campaign Kihapiilani became acquainted with our crater, across which he later constructed the paved trail, another monument to his reign.

Some believe that the trails paved with waterworn rocks were built by *menehune*, the dwarf race supposed to work secretly at night. Actually, the commoners performed the labor, being pressed into service for the task.

They formed an endless chain from the coast, so that rocks could be passed from hand to hand until carefully fitted into place on the walkway. The spaces between the larger stones were often filled with sand or gravel.

The early Mauians had many names for different parts of the crater. Some places had two names; sometimes one name served more than one place. Thus there is duplication of use of the name Haleakala, for besides being the name of the whole volcano, it is also applied to the peak on the rim west of Kaupo Gap. Or could *haole* confusion have given rise to use for the whole mountain a name that once applied only to a prominence on the rim?



White Hill, Pokaoao.

THE TRADITION OF KAOAO

White Hill, Pakaoao, (see [Numbered Points of Interest, topic 6](#)) is of pale gray andesitic basalt that splits into slabs. On the leeward side are many enclosures built of stone, 3 or 4 feet high, which are believed to have been erected as shelters or bivouacs by the men of Kaoao, a quarrelsome chieftain who sought refuge on the mountain after he was driven out of Kaupo, early

in the 18th century. Dr. Kenneth Emory of Bishop Museum has an unpublished manuscript, in Hawaiian, of a legend given to him on June 22, 1922 by Joseph V. Marciel, an old native of Maui. Copy of the translation by Maunupau of Honolulu was graciously given to me so that the story could be told here.



The South Wall: Haleakala Peak on left, Puu Kumu on right.

The *heiau* of Keahuamanono on Haleakala Peak was built by Kaoao, younger brother of Kekaulike, great king of Maui. The brothers were not friends. Kaoao lived on the mountain, but Kekaulike and his men lived 22 by fishing and raising crops in Nuu, the district west of Kaupo Valley. One day Kaoao sent his men north to find food from Keanae to Hana. After they had departed, Kaoao journeyed to his brother's house, which he found deserted since Kekaulike had gone fishing. Kaoao proceeded to pull and destroy all of his brother's crops, and then returned up the mountain.

Kekaulike was very angry when he discovered all his crops had been destroyed. As he knew whom to blame, he ordered his men to wrap 'ala, sling stones, in ti leaves as if they were potatoes. Armed with these they marched up the mountain, and found Kaoao with his bodyguard only, for his men had not returned from the foray for food. The defenders were soon overpowered, but Kaoao jumped over a cliff in an attempt to escape. Kekaulike found him dying, and quickly put an end to him. When Kaoao's men returned from Koolau they found that their leader had been dead many days.

ARCHEOLOGICAL STUDY

Dr. Kenneth Emory made an extensive archeological survey of Haleakala Crater in 1920. He records 58 stone terraces and platforms, 9 groups of open stone shelters, hundreds of *ahu*, and the paved trail of Kihapiilani. ^[1] (See [Numbered Points of Interest, topic 18.](#))

The huge structure built by Kaoao, as mentioned earlier in this chapter, stands in the saddle above Kapalaoa, due south of Puu Maile. This is west of the highest point on Haleakala Peak. It measures 57 × 36 feet and has an eastern supporting wall 18 feet high. This has the appearance of a *heiau*, possibly used for the worship of Pele. As such, it resembles Oalalauo which was located on the rim of Kilauea Iki in the Kilauea Section of Hawaii National Park. Oalalauo, seen in ruins in 1823, was described by the missionary William Ellis, who, probably the first European to go to Kilauea Crater, has given us the first record of a visit. ^[2]

Since the crater is a place of restricted access, it was used for burial sites, which is quite in keeping with practice elsewhere in the Hawaiian Islands. A curious local custom was the deposit of umbilical cords of Kaupo babies in certain localities, principally in the Bottomless Pit ([Numbered Points of Interest, topic 13](#)), and in Na Piko Haua, a pit 15 feet in diameter and 10 feet deep that is located northeast of Halemau Trail, less than a half mile east of Holua Cabin. The cord was wrapped in a small piece of tapa, or, in recent days, in a scrap of gay calico and tied with string. Sometimes it was placed in a bottle or other container. This was then

carefully stowed in crevices or cast into Bottomless Pit. Reasons given for the practice vary. It was believed that if the cord were destroyed or eaten by rats the child would become a thief. Some claimed that proper disposition made a child strong. Some aver that the custom persists to this day, showing, like belief in the existence of Pele, the durability of ancient superstitions.

On the north wall above Paliku is a rock, Pohaku Palaha or Broad Rock, which is called the “hub of East Maui.” Boundary lines radiating from it mark off the pie-shaped land divisions, *ahupuaa*, that extend in all directions to the shores of the ocean.

It is quite natural that legends, traditions, and superstitions should be woven in and about such a great natural feature as this crater. All prominent places had original Hawaiian names, although some were changed with time and some are now lost. Ka Lua o ka Oo was the residence of Kamohoalii, the brother of Pele and the king of vapor. Between Halalii and Ka Moa o Pele is the rim of a spatter cone, Pa Puaa o Pele, which is 30 feet square with an opening on the northwest side. It protrudes only 10 feet above later volcanic deposits. This was a place of highest *kapu* (taboo). Merely to disturb a single grain of sand within it will bring fog and rain, possibly death. Emory discloses the local belief that a stone structure, 9 × 5 feet, located 45 feet east of the rim, holds the bones of two men and a woman who had violated this *kapu* and who had perished in the ensuing fog. His investigation failed to reveal any burial within the structure. In vaguer vein, it was held that a similar fate would be meted to those disturbing a silversword. Were the National Park committed to a policy of nature protection through fear, this belief would be helpful indeed.

THE HISTORICAL BACKGROUND

Captain James Cook discovered Maui on November 26, 1778, as he sailed southwestward from Alaska on his last voyage. His record for the day gives us the first description of Haleakala: “An elevated hill appeared in the country, whose summit rose above the clouds. The land, from this hill, fell in a gradual slope, terminating in a steep, rocky coast; the sea breaking against it in a most dreadful surf.... On the 30th ... another island was seen to the windward, called, by the natives, *Owhyhee*. That along which we had been for some days, was called *Mowee*.”^[3]

After sailing along the eastern and southern coasts of the island of Hawaii in its two armed ships, *Resolution* and *Discovery*, the expedition landed at Kealahou Bay on January 17, 1779. Captain Cook was worshipped as the incarnation of the god Lono, but he overstayed his welcome, ill-will and violence taking its place. A climax was reached over the theft of a ship's cutter, which was broken up merely for its nails and ironware. On February 14, Cook tried to seize the aged king Kalaniopuu to hold as a hostage until reparation was made. In the scuffle that ensued, the Great Mariner was killed by an *alii*, who thrust quite through his back an iron dagger, a chief article of trade of the Expedition. Upon departure toward the northwest, the survivors reached Maalaea Bay on February 24, on which date the journal remarks about Maui: “This side of the island forms the same distant view as the north-east; ... the hilly parts, connected by a low flat isthmus, having, at the first view, the appearance of two separate islands.”

The ill-fated French explorer, Count Jean Francois de Galaup de la Perouse, arrived with his two frigates on May 28, 1786, in the bay southwest of Haleakala that today bears his name. He recorded: “At every instant we had just cause to regret the country we had left behind us; and to add to our mortification, we did not find an anchoring place well sheltered till we came to a dismal coast where torrents of lava had formerly flowed like the

cascades which pour forth their water in other parts of the island.”^[4] This reference is to the latest flows of Haleakala. See [page 37](#).

On Vancouver’s first exploring expedition to the islands, Edward Bell made the following entry for March 6, 1792 in the log of the *Chatham*: “... the south shore ... had by no means a very inviting appearance,—it was remarkably high and seemed extremely barren;—from the top of the Mountains to the waters edge are deep Gullies or ruts form’d I suppose by the water running down,—and there appeared but little wood on this side (except towards the Top) and as little Cultivation, here and there we saw a few Huts and a small Village, several of which appeared half way up.”

On Vancouver’s next visit, Thomas Manby recorded in the journal for March 10, 1793: “... south side of the island which presented a prospect not very grateful to the eye as the land was high and rugged with frequent mounds of Cinders caused by volcanic eruptions.” On March 14, 1793, the botanist A. Menzies, with some of Vancouver’s crew, climbed a valley back of Lahaina and made botanical observations.

When the first missionaries from New England came to Hawaii, Elisha Loomis, a printer, remarked in his journal for March 30, 1820: “As we double the northern extremity of Owhyhee the lofty heights of Maui are on our right.” The spelling, Maui, is evidently a correction made later, as the original spelling appears elsewhere in early missionary usage. Already by 1822, the members of this expedition had adopted the five vowels and seven consonants of the Romance languages used today in reducing the Hawaiian language to writing and printing. Thus, the confusion of earlier English writers was dispelled.

Lorrin Andrews and Jonathan F. Green, ordained missionaries, and Dr. Gerrit P. Judd, physician, were with the third mission from New England. They arrived in Honolulu on March 30, 1828. They visited Rev. William Richards in Lahaina and toured Maui the following summer. Extracts concerning the trip of Dr. and Mrs. Judd were published in 1880 with an introductory 26 note by Albert Francis Judd, a son. In the preface, dated May 1861, Mrs. Judd states that the sketches were “culled and abridged from a mass of papers” without pretense of writing a history. Under the date of July 1828,

the narrative relates a Fourth of July excursion and includes: “The mountain on the east division is Haleakala (house of the sun), and is the largest crater in the world, but is not in action.”^[5] Unfortunately, the original notes have been lost, and the reference to the mountain by name must have been inserted at a much later date in preparing the manuscript for publication. It would be hard to believe that Mrs. Judd could possibly have started the fiction, “largest crater on earth,” at the early date of 1828. The Judds did not climb the mountain during their visit, and Hawaiians were not in a position to make comparisons among craters of the world.

On August 21, 1828, Richards, Andrews, and Green made the first recorded ascent of Haleakala. They could not have known a name for the mountain, for they refer to it only as “the highest land on Maui” and as “an extinct volcano.” Not until six years later was their account published. The following quotations relate to their trip:^[6] “Mr. Richards had for years been particularly desirous of making the tour of this island for the purpose of examining and improving the schools, etc., but having been alone, it has hitherto been impracticable for him to leave his family for a sufficient length of time. During the present season this object has been accomplished.” Mr. Richards had arrived in Hawaii in 1823, and had taken over the mission in Lahaina shortly afterwards at the request of Queen Keopuolani. He had not mentioned the big mountain in previous correspondence and reports.

“Here (at ‘Kaalimaile,’ perhaps the Haliimaile of today) we tarried overnight, intending, in the morning, to ascend the mountain, near which we were, and sleep on the highest land on Maui. We were told by the natives, that the way was long, but the ascent very easy. We suppose no English travellers had ever ascended this mountain.

“21. We rose early, and prepared for our ascent. Having procured a guide, we set out; taking only a scanty supply of provisions. Half way up the mountain, we found plenty of good water, and, at a convenient fountain, we filled our calabash for tea. By the sides of our path, we found plenty of ohelos, (a juicy berry, very palatable,) and, occasionally, a cluster of strawberries. On the lower part of the mountain, there is considerable timber; but as we proceeded, it became scarce; and, as we approached the summit, almost the only thing, of the vegetable kind, which we saw, was a

plant which grew to the height of six or eight feet, and produced a most beautiful flower. It seems to be peculiar to this mountain, as our guide and servants made ornaments of it for their hats, to demonstrate to those below, that they had been to the top of the mountain.

“It was nearly 5 o’clock, when we reached the summit; but we felt ourselves richly repaid for the toil of the day, by the grandeur and beauty of the scene, which at once opened up to our view. The day was very fine. The clouds, which hung over the mountains on West Maui, and which were scattered promiscuously, between us and the sea, were far below us; so that we saw the *upper side* of them, while the reflection of the sun painting their verge with varied tints, made them appear like enchantment. We gazed on them with admiration, and longed for the pencil of Raphael, to give perpetuity to a prospect, which awakened in our bosoms unutterable emotions. On the other side, we beheld the seat of Pele’s dreadful reign. We stood on the edge of a tremendous crater, down which, a single misstep would have precipitated us, 1,000 or 1,500 feet. This was once filled with liquid fire, and in it, we counted sixteen extinguished craters. To complete the grandeur of the scene, Mouna Kea, and Mouna Roa lifted their lofty summits, and convinced us, that, though far above the *clouds*, we were far below the feet of the traveller who ascends the mountains of Hawaii. By this time, the sun was nearly sunk in the Pacific; and we looked around for a shelter during the night. Our guide and other attendants we had left far behind; and we reluctantly began our descent, keeping along on the edge of the crater.

“After descending about a mile, we met the poor fellows, who were hobbling along on the sharp lava, as fast as their feet would suffer them. They were glad to stop for the night, though they complained of the *cold*. We kindled a fire, and preparations were made for tea and lodgings. The former we obtained with little trouble. We boiled part of a chicken, roasted a few potatoes, and, gathering round the fire, we made a comfortable meal; but the place of lodging, we obtained with some difficulty. At length, we spread our mats and blankets in a small yard, enclosed, probably, by natives, when passing from one side of the island to the other. We were within twenty-feet of the precipice, and the wind whistled across the valley, forcibly reminding us of a November evening in New England. The thermometer had fallen from 77 to 43* (*The next morning, the thermometer

stood at 40.), and we shivered with the cold. The night was long and comfortless.

“22. Early in the morning, we arose, and reascended the mountain, to its summit, and contemplated the beauties of the rising sun, and gazed a while longer, on the scenery before us. There seemed to be but two places, where the lava had found a passage to the sea, and through these channels, it must have rushed with tremendous velocity. Not having an instrument, we were unable to ascertain the height of the mountain. We presume it would not fall short of 10,000 feet.* (*This, I believe, is the height at which it has generally been estimated.) The circumference of the great crater, we judged to be no less than fifteen miles. We were anxious to remain longer, that we might descend into the crater, examine the appearance of things below, and ascend other eminences; but as we were nearly out of provisions, and our work but just commenced, we finished our chicken and tea, and began our descent.

“Nothing remarkable occurred, on our way down....”

The United States Exploring Expedition under the command of Lieutenant Charles Wilkes, USN, visited Hawaii in 1840 and 1841. On February 15, 1841, Wilkes dispatched Messrs. Pickering, Drayton, and Breckenridge from Hilo to explore Maui. They were joined at Lahaina by Rev. Andrews, his son, four students of the seminary, and six *kanakas* to carry their food. At Wailuku they were joined by Mr. Bailey (see [page 40](#)). They spent the 29 night at an elevation of 1,692 feet on the sugar plantation of Lane and Minor, two Bostonians. The story of their ascent, which is the second [7] recorded, is told by quoting from their report:

“The next day, the party set out at an early hour, in hopes of reaching the summit, but it began to rain violently, in consequence of which they took shelter in a large cave, at an altitude of eight thousand and ninety feet. Here many interesting plants were found, among which were two species of *Pelargonium*, one with dark crimson, the other with lilac flowers; the *Argyroziphium* began to disappear as they ascended, and its place was taken up by the silky species, which is only found at high altitudes. From the cave to the summit they found shrubby plants, consisting of *Epacris*, *Vaccinium*, *Edwardsia*, *Compositae*, and various rubiaceous plants.

“On their arrival at the edge of the crater, on the summit, the clouds were driving with great velocity through it, and completely concealed its extent. The height, as ascertained by the barometer, was ten thousand two hundred feet. The driving of the sleet before the strong gale soon affected the missionaries and native students, the latter of whom for the first time, felt the effects of cold. The limit-line of woods was ascertained to be at six thousand five hundred feet.

“Some sandalwood bushes were noticed about five hundred feet above the cave. Above the cave the ground assumed a more stony appearance, and the rock became now and then more visible, which had not before been the case. Where the rock was exposed it was found to be lava more or less vesicular, but no regular stream was observed. The surface of the lava appeared to be more thickly covered with earth than that of Mauna Kea, and consequently a greater proportion of soil existed, as well as a thick coating of gravel. Near the summit, bullock-tracks were observed, and likewise those of wild dogs, but no other animals were seen except a few goats.

“The crater of Haleakala, if so it may be called, is a deep gorge, open at the north and east, forming a kind of elbow: the bottom of it, as 30 ascertained by the barometer, was two thousand seven hundred and eighty-three feet below the summit peak, and two thousand and ninety-three feet below the wall. Although its sides are steep, yet a descent is practicable at almost any part of it. The inside of the crater was entirely bare of vegetation, and from its bottom arose some large hills of scoria and sand: some of the latter are of an ochre-red colour at the summit, with small craters in the centre. All bore the appearance of volcanic action, but the natives have no tradition of an eruption. It was said, however, that in former times the dread goddess Pele had her habitation here, but was driven out by the sea, and then took up her abode on Hawaii, where she has ever since remained. Can this legend refer to a time when the volcanoes of Maui were in activity?

“The gravel that occurred on the top was composed of small angular pieces of cellular lava, resembling comminuted mineral coal. The rock was of the same character as that seen below, containing irregular cavities rather than vesicles. Sometimes grains of chrysolite and horn-blende were disseminated. In some spots the rock was observed to be compact, and had the appearance

of argillite or slate: this variety occurred here chiefly in blocks, but was also seen in situ. It affords the whetstones of the natives, and marks were seen which they had left in procuring them.

“Of the origin of the name Mauna Haleakala, or the House of the Sun, I could not obtain any information. Some of the residents thought it might be derived from the sun rising from over it to the people of West Maui, which it does at some seasons of the year.

“Having passed the night at the cave, Mr. Baily (*sic*) and young Andrews preferred returning to the coast, rather than longer to endure the cold and stormy weather on the mountain.

“Our gentlemen made excursions to the crater, and descended into it. The break to the north appears to have been occasioned by the violence of volcanic action within. There does not appear any true lava stream on the north, but there is a cleft or valley which has a steep descent: here the soil was found to be of a spongy nature, and many interesting plants were found, among the most remarkable of which was the arborescent Geranium.

“The floor of the crater, in the north branch, is extremely rough and about two miles wide at the apex, which extends to the sea. In the ravines there is much compact argillaceous rock, similar to what had been observed on Mauna Kea, retaining, like it, pools of water. The rock, in general, was much less absorbent than on the mountains of Hawaii.

“Mr. Drayton made an accurate drawing or plan of the crater, the distances on which are estimated, but the many cross bearings serve to make its relative proportions correct. Perhaps the best idea that can be given of the size of this cavity, is by the time requisite to make a descent into it being one hour, although the depth is only two thousand feet. The distance from the middle to either opening was upwards of five miles; that to the eastward was filled with a line of hills of scoria, some of them five or six hundred feet high; under them was lying a lava stream, that, to appearance, was nearly horizontal, so gradual was its fall. The eastern opening takes a short turn to the southeast, and then descends rapidly to the coast.

“At the bottom were found beds of hard gravel, and among it what appeared to be carbonate of lime, and detached black crystals like augite, but chrysolite was absent.

“From the summit of the mountain the direction of the lava stream could be perceived, appearing, as it approached the sea, to assume more the shape of a delta.

“From the summit the whole cleft or crater is seen, and could be traced from the highest point between the two coasts, flowing both to the northward and eastward. Volcanic action seems also to have occurred on the southwest side, for a line of scoria hills extends all the way down the mountain, and a lava stream is said to have burst forth about a century ago, which still retains its freshness. The scoria hills on the top very much resemble those of Mauna Kea, but the mountain itself appears wholly unlike either of the two in Hawaii, and sinks into insignificance when compared with them.

“Although I have mentioned lava streams on this mountain, yet they are not to be understood as composed of true lava, as on Mauna Loa; none of the latter were seen except that spoken of on the southwest side, and none other is believed to exist. No pumice or capillary glass was at any time seen, 32 nor are they known to exist on this island. On the wall of the crater, in places, the compass was so much affected by local attraction as to become useless.

“Near the summit is a small cave, where they observed the silkworm eggs of Mr. Richards, which were kept here in order to prevent them from hatching at an improper season. The thermometer in the cave stood at 44°; the temperature at the highest point was 36°, and in the crater 71°. After three days’ stay, the party returned to the establishment of Messrs. Lane and Minor, and thence to Wailuku. They were much gratified with their tour.”

The name Haleakala can thus be regarded as having been formally introduced by members of the Wilkes Expedition. As the fame of the beauties and wonders of the mountain spread, visitors from all parts of the globe came to make the arduous climb to the summit. Most found shelter from the elements in natural caves. Big Flea and Little Flea caves, a quarter of a mile from the summit, are often mentioned in early accounts. That their

accommodation was not highly relished can be seen from a description by Damon in 1847: "... which did not hold out many attractions, and I have good reasons for believing it already possessed tenants that would sharply contend for occupancy with any way-faring and luckless wight."^[8] In tales of early visits, literature—especially the Bible—was gleaned for phraseology that might help portray emotions felt; fantastic similes and metaphors were drawn to transmit comprehension of the scene. On a visit in 1853, G. W. Bates mentioned: "From the point where I stood a huge pit,^[9] capable of burying three cities as large as New York—opened before me." True, New York then lacked its present colossal stature, but a milder expression, "could hold the whole of New York City" still is in use today. For information, the following areas are given from *Thrum's Annual* and the *World Almanac*: area of Haleakala "crater," 19.0 sq. mi.; area of Maui, 728 sq. mi.; area of Manhattan borough, 31.2 sq. mi.; area of New York City, 381 sq. mi. Discomforts, silversword, sandalwood, wild dogs, cattle, goats, the weather, and personal impressions form much of the subject matter of early essays.

The early residents of Maui recognized the value of the mountain as a scenic feature and tourist attraction. Their first move was for better overnight shelter on the mountain. C. W. Dickey in 1894 raised \$850 by popular subscription for material with which to build a simple shelter at Kalahaku Lookout. H. P. Baldwin and the sugar plantations furnished labor and pack animals. The long trip of 25 miles to the location had to be made on foot or by saddle, and required a full, tiring day; all building material except rock had to be transported by pack stock. In painfully characteristic manner, many of those for whose benefit the sweat and toil were expended proved unworthy since they roughly abused the structure. Windows were broken, timber in the floor and walls was ripped out and used for firewood, and garbage and filth accumulated. A tropical storm added to the damage by unroofing the house; Worth Aiken raised \$1,500 for its renovation and repair. In 1914-15, the cabin was improved with a concrete floor, metal doors, and metal shutters. Two additional dormitories were added in 1924-25 at a cost of \$11,000 and operation of the building was turned over to E. J. Walsh, manager of the Grand Hotel, Wailuku. Usefulness dropped with the opening of the Haleakala Road, so that on September 24, 1934 its custodian, the

Maui Chamber of Commerce, transferred ownership to the National Park Service. The structures were razed in 1957, but plans are underway to replace them with a modern observatory in which people may look at the scene in glass-enclosed comfort.

In the movement to create a National Park in Hawaii in the early part of the 20th century, the idea developed that it should consist of the craters of Kilauea, Mauna Loa, and Haleakala. The citizens of Maui gave full approval for including their beloved mountain. Hawaii National Park, composed of these three sections, was established by Act of Congress on August 1, 1916, but formal dedication was delayed until 1921. Following improvement of Halemau Trail in 1929, a permanent ranger position was set up for continuous attention to the area. Today it is administered by an assistant superintendent who has a staff of two rangers and a naturalist.

The building of a road to the summit was fulfillment of a promise to the people of Maui when the park was created. The first step had to be 34 construction by the Territory of a highway from Pukalani Junction to the park boundary near Puu Nianiau. This was completed at a cost of \$504,000 in April 1933 after 39 months of work. As with similar projects elsewhere in Hawaii, superstitions of long-ago reappeared. Although not antagonistic to progress, Hawaiians raised a cry that all effort was futile; the chicken god, Kalau-heli-moa, would conspire and never permit the project to be completed. Every mishap was attributed to this nemesis.

The Park Service completed its commitment soon afterwards at the cost of \$376,000. The road was armor-surfaced in the fall of 1935. Extensive, appropriate dedication ceremonies were held for the opening.

With the establishment of a Civilian Conservation Corps (CCC) camp in Kilauea, a strip camp of 25 men was set up near Puu Nianiau cinder cone. Part of the time, a tent camp was also established within the crater. Many improvements became possible through this undertaking. The observatory building on the summit was constructed in 1936; the three shelters within the crater, Kapalaoa, Paliku, and Holua, were built a year later. Prior to their completion, overnight shelter was sought in caves, the best known being Bubble Cave (see [Numbered Points of Interest, topic 15](#)) and Holua Cave which is in the pali wall behind the cabin.

The CCC camp was abandoned in April 1941, and its structures were turned over to the Army that greatly improved them. During the period of its occupancy, the Army constructed for radar installation the ugly concrete block house that still protrudes on the summit of Red Hill. With evacuation of the military, the CCC quarters were adapted for the service of a concessioner to supply meals and lodging within the park.

Normal travel to Haleakala was interrupted for a year when the section was closed following the attack on Pearl Harbor, December 7, 1941. Annual travel had passed the 25,000 mark in 1939. Travel for the year 1958 reached a high of 56,940 visitors.

The summit of Haleakala attracted early consideration as a site for various scientific studies. It is a prime target for many scientific planners, because the high mountain is situated in the middle of a broad ocean, yet the summit is readily accessible by road. Often this eagerness has obscured possible use of other suitable sites and has clashed with the basic purpose of the area as a National Park. The National Park Service strives to keep the scene as primitive as possible, assiduously blending buildings and structures into the landscape, whereas the non-conformist gives no thought to this but follows the easiest way. Too often economy of construction and operation, together with careless housekeeping, invert an attraction to a repulsion which even the splendor of the scene cannot offset. Skyline buildings, obtrusively strung wires, thoughtlessly gouged land, and abandoned debris are not conducive to aesthetic experience, whatever their purpose or whoever the offender.



Haleakala holiday.

The earliest scientific study associated with the mountain concerned weather. It is said that the Hawaiian Islands are situated near a critical area in the Pacific which is a birthplace of weather. The summit offers an ideal place for detection and observance of the formation of high clouds. Much additional research is needed to provide a steady flow of data for successful and safe operation of air transportation. Many crashes have been blamed on lack of weather data. The task is not always simple, as statistical readings may be confusing. Those made on Red Hill, for example, are influenced by local circulation set up by the heating of several square miles of black, barren lava and cinders. 36

The Federal Aeronautics Administration maintains a station a mile beyond Red Hill. In the earlier fifties cosmic radiation was studied with a huge revolving truss located back of Red Hill. The military has set up apparatus and teams from time to time for experiments with radio, radar, and radiation. Finally, for the International Geophysical Year, Haleakala summit was

chosen for one of the important satellite tracking stations supervised by the Smithsonian Institution of Washington.

IMPORTANT DATES

500 A. D., ca.—Hawaii discovered by Hawaii-loa, Polynesian fisherman-navigator who, tradition says, came from Kahiki (Tahiti?), an island to the south. He made several round trips, bringing with him a large company of retainers.

1100 ca.—After a wave of navigation, intercourse with Tahiti ended.

1300 ca.—According to an ancient chant, *mele*, Kalaunuiohua, *moi* of Hawaii, conquered Maui. *Moi*, in 19th century Hawaiian, signifies the supreme ruler or head chief, now usually termed *king*.

1500—Piilani, king of Maui. He was succeeded by Lonoapii who in turn was overthrown by his brother Kihapiilani and his brother-in-law, King Umi of the Big Island. Bloody battles stretched from Kauiki to the sands of Waihee.

1555—Possible discovery of Hawaii by Juan Gaetano, Spanish navigator. He prepared a manuscript chart now in the Spanish archives which contains a group of islands in the latitude of Hawaii but whose longitude is 10 degrees too far to the east. What corresponds to Maui is called *La Desgradiada*, the unfortunate. The largest, most southerly island, which should be the present Hawaii is labelled *La Mesa*, the table. Three other islands, appearing to be 37 Kahoolawe, Lanai, and Molokai, are called *Los Monjes*, the monks.

1736—King Kekaulike died and was succeeded by Kamehamehanui.

1737—Alapainui, *moi* of Hawaii, invaded Maui via Kaupo. He took with him two young princely half-brothers, Kalaniopuu and Keoua. Keoua was father of Kamehameha I. Since Alapainui found his adversary, Kekaulike, dead, he made peace with the nephew, Kamehamehanui. The two joined forces to repel the invader, Kapiiohokalani of Oahu,

in bloody, obstinate battles that ended in the rout of the Oahu army at Kawela, Molokai.

1738—At Keawawa, West Maui, Alapanainui and Kamehamehanui decisively defeated Kauhi, the latter's brother and usurper of power.

1750 ca.—The calculated date of the most recent activity of Haleakala, the Keoneoio flow above La Perouse Bay. The flow originated at Kaluaolapa at an elevation of 575 feet, and from vents one mile further northeast at an elevation of 1,550 feet. The method of dating is interesting. In 1841, Rev. Edward Bailey of Wailuku inquired about the eruption and was informed by Hawaiians that it happened at the time of their grandfathers. In 1906, Lorrin A. Thurston was told by a Chinese-Hawaiian cowboy, Charles Ako, that his father-in-law's grandfather at the time of the event was just old enough to carry "two" coconuts 4 or 5 miles from the sea to the upper road at an elevation of 2,000 feet. Since Hawaiians counted coconuts by fours, "two" probably refers to a total of eight nuts. Mr. Bailey was told that a woman and child were trapped by the flow but escaped after it cooled. By 1922, 80 years later, this tale had grown into a neo-myth about a husband and wife with their two children. The mother and her young daughter fled *mauka*, but were seized by Pele, and turned into the two lava columns that stand beside the vent at Kaluaolapa. The father and son, plunged into the sea and started swimming toward Kahoolawe. Pele cast rocks after them and turned the two to stone. The two rocks, a big and a little one, can be 38 seen today rising out of the sea several hundred feet out from shore as proof of the tale. Mr. Thurston's estimate of the date of the eruption is 1750 while J. F. G. Stokes, Hawaiian ethnologist, favors a later date, possibly 1770.

1754—Kalaniopuu, warlike king of Hawaii, captured the fortress Kauiki and held it successfully for more than 20 years.

1765—Kamehamehanui died and was succeeded by his brother, Kahekili.

1768—Queen Kaahumanu was born at Kauiki. She became the favorite wife of Kamehameha I.

- 1775—Kalaniopuu was defeated by Kahekili at Kaupo.
- 1776—Kalaniopuu invaded Maui at Maalaea; his army was annihilated on the sand hills near Wailuku.
- 1777—Kalaniopuu took Lanai but again was repelled when he tried to invade Maui.
- 1778, November 26—Captain James Cook, Royal British Navy, discovered Maui.
- 1781—Kahekili reconquered East Maui. He recaptured the fort at Kauiki by cutting off the water supply. To show contempt, he baked the bodies of the defenders in earth ovens.
- 1786—Kamehameha I sent an expedition to recapture East Maui. It was defeated at Kipahulu by Kalanikapule, the son of King Kahekili.
- 1786, May 28—La Perouse visited Maui and camped on Keoneoio lava flow.
- 1790—Olowalu massacre. The ship, *Eleanor*, under Captain Simon Metcalf, treacherously opened fire on native boats following a truce made after one white sailor had been murdered. More than a hundred natives were slaughtered.
- 1790—Conquest of Maui by Kamehameha I after landing at Hana. He decisively defeated Kalanikapule, in the Battle of Iao Valley or Kepaniwai.
- 1793—Vancouver visited Maui on his second expedition. He tried to bring about an end to the wars and to establish a lasting peace between Maui and Hawaii.
- 1795—Maui was subdued by Kamehameha I without a battle.
- 1819—Kamehameha I, king of all Hawaii, died. Abolition of the *kapu* system by Kamehameha II, incited by his guardian, Queen Kaahumanu.

1823—The Christian mission at Lahaina was founded by Rev. William Richards and A. S. Stewart. On September 16, Queen Keopuolani, a wife of Kamehameha I and a devout Christian, died at Lahaina. She was buried with services by Rev. William Ellis.

1824—At Lahaina, Queen Regent Kaahumanu orally proclaimed a law forbidding desecration of the Sabbath, fighting, murder, and theft.

1825—The English frigate *The Blonde* anchored off Lahaina with the bodies of King Liholiho (Kamehameha II) and his queen, Kamamalu. They had died from measles while on a visit to London.

1825—The crew from the Whaler *Daniel* attempted to demolish the home of Rev. Richards, Lahaina.



Sliding Sands Trail.

- 1826—Mosquitoes from Mexico were introduced at Lahaina by the SS *Wellington*.
- 1827—The Whaler *John Palmer* fired on the home of Rev. Richards.
- 1829—Ascent of Haleakala by a missionary party.
- 1834—First Hawaiian newspaper *Lama Hawaii* was published at Lahainaluna Mission School.
- 1839—An Hawaiian “Bill of Rights” was signed at Lahaina by Kamehameha III. It afforded protection to all people and their property while they conformed to the laws of the kingdom.
- 1841—Haleakala Crater was visited by Pickering and Breckenridge of the United States Exploring Expedition under Captain John Wilkes, U. S. Navy.
- 1841-1849—Peak of whaling industry in Hawaii. Lahaina was visited by 596 Whalers in 1846.
- 1850—David Malo, Hawaiian antiquarian and teacher at Lahainaluna School, Lahaina, conducted Rev. William P. Alexander and Curtis Lyons from Kaupo through Haleakala Crater, to Makawao, “a trip never before undertaken by white men.”
- 1876—S. F. Alexander and H. P. Baldwin started construction of the Hamakua Ditch, first big irrigation project in Hawaii.
- 1890—First pineapples were planted at Haiku.
- 1893—Overthrow of the monarchy and establishment of the Republic of Hawaii. Queen Liliuokalani was the last reigning sovereign.
- 1898, August 12—Hawaii was annexed to United States by joint legislation of Congress. President Dole was appointed first governor.
- 1916—Hawaii National Park was established by Act of Congress on August 1 with Haleakala Crater forming the Section on Maui.

1921—Hawaii National Park was formally opened.

1929, November 11—Establishment of commercial air service between the islands.

1930—First permanent park position (ranger) was established to give continuous service at Haleakala.

1931—First permanent park naturalist was appointed for Hawaii National Park, although temporary, summer interpretive services were started in the late twenties by the employment of Otto Degener, formerly botanist at the University of Hawaii. Dr. Degener, presently writing Book 6 of his “*Flora Hawaiiensis*,” kindly supplied many of the scientific plant names for this guide. 41

1935 February 23—Dedication ceremonies of Haleakala Road in Hawaii National Park.

1937—Kapalaoa, Paliku, and Holua cabins were constructed.

1941—Haleakala closed to travel for military reasons.

1952-3—Present exhibits were installed in Summit Observation Station.

1958—Permanent naturalist position established for Haleakala.

1959—Hawaii becomes the 50th state in the Union.

GEOLOGY

THE ORIGIN OF THE SCENIC FEATURES

The Island of Maui was built by two volcanoes. That forming West Maui is deeply dissected into several high peaks. The old summit crater now is encompassed by the head of Iao Valley. Mt. Kukui, the highest point, has an elevation of 5,788 feet.

East Maui is built of three series of lava products from Haleakala Volcano whose flows extended westward across the present isthmus to come to rest against the base of West Maui Volcano. These represent three great periods of activity, the latter two being separated by a long interval of quiescence that was characterized by intense erosion and mild, if any, eruptions. Geologists designate these three periods by the names Honomanu, Kula, and Hana. The mountain was built over three rifts, northerly, easterly, and southwesterly, each extending about fifteen miles.

In earliest (Honomanu) time, about the beginning of the Ice Ages, a symmetrical shield like Mauna Loa was built of pahoehoe and aa basalts 8,500 feet above present sea-level. During the next (Kula) cycle, eruptions were more explosive in nature; flows were composed of more viscous andesite between which layers of ash and soil accumulated. Big cinder cones and extensive ash beds were formed at this time. Like Mauna Kea today, the Honomanu dome was capped by a craterless mound of cinders, 2,500 feet high, that was studded with many lesser cones. The summit was a mile east of the present top on Red Hill and a thousand feet higher than it is today.

As Kula eruptions declined and grew less frequent, running water cut deeply into the sides of the mountain and excavated four great valleys, Keanae, Kaupo, Kipahulu, and Waihoi, that had broad heads, thousands of feet deep. Numerous lesser valleys were later to be buried more or less by lava flows. Most of the eastern summit ridge was worn away; Kaupo and Keanae Valleys met near the summit and fused into a great depression like that near

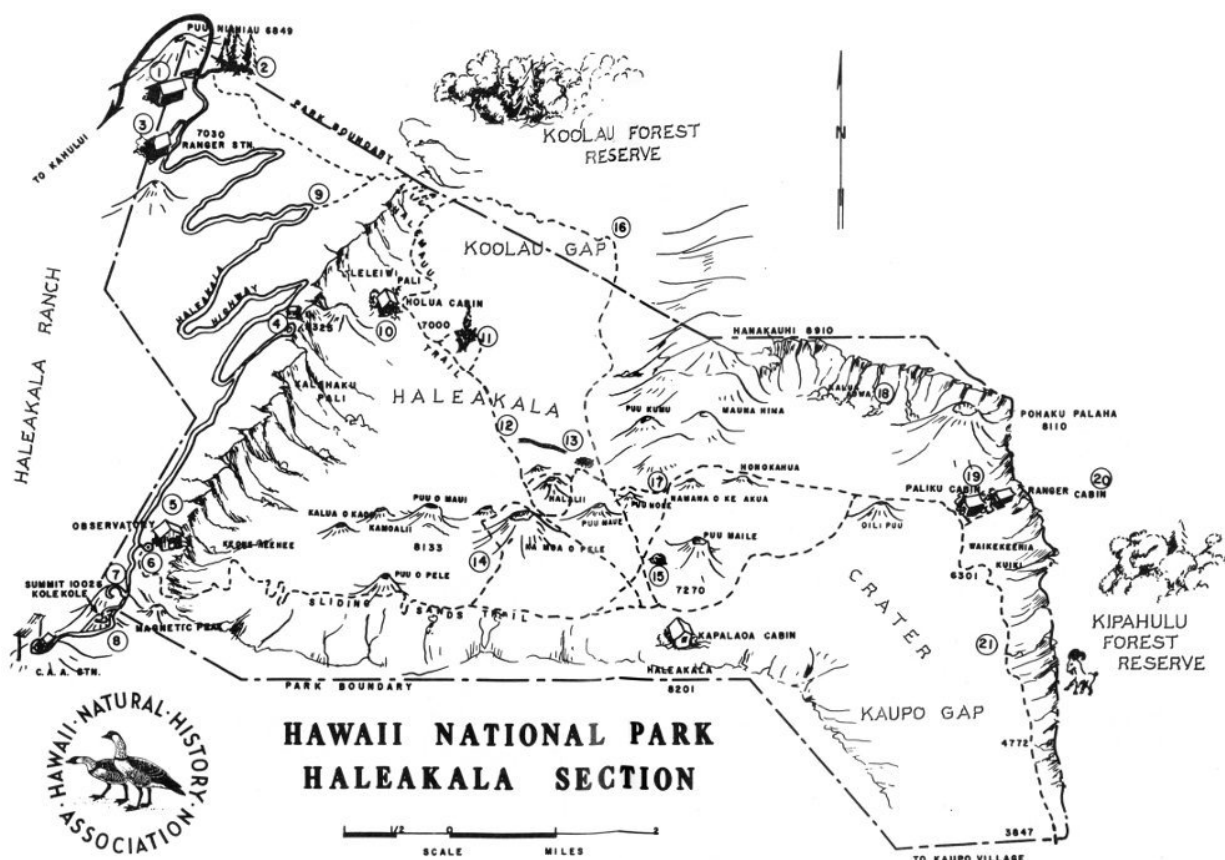
the head of Iao Valley today. At one time, a great flow of mud, probably triggered by an earthquake, swept all before it as it moved down Kaupo Valley into the sea. Its remnants today are 350 feet deep at Puu 43 Maneoneo near the coastal road. A similar mass movement of rock on soft mud was started by an earthquake on April 2, 1868 at Wood Valley, west of the Kilauea Section of Hawaii National Park; the flow, in its precipitous descent, buried a village with 31 people and more than 500 head of stock.

In recent times, volcanism again quickened at Haleakala, giving the third (Hana) series of volcanics. This veneered the east and west slopes of the volcano, covered the floor of the depression, and pushed great lava flows through Koolau and Kaupo Gaps to the sea. Large flows and cones mask the divide that delimited the two great valleys. During Hana time, the northern rift alone remained inactive. The most recent activity, dated by Hawaiian legend as 1750, is represented by two bare, black flows above La Perouse Bay, the southwest corner of the island.



The Recent Cinder Cones.

Haleakala Crater, 7 miles long and $2\frac{1}{2}$ miles wide, is locally proclaimed the largest extinct crater on earth, but the claim like the name is inaccurate. Nevertheless, it possesses a most unusual geological origin and beauty that give it a worthy place among the National Parks.



HAWAII NATIONAL PARK HALEAKALA SECTION

[High-resolution Map](#)

NUMBERED POINTS

1. PARK ENTRANCE; SILVERSWORD INN
2. HOSMER GROVE CAMPGROUND AND PICNIC AREA
3. ADMINISTRATION BUILDING, HAWAII NATIONAL PARK, HALEAKALA SECTION
4. KALAHAKU OVERLOOK; SILVERSWORD PLANTS
5. OBSERVATORY
6. WHITE HILL, START OF SLIDING SANDS TRAIL
7. RED HILL; SUMMIT OF HALEAKALA
8. SKYLINE DRIVE
9. START OF HALEMAUU TRAIL
10. HOLUA CABIN
11. SILVERSWORD LOOP
12. ANCIENT HAWAIIAN TRAIL
13. BOTTOMLESS PIT
14. KA MOA O PELE TRAIL
15. BUBBLE CAVE
16. WAIKAU CABIN

- 17. VOLCANIC DIKES; ANCIENT DIVIDE BETWEEN KOOLAU AND KAUPU RIVER VALLEYS
- 18. LAUULU TRAIL
- 19. PALIKU CABIN
- 20. KIPAHULU VALLEY
- 21. KAUPU TRAIL
- KIPAHULU FOREST RESERVE
- KOOLAU FOREST RESERVE
- HALEAKALA RANCH

THE GEOLOGICAL INTERPRETATION

The summit depression of Haleakala stimulates speculation, and competent geologists have come up with widely differing hypotheses regarding its origin. In the account of their visit (see The Historical Background, [p. 29](#)), the first foreign visitors naturally used the term “crater,” which has been in vogue ever since. Pickering and Drayton of the Wilkes Expedition remark, [\[10\]](#)

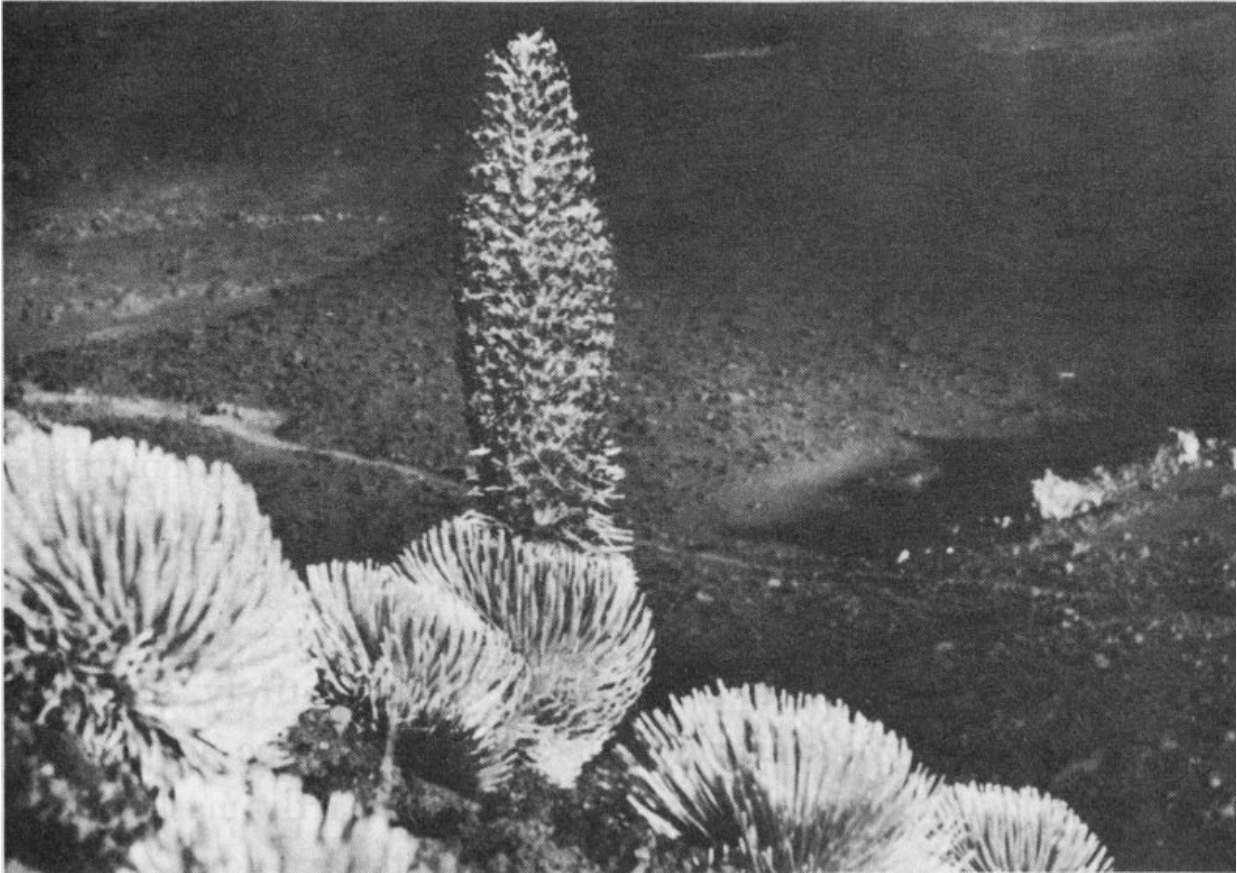
“The crater of Haleakala, if so it may be called, is a deep gorge.”

Drayton’s sketch was the first published map of the crater. James Dana, the great geologist with the Expedition, sailed past the mountain and later wrote a physiographic description based on notes made by Pickering and Drayton. In the official report, he expressed the idea, suggested by the crude map, that the mountain has been ripped apart by mighty convulsions that attended the most recent activity, so that the northeastern (Hana) part was separated along a zigzag crack from the rest of the mountain by the width of Keanae and [\[11\]](#)

Kaupo Valleys. During the great eruptions that attended the rending, lava covered the floor and poured in great floods through Koolau and Kaupo Gaps.

W. D. Alexander, who surveyed the crater in 1869, believed: “... this is a real terminal crater, and not merely ‘a deep gorge open at the north and east’ or a caldera. I have indeed heard the theory proposed that the mountain is but a wreck of a complete dome with a small terminal crater, the whole top of which has fallen in and been carried away, as is supposed to have been the case with some of the volcanoes of Java, and the caldera of Palma.” [\[12\]](#)

C. E. Dutton, volcanologist of the United States Geological Survey, objected to Dana's explanation and wrote that the depression is "strictly 47 homologous" to Kilauea Crater, that is, a collapsed caldera. ^[13] He assumed that this had been tapped by the upper ends of Keanae and Kaupo drainages. In 1887, Dana had opportunity to make a quick trip through the crater and down Kaupo Valley, so that he tempered his earlier opinion and ^[14] decided that Keanae and Kaupo valleys might be graben. Reginald Daly of Harvard rejected the hypothesis that the depression was like Kilauea Crater, since arcuate faults so prominent at Kilauea are apparently absent at Haleakala. ^[15] In his paper on petrography, Whitman Cross stated, "What is commonly called the crater of Haleakala appears to me to be, in some part at least, a result of erosion." ^[16] At about the same time, Sidney Powers stated his belief that Kipahulu and Waihoi Valleys are graben, but he based his ^[17] opinion on "authentic reports" and does not claim that he saw the valleys.



Ahinahina (silversword).

H. T. Stearns analyzed carefully all the supporting evidence and 48 objections to the various viewpoints.^[18] Of all of his profession, he made the most thorough field surveys, with the conclusion that the “crater” is chiefly erosional, affected by the recession of two great amphitheater-headed valleys, instead of by collapse, sliding away of the side of a cone, or explosion. Small craters may have existed at the time, but not a large one resembling the present depression. He believed that the big size results from the fact that the heads were offset and not in a straight line. The shape of the depression is what would result upon the fusion of two amphitheater heads similar to those of Waikoi, Kipahulu, and Manawainui valleys of today. Each of these is a typical Hawaiian valley, narrow at the base, but with a broad amphitheater at the head. Kipahulu is separated from Waihoi and from Kaupo Valleys by narrow divides. It can be assumed that a similar divide once separated the amphitheaters at the heads of the early Kaupo and Keanae drainages. Stearns further believes that once the rift zone was reached,

stream erosion was greatly accelerated because of the loosely knit structure, the presence of many weak cinder cones, and the dike complex. This complex, as in other places in Hawaii, must have yielded perennial spring water to accelerate erosion. His summary is as follows:

“No stratigraphic or constructural evidence was found to support the hypothesis that Haleakala Crater is a true caldera, that it was formed by renting, or that Keanae and Kaupo valleys tapping this depression are grabens. Instead, detailed mapping and the examination of water tunnels show that Haleakala dome has been eroded by a number of great valleys. The hypothesis is presented that the so-called “crater” of Haleakala is chiefly, if not entirely, the result of the coalescence of the amphitheater heads of Kaupo and Keanae valleys and that renewed volcanic activity has partly masked their former divide and partly filled these valleys with lava flows.”

HALEAKALA PLANTS

Haleakala rises above the belt of warm trade winds into the cold, dry climate of the Alpine Zone. Temperatures at night may drop below freezing even in the warmer part of the year; the growing season is short and life is severe. The sparse plants that can live here crouch closely to the ground, diffusing or forming compact rosettes. All they have in which to grow are porous rocks and loose cinders that cannot hold moisture, lack organic matter, and do not yield a firm base for rooting. Species extending to lower elevations are here depauperate from wind and cold, although elsewhere they may attain a sturdy stature, even tree-size. Few seedlings are seen; individuals whose life span has finished remain conspicuous in death. Neither trees nor mats of shrubs are seen, but mosses and lichens find this Alpine desert to their liking. Oddly enough, one looks in vain for brilliant blossoms, such as one has come to expect in the high Sierra and the Rockies. The summit is a biological island on an oceanic island which, geological evidence indicates, never was part of a continent. The Hawaiian Chain was never near land of appreciable height and size.

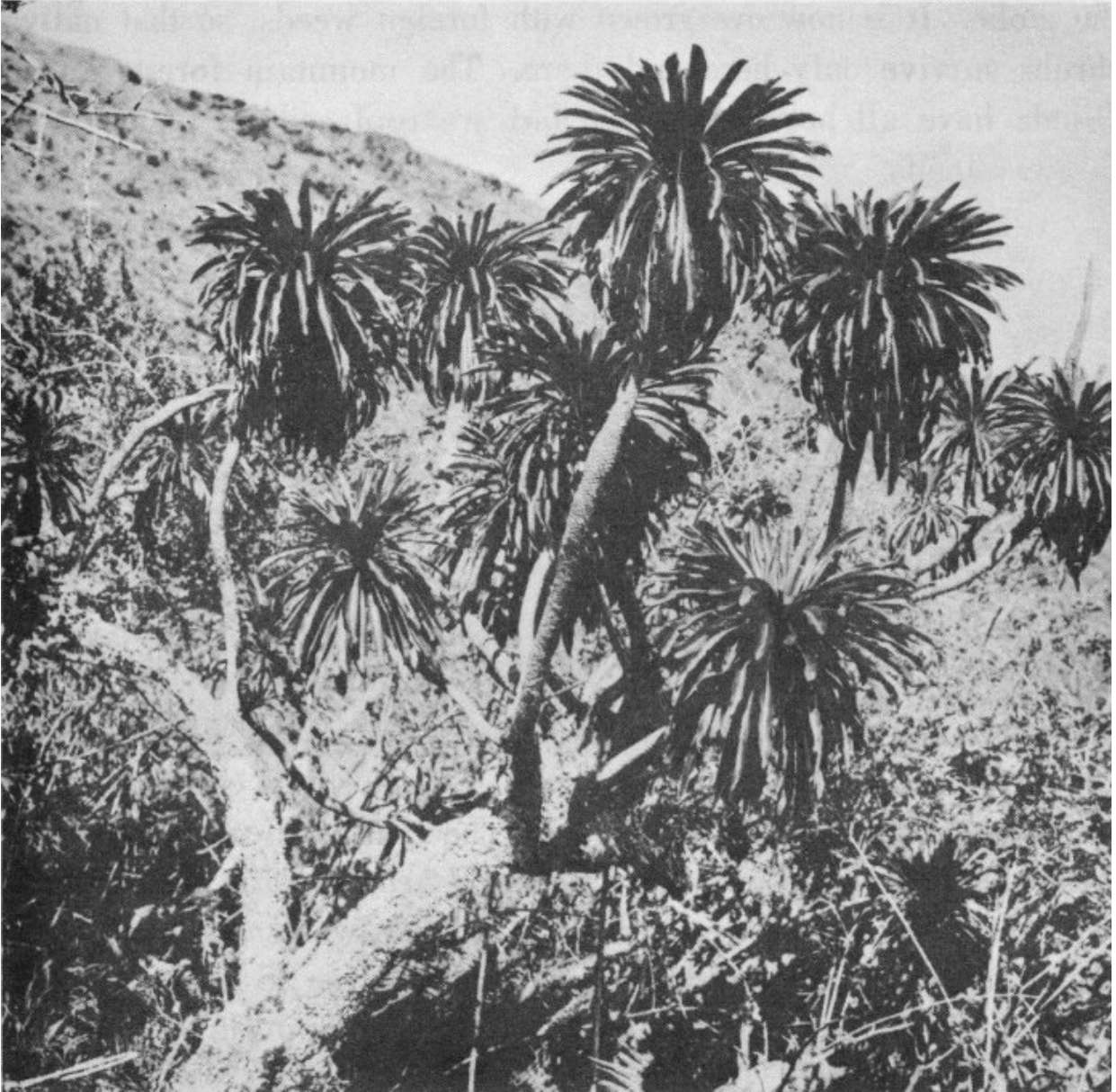
How the Alpine plants came to Hawaii is uncertain since they have descended from ancestors so remote that past relationships are vague. The great Swedish botanist, Carl Skottsberg, believes that they are derived from a flora that grew on summits higher than any that exist today. Such assumption adds to the problem, as little else leads one to believe that elevations ever significantly exceeded 14,000 feet, i.e., the highest that exist today. In a report for the Fifth International Botanical Congress (Proceedings, pp. 91-97, Cambridge, 1931), Skottsberg listed 13 species found exclusively in the Alpine Zone, all of which are endemic, native to no other place than Hawaii's highest volcanoes. Three other species listed as occurring in the Alpine Zone extend their range downward into the next lower Subalpine Zone, for which he has listed a total of 23 species, 20 of which he labels endemic. These adjoining zones are not sharply delimited, but blend into each other at 9,000 feet.

As elevation decreases and conditions for growth become better, the number of plant species increases. The Subalpine Zone with lower limits just above the park entrance has some of the most interesting plants found in the islands. Trees are absent but some of its shrubs extend their range to lower elevations and grow big, sometimes even to tree-size. Most Subalpine plants have small leaves or leaflets, indicating that lasting moisture is still scant. Occasional snowfall seldom remains long and never piles into drifts, which would conserve moisture.

Most of the interior of the crater is bare or thinly covered with vegetation. The uncongenial climate of the summit spills into it and extends throughout its length, though the elevation is much lower. Extensive aa flows, ash, and cinder cones cover the floor. Yet the crater has perhaps become best known through the presence of one of its plants, the silversword. Also, the lips of Koolau and Kaupo Gaps are botanically distinctive in contrast to the barrenness elsewhere. This results from the fog and rain that sweep through them in late morning, only to dissolve upon mixing with the warmer air inside. As the day progresses, clouds push further and further inward, until, rarely, the whole depression may become filled. Koolau especially is a treasure-trove for the plant-lover. Below 6,000 feet, quite outside the park, it becomes impenetrable jungle surpassed only by that in Kipahulu Valley, a few miles east. Kaupo Cap is comparatively dry, but it supports a sparse scrub cover of great interest. The lower parts of Kaupo Valley are grazing

land. Paliku is the only place within the park in which vegetation is lush. As it climbs Leleiwi Pali, Halemau Trail is bordered by **OHELO** and ferns, among them **AMAUMAU** which is pleasing to the weary hiker's eye.

In the northwestern angle of the park not far from the inn, Nianiau Crater has had a renowned floral character and history. Today it is overgrazed and drab. Dr. Joseph Rock, Hawaii's famed dendrologist, discovered a curious tree *lobelia*, *Clermontia haleakalensis*, growing within it. He happily described the plant as "antediluvian in appearance." This most primitive member of a distinctive Hawaiian floral group had a robust trunk from which clumsy, stubby branches shot off, each crowned with a feather-duster of long, thick, strap-shaped leaves like some pompous dictator's headdress. **MAMANE** and **AKALA** grew thickly around it; rarer associates included tree geraniums, *Neurophyllodes* sp., with flowers like violets, tree *Railliardia*, shrubby Hawaiian buttercups, and greensword, *Argyroxiphium virescens*, that is threatened with extermination by grazing to an extent greater than of its relative, the silversword.



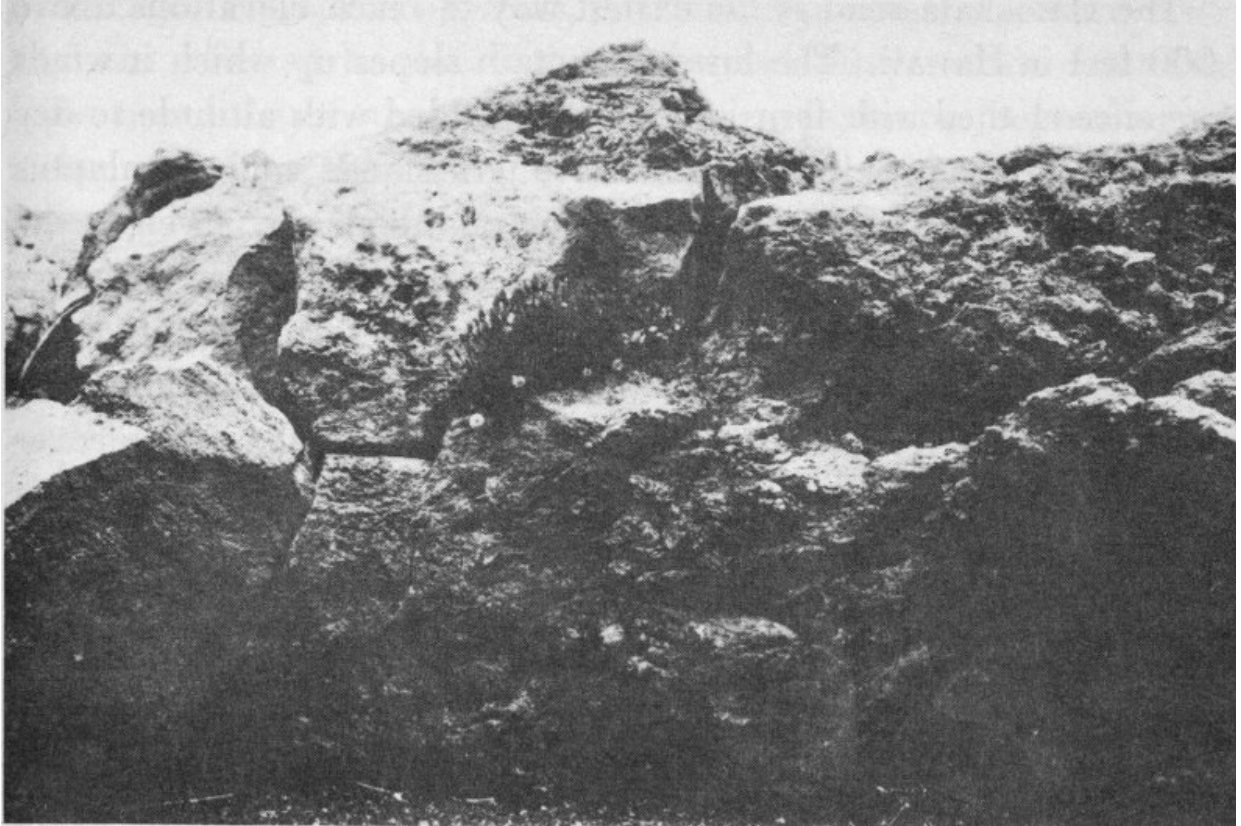
BISHOP MUSEUM, J. F. ROCK COLLECTION
***Clermontia Haleakalensis*.**

Even though they are outside the park, some plant communities on the outer slopes of the mountain should be mentioned, because they are related to park forms and carry a compelling interest. The trail from Olinda to Waikamoi is through a transition forest between the wet and the dry that once drew botanists from many parts of the globe. It is now overgrown with foreign weeds, so that native shrubs survive only here and there. The

mountain forests above Olinda have all been destroyed and are replaced by plants from faraway lands.

Although it now lies devastated by change, one of the richest botanical regions in Hawaii comprises the forbidding lava fields of Auwahi on the southwest slope of Haleakala. Fifty species of native trees once thrived in its fabulous mixed forests. The only known specimen of Maui hibiscadelphus, *H. wilderianus*, a relative of the hibiscus, was found here. It had a curved corolla that opened only slightly at the top. Its congener, *H. giffardianus*, once equally rare, still survives in Kipuka Puauulu at Kilauea, where a few vigorous plants, started by air-layering, receive tender care from the National Park. The last known Maui specimen of MAHOE, *Alectryon macrococcus*, grows in an Auwahi gulch. This tree has large double fruits (*mahoe* means *twins*) that split open to expose a shiny, chestnut-brown seed clasped in a brilliant scarlet aril. The ALANI, *Pelea multiflora*, of the lava fields is festooned with a lichen, *Usnea australis*, that appears to prefer it to all other trees. Haleakala sandalwood, *Santalum haleakalae*, with attractive red flowers grows to be a tree 25 feet tall. Other famous native trees include 'OHE'OHE, *Tetraplasandra kauaiensis*; 'OHE, *Tetraplasandra meiantra*; A'E, *Fagara* sp.; 'ALA'A, *Planchonella auahiensis*, with golden fruits; HO'AWA, *Pittosporum terminalioides*; OLOPUA, *Nestegis sandwicensis*; A'IA'I, *Pseudomorus sandwicensis*; MEHANE, *Antidesma pulvinatum*; and KAUILA, *Alphitonia ponderosa*, whose hard and durable timber was used in sacred structures.

Many unusual plants occur in the steep valleys to the north and east outside the park. Since the dense jungles in which they grow are inaccessible except to hardest botanists, they are seldom seen. Here grow 'APE'APE, *Gunnera petaloidea*, a plant with geranium-like leaves three or more feet in diameter; a delicate, rare, native begonia, *Hillebrandia sandwicensis*; several exquisitely flowering lobelias. The summit bogs on Kukui and Mt. Eke in West Maui have a curious, distinctive flora that includes three species with origins ascribed to the Antarctic: *Orebolus furcatus*, a sedge; *Acaena exigua* of the Rose Family; and *Lagenophora mauiensis*, a composite.



BISHOP MUSEUM, J. F. ROCK COLLECTION
Tetromalopium.



BISHOP MUSEUM, J. F. ROCK COLLECTION
Begonia.

The Haleakala road is the easiest way to reach elevations above 7,000 feet in Hawaii. The lower mountain slopes up which it winds were once clothed with fern jungles that yielded with altitude to dry forest. All is now altered. Extensive grasslands and eucalyptus groves today leave the lasting impression on the visitor. Even on the heights, cattle, goats, introduced plants, insects, and other agents have wrought permanent change. Rare and interesting forms of native life have been exterminated or are well on the way to extinction. The National Park is trying hard to save what is left of the native cover within its boundaries and to restore the former scene wherever it is possible to do so.



BISHOP MUSEUM, J. F. ROCK COLLECTION
'Ape'ape.

Plants of the distinctive Haleakala environment show differences, some slight, some considerable, from close relatives elsewhere. Examples are the **OLAPA** at Paliku, the silversword, and the **KUPAOA**. The differences have been fashioned by combinations of factors. The unstable, permeable ash and cinders have scant soil and little available mineral matter. They cannot hold water nor do they yield secure anchorage. The effects of winds, [55] isolation, exposure, and nature of terrain are reflected by the distorted shapes. Silversword, *Artemisia*, *Bidens*, and many other plants well show adaptation to peculiar environment. Why are some species found nowhere else? How did the plants get here in the first place and how have they changed with the passing of time? What has been the impact of exotics? What use did the Hawaiians make of the plants? To the hurried and casual visitor, the flora of Haleakala may appear drab and uninteresting. With better acquaintance, it becomes a stimulating study indeed.

PLANT NOTES

THE FERNS. The Sliding Sands Trail drops from White Hill on bare slopes of red and gray Cinders. As it levels below Puu o Pele, a lush, green carpet spreads along the south wall of the crater. It is a surprise to discover that ferns compose the verdure, for several kinds find the shelter of the cliffs agreeable. Haleakala's **KA'UPU**, *Polystichum haleakalense* Brack., a rather coarse, low plant with scaly stems, grows among the shrubs. The pellucid polypody, **'AE**, *Polypodium pellucidum* Kaulf., and the maidenhair spleenwort, **'IWA'IWA**, *Asplenium adiantum-nigrum* L., grow here, but they are common elsewhere as well. Iwaiwa is a small plant with shiny, slender stems and stiff, triangular fronds that thrives in the brush. On Leleiwi Pali it is a tiny thing, clinging to the rocks. This fern is known in mountainous regions in many countries. Another maidenhair spleenwort, **'OWALI'I**, *Asplenium trichomanes* L., makes its home on barren lava above 5,000 feet on the inner slopes of the crater. Its small, opposite, rounded or ovate frond segments, *pinnae*, grow on wiry stems that form dense clumps. Look for it along Halemauu Trail on Leleiwi Pali. It is found in the temperate zones and on high mountains in many parts of the world.

Bracken, **KILAU**, *Pteridium aquilinum* var. *decompositum* (Gaud.) Tryon, a stiff cosmopolitan, is among the best-known ferns. It grows in grassy spots inside and outside the crater. Foraging pigs, seeking its tasty rootstocks, often uproot it in the forests. The young leaves of **AMA'UMA'U**, *Sadleria cyatheoides* Kaulf., add a touch of red along Halemauu Trail on Leleiwi Pali. At Paliku, several moisture-loving ferns grow on logs, rocks, and 56 moss-covered trees. Large, dense ae are quite unlike the frail specimens on Leleiwi Pali. The following can be seen in damp spots and on tree trunks: Maui's paddle, **'EKAHA**, *Elaphoglossum reticulatum* (Kaulf.) Gaud., with paddle-shaped blades 6-12 inches long; **'EKAHA-'AKOLEA**, *Pleopeltis thunbergiana* Kaulf., a small fern with tough, pale, elliptico-oblong (paddle-shaped) fronds; and **MOA**, *Psilotum complanatum* Sw., belonging to a small group of tropical plants, **PSILOTALES**, which reproduce

by spores, but are distinct from true ferns, clubmosses, and the better-known orders.

Swordfern, NI'ANI'AU or 'OKUPUKUPU, *Nephrolepis exaltata* (L.) Schott., is widely distributed at Haleakala as it is elsewhere in Hawaii. Cliffbrake, KALAMOHO LAULI'I, *Pellaea ternifolia* (Cav.) Link, grows among rocks in dry, sunny locations at higher elevations above the park entrance and inside the crater. It is common. The bluish-green pinnae of this short, slender fern grow on opposite sides of the dark, wiry stems. They are cleft into three linear segments.

THE NATIVE GRASSES. Several native grasses grow above the park entrance: *Trisetum glomeratum* (Kunth) Trin., *Deschampsia australis* forma *haleakalensis* Skottsb., and *Agrostis sandwicensis* Hillebr. The *Trisetum*, also common inside the crater, is called mountain pili, PILI being the lowland grass known in many tropical regions and used for thatching houses in Hawaii. The Hawaiian name for *T. glomeratum* is PILIUKA, upland pili. In some places it is also called HE'U PUEO, the hoot of an owl.

All three grasses are tufted, i.e., bunch grasses. The *Deschampsia* has tough, wiry blades (leaves) with shiny, open panicles of flowers and seeds. The *Agrostis* has stiffly upright blades and culms (jointed stems) with spike-like panicles. The pili has flat blades usually covered with soft hairs. The panicles are contracted or spike-like.

THE SEDGES. [Fig. 1](#). Sedges are generally grasslike wind-pollinated herbs that grow in tufts or bunches. They are often the dominant plants in cold marshes, especially in the Arctic. They have little economic value and grazing animals find them unpalatable. Some have tough, pliable stems that are woven into mats and baskets; some have fruiting spikes that are attractive in dry bouquets. Most have 3-angled stems around which the blades are ranked. The inconspicuous, green flowers are crowded in tight, flattened spikes, often grouped on top of a slender, grasslike stem. Hawaii has a dozen or more native genera with many of its species widespread in the world.

The species common on the upper slopes of Haleakala and on the crater floor, *Gahnia gaudichaudii* Steud., bears shiny, ebony fruits. These can dangle, suspended by the wilted, threadlike stamens for more than a year from the fruiting stalk. Another sedge, *Carex macloviana subfusca* (W. Boott) Kukenth., grows in clumps along the south wall. The fruiting stalks bear cylindrical or ovoid clusters of 4-9 spikelets, each ½-1 inch long. This sedge grows from Lapland and Greenland to northern South America, but in Hawaii it has been found only on Haleakala and Kohala Mountains above 4,000 feet. The Hawaiian variety was first described from a specimen collected at Lake Tahoe, California. The interesting *Oreobolus furcatus* H. Mann, mentioned as occurring on West Maui, also grows in Koolau Gap.

RUSH, *Luzula hawaiiensis* (O. Ktze.) Buch. A visitor brought me a small tufted plant with grass-like leaves covered with soft, silky hairs. He was all excited, believing he had found a young silversword on the cinder flats near the summit of Mt. Hualalai. The plant was the endemic rush that lives in wet places above 3,000 feet in the mountains of Maui, Hawaii, and Kauai. In Haleakala Crater, I found it on Leleiwi Pali and near Paliku, where others might mistake it for an immature silversword or greensword. The Hawaiian general name for grasses, sedges, and rushes is MAU'U.

PAINIU, *Astelia degeneri* Skotts. This plant is reported by Degener to be growing within the crater. ^[19]

MAU'U-LA'ILI, *Sisyrinchium acre* H. Mann. A native member of the Iris Family with grass-like leaves that grows on old lava flows between altitudes of 3,500 and 7,500 feet on Maui and Hawaii. It is an attractive plant that rarely exceeds 12 inches high. In July and August it bears yellow flowers ¾ inch across that last only a few hours. In the crater it may be found in Koolau and Kaupo Gaps. The leaves, bound tightly around the wrists and ankles, stain the skin a blue color that lasts several days. This stain was regarded as proof that a person had been to the crater. A number of other species of the genus are mainland wildflowers commonly called blue-eyed grass. 58

ORCHIDS. [Fig. 2](#). The Orchid Family of over 15,000 species is second only to the Composite Family in size, yet only three species of three genera are native in Hawaii, a land connected in thought with an exuberance of gay orchids. Moreover, the three endemic species are characterized by small size, relative rarity, and inconspicuous flowers. Twayblade, *Liparis hawaiiensis* H. Mann, grows in open woods on the ground and on moss-covered trees on the flanks of the volcano and at Paliku. The rarest of the three natives, *Habenaria holochila* Hillebr., has been found growing 1-2½ feet high in deep moss in fog-swept Koolau Gap. It bears its dull-greenish, inconspicuous orchids on a tall, many-flowered spike.

'ALA 'ALAWAINUI, *Peperomia* sp. A small herb with succulent leaves found at Paliku. It is a member of the Pepper Family, *Piperaceae*, to which [AWA](#), *Piper methysticum*, belongs. There are many *Peperomias* native to Hawaiian forests. Some have leaves with gay red undersides.

SANDALWOOD, ['ILIAHI](#), *Santalum haleakalae* Hillebr. [Fig. 3](#). A small but striking tree found above the park entrance, in Koolau Gap, and along the Kaupo Trail below Paliku. It has leathery, dark green leaves so that it stands out in the vegetation and is readily distinguished from afar. In mid-summer, corymbs of four-pointed, deep-red or vermillion flowers appear on the ends of branches. The dry heartwood has the fragrance which is associated with the name sandalwood. Degener tells the detailed story of sandalwood trade that flourished in the islands for fifty years beginning in 1790. [\[20\]](#)

SHEEP SORREL, *Rumex acetosella* L., an abundant, well-known, introduced weed found both inside the crater and out. A native species called [PAWALE](#), *Rumex giganteus* Ait., is more interesting. It grows as a stocky undershrub on barren lava flows and in rock crevices inside the crater. In Koolau Gap, it becomes a sprawling vine. It is said that a mixture of an 59 extraction from the boiled bark and [AWA](#) was used by Hawaiians for skin diseases. This was tried in vain as a cure for leprosy when the disease first appeared in the islands in 1840.

HAWAIIAN BUTTERCUP, **MAKOU**, *Ranunculus hawaiiensis* A. Gray. Like the native violets and geraniums, members of this genus, well-known on the mainland, either become bushy or spread as a woody vine in Hawaii. Two native species occur on Haleakala, but the yellow-flowered, sprawling *R. mauiensis* A. Gray is reported only outside the park. The larger flowered, erect Hawaiian buttercup, however, was reported by J. F. Rock to be abundant formerly in Puu Nianiau Crater. It is a coarse, hollow-stemmed, hairy herb with compound leaves divided into three sharply-toothed, irregular leaflets. Within Haleakala Crater it grows on moist, grassy slopes in Koolau Gap.

HO'AWA, *Pittosporum confertiflorum* A. Gray. A small tree, sometimes becoming 25 feet tall. The large leaves, shiny green on top, brown hairy underneath, are crowded like whorls on the ends of the branches. In the centers of these, dense clusters of fleshy, cream-colored flowers appear in late summer. Wrinkled fruits, resembling English walnuts, hang on the trees throughout the year. Very few trees grow in the crater, but they are more numerous above 4,000 feet along the Kaupo Trail and at Ulupalakua. The genus of some 200 species is widespread; it has many species native to Hawaii. Some kinds are well-known garden plants. *Pittosporum tobira* (Thunb.) Ait., a native of China and Japan, is a favorite shrub in California.

HAWAIIAN HAWTHORN, **'ULEI**, *Osteomeles anthyllidifolia* Smith (Lindl). A spreading shrub with compound leaves and fragrant, small, white flowers, like apple blossoms, that may appear throughout the year. The fruit is white and contains five stony seeds. Plants growing on ash flats are very small. The strong but pliable wood of ulei was used for digging sticks, fish spears, and hoops to keep the mouths of fishing nets open.

HAWAIIAN RASPBERRY, **'AKALA**, *Rubus hawaiiensis* A. Gray. [Fig. 5.](#) A shrub with attractive pink flowers abundant at Paliku. It is found also 60 in Koolau Gap at the foot of Leleiwi Pali and elsewhere. The fruit, agreeable but somewhat bitter, is remarkable for its large size. It ripens about the Fourth of July or later. A trailing native, *R. macraei* A. Gray, sprawls in foggy Koolau Gap. Its large, dark fruits are bitter.

MAMANE, *Sophora chrysophylla* Salisb. [Fig. 4](#). A common native shrub or small tree of the Bean Family both inside and outside the crater up to tree line. It is recognized by more or less downy, narrow, compound leaves, racemes of yellow flowers, and twisted pods that have four wings and are constricted between the seeds. Goats eat it greedily and quickly exterminate it in an area. The hard, durable wood was used by Hawaiians in many ways; today it is a principal firewood for the crater cabins. A tree in full bloom is a beautiful object. The height of the flowering season is mid-spring.

GERANIUM, [NOHOANU](#), [HINAHINA](#). [Fig. 6](#), [7](#). *Neurophyllodes tridens* (Hillebr.) Degener & Greenwell is common above the park entrance and on the south wall within the crater. It sometimes becomes three feet tall and is readily identified by its silvery leaves, each of which has three small teeth on the end. The white flowers have purplish veins especially toward their centers. The blooming season is July to October. The silvery aspect of the plant, like the silversword, is imparted by a mesh of fine white hair that reflects the light of a passing car as effectively as the glass-beaded paint of directional signs.

Two native geranium relatives grow in Koolau Gap, *N. ovatifolium* (A. Gray) Degener & Greenwell and its variety *superbum*. The latter is common on the trail to Waikau not far from the foot of Leleiwi Pali.

One of the common plants growing on the crater floor is the exotic pink-flowered *G. carolinianum* var. *australi* (Benth.) Fosberg whose pointed fruiting bodies give it the common name, *cranesbill*.

HAWAIIAN HOLLY, [KAWA'U](#), *Ilex anomala* Hook and Arn. The forest growing on the talus behind the Paliku cabins is the finest within the crater. A striking tree of this association is the Hawaiian holly that has dark, shiny, oval leaves with conspicuous networks of slightly depressed veins that make identification easy. Dense panicles of small, white flowers are followed by shiny, black drupes, like Christmas holly "berries." The genus is that of the English holly. Curiously, the scientific name has been locally corrupted to *ileck*.

OLOMEA, *Perrottetia sandwicensis* A. Gray. A native shrub or small tree belonging to the same family as the bittersweet of the continent. The numerous tiny, round, red fruits, borne in panicles, suggest the relationship. The bright red venation of leaves and petioles make the plant easily recognized. I know of no plants within the park, but found many trees a thousand feet below the park boundary in Keanae Valley.

A'AL'II, *Dodonaea eriocarpa* Smith. [Fig. 8](#). This is a common shrub in several varieties along the highway and in the western end of the crater. At Paliku and Kaupo Gap it becomes a tree up to 20 feet high and 8 inches in diameter. Its flowers are inconspicuous but clusters of dry, reddish fruit-capsules contrast, flower-like, with the surrounding green foliage. The fruits, abundant from July to September, are used for leis and dry bouquets. The hard brown heartwood was used for spears, *pololu*, daggers, *pahoa*, and other implements.

BEGONIA—**PUAMAKANUI**, *Hillebrandia sandwicensis* Oliv. (See illustration [p. 53](#).) The only native begonia, found in wet ravines often by waterfalls. It grows profusely at Koolau a mile below Holua Cabin at the foot of the rain-drenched *pali*. This succulent herb has a tuberous rhizome, unbranched, slender stems, and hairy, toothed leaves 4-10 inches in diameter. From June to August it bears sprays of bright, pink and white flowers. This is one of the floral treasures of Hawaii.

TARWEED. *Cuphea carthagenensis* (Jacq.) McBride. A low, sticky, hairy perennial from tropical America widely spread at lower elevations in the park. It has red or green branches, small ovate leaves, and tiny but not unattractive pink flowers $\frac{1}{4}$ inch across. Plants in rock crevices on cliffs are tiny; on the crater floor, they may become a foot high and form a dense shrubby mat over a sizeable area. The plant belongs to the **LYTHRACEAE** or Crepe Myrtle Family.

'OHI'A LEHUA, *Metrosideros collina* (Forst.) A. Gray. [Fig. 9](#). This, the commonest tree in the islands, consists of a swarm of hybrids of which the parentage is still unknown. It is scattered within the park to tree line. It is abundant in the eastern end of the crater and at Paliku. The beautiful flowers, mostly red, a few yellow, may appear throughout the year. Here spring seems to be the best season for them.

EVENING PRIMROSE, *Raimannia odorata* (Jacq.) Sprague & Riley. A slender, erect, hairy South American herb introduced forty years ago. It is a prolific bloomer, is widespread, and, in blooming season, the most conspicuous flower both at Park Headquarters and within the crater. The large, sulphur-yellow flowers appear at night, but wilt within the following day, turning reddish as they do so.

'APE'APE—*Gunnera petaloidea* Gaud. (See illustration [p. 54](#).) A huge-leaved forest perennial with a massive prostrate stem which stands erect 3 or 4 feet at the tip. In the center of the leaves a tall stalk rises that bears hundreds of small yellow-brown flowers. I know of no plants within the crater, but some grow within a mile of the park boundary below Koolau Gap, as well as in adjacent wet valleys.

OLAPA, *Cheirodendron trigynum* (Gaud.) Heller. [Fig. 10](#). This tree occurs in several varieties. The variety *oblongum* Sherff grows 30 feet tall around the cabins at Paliku, also at Ulupalakua. The variety *mauiense* Levl. common at Olinda, also grows in Kaupo Gap. The genus is Hawaiian but has a lone representative in the Marquesas. It is widespread in deep soils in all of the islands. The leaves are compound with 3 or 5 leaflets which, at Paliku, have reddish petioles. The panicles of small, green flowers are followed by black drupes. The leaves, bark, and fruit are said to have yielded a blue dye for staining tapa.

'OHELO, *Vaccinium reticulatum* Smith. [Fig. 11](#). Hikers in the crater are grateful for the widespread ohelo bushes that yield pleasant fruits to be

nibbled along the way. In areas rich in moisture, like Koolau Gap, large, maroon bell-shaped flowers droop from the axels of the leaves. The berries found in the eastern end of the crater appear to be largest. The bearing season appears at its height in mid-autumn. The plants along the Halemau Trail constitute a distinct species, *V. berberidifolium* Skottsb.

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PUKIAWE, *Styphelia tameiameia* (Cham.) F. Muell. [Fig. 12](#). A most abundant shrub, both inside and outside the crater, but near the top it is replaced by a trailing shrub that has been classified as *S. douglasii* (A. Gray) Hochr. The berries of this plant are white, pink, red, or mahogany brown; they are most abundant in winter, but some may be seen on the shrub throughout the year.

KOLEA, *Suttonia lessertiana* (A.DC.) Mez; syn. *Rapanea lessertiana* (A.DC.) Degener and Hosaka. [Fig. 13](#). This variable tree grows up to 50 feet tall as one of the common trees around Paliku cabins. The thick leaves with short petioles, crowded near the ends of thick branches, have a beautiful roseate hue when young. The branches are studded with spurs on which grow small 5-parted flowers in clusters of three or more that are followed by dark, purplish-red or black fruits up to ¼ inch in diameter, often so numerous that the branch is completely hidden. Hawaiians made a red dye for tapa from the sap and bark. The crimson sap bleeds freely from a cut made deeply into the bark of a living tree.

SELFHEAL, *Prunella vulgaris* L., a common weed of Eurasian origin, is widespread in rocky, scrub cover, and at Paliku. A dense cluster of small, lipped, blue to purple flowers appears on the end of each upright stem.

PUA'AINAKA, *Stenogyne rotundifolia* A. Gray. This endemic long-branched shrub of the Mint Family is found only on Haleakala. Within the crater and on upper slopes it is trailing and is relatively rare, being most abundant in Kaupo Gap. The attractive, pale purple flowers in whorls of six are 1½ inches long and are covered with silky, white hair. They appear in late summer. Outside the crater, *S. haliakalae* Wawra is abundant in forests

as a large, diffuse shrub that often forms a dense mat over surrounding shrubs. Another mint, *Stenogyne crenata* A. Gray, was collected by Skottsberg among shrubs on the south wall of the crater.

GROUNDCHERRY, *Cape Gooseberry*, **POHA**, *Physalis peruviana* L. A South American perennial herb, widely scattered throughout Hawaii, 64 well-known for its round, orange, many-seeded, husk-enclosed fruits that are edible raw or preserved. It grows extensively in Kaupo Gap. The large yellow flowers with brownish spots near the center appear from June to late fall in this area.

PLANTAIN, **LAUKAHI**, *Plantago* sp. Plantain is a hardy, cosmopolitan, stemless weed forming a rosette of broadly oval leaves, 1-10 inches long, near the ground. The tiny flowers and seed capsules are borne as cylindrical heads at the ends of tall stalks. The plant is widespread along the side of the road as well as within the crater. Of several hundred species of *Plantago*, four or more are endemic to Hawaii. One of these, *P. princeps* Cham. & Schl., is a shrub several feet high with tufts of long narrow leaves at the ends of the branches. It grows on cliffs at Kaupo Gap. Other plantains with thick leaves, silky underneath, creep on the ground in Koolau Gap.

KUKAENENE, *Coprosma ernodioides* A. Gray. [Fig. 15](#). A common woody shrub with long trailing branches that send up short, erect, densely foliose branchlets at each node. The awl-shaped leaves are rigid and dark-green. The fruits are shiny black drupes which are a favorite food of the native goose, *nene*.

PILO, *Coprosma montana* Hillebr. [Fig. 14](#). One of the commonest shrubs throughout the crater and from Park Headquarters to 9,000 feet outside the crater. It is a small tree up to 20 feet tall in Kaupo Gap. As a shrub the ascending tips look like jets shot up from densely foliose branchlets. The alternate, small, thick leaves have conspicuous nerves impressed on the upper face. Below each pair of leaves is a pair of triangular bracts, stipules, with cilia on the upper border. The greenish, inconspicuous flowers are

followed by showy, bright orange, yellow, or red fruits which make the plant a subject attractive to color photographers in fall.

MANONO, *Gouldia terminalis* (H. & A.) Hillebr. A shrub or small tree growing on the talus above the Paliku cabins. It has shiny, opposite leaves and dense terminal clusters of greenish, four-lobed, cup-shaped flowers that are followed by small black berries. It blooms in late summer. The genus is one of three in the Coffee Family that are endemic to Hawaii.

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CATCHFLY, *Silene struthioloides* A. Gray. [Fig. 16](#). A plant that is typical only of arid Cinders and ash on East Maui and on the island of Hawaii. With the silversword as its companion in Haleakala Crater the plants are found at the bases of barren cones. They show neat adaptation to their stark home. The Haleakala plants, as illustrated, are low and compact, but those growing at Kilauea Crater bear only a few awl-shaped leaves and resemble dead twigs. The thick tap roots are sweet and edible. About 250 species belong to this genus, a member of the Pink or Carnation Family. A well-known introduced weed, the English catchfly, *Silene anglica* L., was reported by Degener at 10,000 feet on Haleakala.

'OHA, *Labelia grayana* E. Wimm. A low plant with woody trailing stems with knobby leaf scars and ending with a crowded arrangement of silvery, linear leaves, 4-8 inches long and crowned with densely flowered racemes, 6-15 inches long. The flowers are lilac-blue with a satiny sheen. The plant is not uncommon on wet *pali* from 5,000-7,000 feet at Paliku, Kaupo and Koolau Gaps, and in the northwestern end of the crater. It is a glorious plant worth hunting for and going miles to see.

NAUPAKA. *Scaevola chamissoniana* Gaud. This shrub was noted only on the east side of Kaupo Gap. This is a varying species found up to the 6,000-foot elevation. It is not common and blooms in summer. The white flowers with purplish streaks are slit to the base on the upper side. They look like flower-halves rather than complete corollas. There are several legends about the peculiar flower, each dealing with lovers separated from each other. In a song composed about it, the lovers were forceably parted, so the girl divided

a perfect corolla, giving one half to her lover while keeping the other half herself. One of the lovers carried the flower to the sea, *naupaka kahakai*, the other to the mountain, *naupaka kuahiwi*, where the plants are found today.

MAUI WORMWOOD, *Artemisia mauiensis* (A. Gray) Skottsb. [Fig. 17](#). Typical of Maui and found only on Haleakala, this hoary ornamental shrub, usually 2-3 feet high, perches on cliffs usually above the reach of man. It has a densely-branched crown with silvery leaves that are aromatic and 66 bitter. The leaves are composed of thin segments that are covered with a mat of cottony hair, giving the plant a silvery appearance. The small orange flowers are borne in terminal panicles.

The Hawaiians call the wormwood **AHINAHINA**, applied also to silversword, geranium, and other gray plants. The basic word refers to the color of silvery-gray hair, the connection being obvious. Hawaiians use the pounded leaves to relieve asthma. The genus is large, having some 250 species that are generally found in arid regions. The sagebrush of the western states, *A. tridentata* Nutt., is the best known to most park visitors.

KO'OKO'OLAU, *Bidens* sp. Like *Artemisia*, the genus is a huge one with over 200 species and belongs to the Composite Family which includes dandelions, daisies, and sunflowers. E. E. Sherff of the Chicago Museum of Natural History, a specialist on the genus, lists sixty species native to Hawaii. Native **KO'OKO'OLAU** are shrubby and often of great beauty. This is true of *B. campylotheca pentamera* Sherff which sprawls over the vegetation in Koolau Gap. It has fern-like leaves and large, pretty, yellowish flower-heads. Hawaiians use the tips of young plants for tea, often in preference over imported tea.

Besides the native varieties, three introduced species grow in the islands, including beggar ticks or Spanish needles, *B. pilosa* L. It is a nuisance, as the three-pronged fruits that give it the common name readily attach themselves to clothing as well as to fur of passing animals.

KUPAOA, **NA'ENA'E**. [Fig. 18](#), [19](#). Several kinds of composite shrubs are called by these names, both of which mean fragrance or perfume. They were

used for scenting tapa. They belong to the endemic genera *Dubautia* and *Railliardia*, both of which have species found in the park. *Dubautia plantagiens* var. *platyphylla* Hillebr. Gaud., a shrub at 6,000 feet in Kaupo and Koolau Gaps, has linear leaves 4-8 inches long, with 7-13 conspicuous nerves. It is a small tree below Nianiau Crater. It is a handsome sight when in flower; the flower-heads are yellow. In general appearance it is much like that of *Railliardia platyphylla* A. Gray which grows in cinders and ash, 67 mostly inside the crater. At 8,000 feet, *R. platyphylla* becomes a straggling shrub. The commonest member of this group, *Railliardia menziesii* A. Gray, is a shrub in the crater and on the rim from 8,000-10,000 feet. Between 6,000 and 7,000 feet it is a tree up to 20 feet tall. The dark green, pointed, linear, fleshy leaves are ranked in vertical rows of four on upright stems and branches. Its dark yellow flower-heads are borne in panicles. *R. scabra* DC., found in Kaupo Gap and on Leleiwi Pali, does not have the regular leaf arrangement.

PAMAKANI-HAOLE, *Eupatorium glandulosum* HBK. *Eupatorium* is another huge genus of the Composite Family with several hundred species, mostly from tropical America. Five species have been introduced into Hawaii of which two are bad pests. *E. glandulosum*, a native of Mexico, spread rapidly on Maui, crowding out desirable plants and making pasturelands worthless. A parasitic insect, *Procecidochares utilis* Stone, was introduced in 1944 to combat it. This insect belongs to the order **DIPTERA**, the flies, gnats, midges, and mosquitoes, and to the family **TRYPETIDAE**, that includes the Mediterranean fruitfly and the common “apple-worm” which is actually the larva of a fly. The trypetids infest living plants, frequently causing galls, and have piercing ovipositors, often prominent, with which females deposit eggs beneath the skin of their host.

It is not expected that the studied introduction of a parasite will result in the extermination of a host, but rather that it will check unbridled increase and spread. The method is termed “biological control.” It has been successfully adopted against several menaces, such as cactus, lantana, and a fern-weevil, *Syragrius fulvitarsis* Pascoe, that kills amaumau ferns in the Kilauea Crater area.

Pamakani abound throughout the crater, even clinging as tiny starvelings in small cracks in cliffs, like those along Halemauu Trail. Every plant shows swellings in which the little maggots live, and exit holes through which the new adults emerge. Many plants bear only few leaves and fight tenaciously for survival. Big plants may be seen along Kaupo Trail. *Pamakani* means *wind-blown*, in reference to the method of seed dispersal.

HAIRY CAT'S-EAR, "Dandelion," *Hypochoeris radicata* L. A common composite, native to the Mediterranean, found in abundance 68 throughout the park. Its narrow leaves with yellow hairs form flat rosettes. A branching, leafless stalk up to a foot or more tall bears yellow flower-heads, an inch in diameter, that resemble the well-known dandelion, *Taraxacum officinale* Weber, by which common name many call the cat's-ear. The hairless, *H. glabra* L. is a smaller plant with smooth leaves and flower-heads ¼ inch in diameter. The cat's-ear is a favorite food of the Hawaiian goose, *nene*. *Gosmer* is a common name used locally.

WOOD GROUNDSEL, *Senecio sylvaticus* L., a native of Europe, is a branching, weedy herb, 1-2 feet high, abundant along the foot of Kalahaku Pali. It has irregularly lobed leaves and small yellow flowers in a tight flower head, ⅓ inch long and ½ inch in diameter.

TETRAMALOPHIUM. [Fig. 20](#), illustration [p. 53](#). This endemic genus has a dozen species, two of which are a pride of Haleakala. The small leaves are narrow and crowded at branch ends. The showy flowerheads of white, pinkish, or lavender ray florets surround a disk of purplish central florets. *T. humile* (A. Gray) Hillebr. is found inside and outside the crater between 6,000-9,000 feet. It is a small, low, shrubby plant with narrow, spoon-shaped leaves quite covered with sticky, curly hair. The plant growing in cracks between rocks at the very summit is stiffer and considerably different.

SILVERSWORD, AHINAHINA, *Argyroxiphium sandwicense* DC. Illustrations on [cover](#) and pages [iv](#), [47](#). As famous as the crater itself, and almost as well-known, the silversword is regarded as typical of Haleakala, although its natural range embraces Mauna Loa, Mauna Kea, and Hualalai from 7,000 to

12,000 feet. A lustrous silvery down thickly covers all leaves and makes the plant exceedingly beautiful. It is highly evolved to withstand the extreme dryness of cinder cones and the intense sunlight of lofty elevation. Surprising to most visitors on first acquaintance is the fact that it bears no relationship to the yucca of the Lily Family, but belongs to the **COMPOSITAE** along with sunflowers, asters, and chrysanthemums. Like *Dubautia* and *Railliardia*, it has no close relatives outside of the islands. *Wilkesia*, endemic to Kauai, is so closely similar that some authorities class it in the genus *Argyroxiphium*.

The Haleakala silversword has a short, simple, woody stem 2-3 inches 69 in diameter crowded with thick, dagger-like leaves arranged spirally around it. After growth from 7 to as much as 20 years, a foliose raceme 3-8 feet high develops on which 100-500 flower-heads nod. Each has a central disk of hundreds of bright yellow florets surrounded by a score of short reddish-purple ray florets. The flowering season is from June through October. The whole plant dies after flowering but once.

The greensword, *A. virescens* Hillebr., a much rarer plant with green leaves, once grew from 6,000 to 9,000 feet. Today it has all but vanished; a few plants are still to be found in Koolau Gap just outside the park, and between rocks on the edges of cliffs at Kaupo Gap.

Visitors once gathered specimens of silversword as evidence that they had been up the mountain, even as the mountaineer gathers edelweiss in Switzerland. Thoughtless people uprooted the silvery globes merely to watch them tumble down the slopes. Goats eat the growing heart of the plant. Grazing stock are incompatible with its existence. It has many insect parasites (see [The Insect Life](#)). All of these, singly and in combination, threaten the existence of the species.

The outer slopes of the mountain had thousands of silverswords according to early accounts, while today they are rare. In the crater, some of the cones were so thickly covered that they appeared to be bathed in moonlight. Except for strong, enlightened action, this beautiful plant might have completely vanished from the mountain.

SUMMARY LISTS

The following are incomplete lists of plants reported at various places within the park. They may serve as the start or a check for those wishing to know something of the plants in a certain area. The names given are those most commonly used locally or those used to head the corresponding sections above.

COMMONEST SHRUBS ALONG THE HIGHWAY above Headquarters. Sandalwood, mamane, geranium (*N. tridens*), aalii, ohia, ohelo, pukiawe, mountain pilo, kupaoa.

WIDESPREAD below 8,500 feet. *Stereocaulon* lichen (mostly on barren lava), pellucid polypody, maidenhair spleenwort, cliffbrake, swordfern, bracken, sedge (*Gahnia*), sheep sorrel or dock; ulei, mamane, aalii, tarweed, ohia, evening primrose, ohelo, pukiawe, selfheal, plantain, kukaenene, mountain pilo, kupaoa, pamakani, hairy cat's-ear, horseweed.

SUMMIT FLORA (above 8,500 feet). Mountain pili (grass), trailing pukiawe, *tetramalopium sp.*, kupaoa, common dandelion.

CRATER FLOOR on ash or barren lava. Mountain pili, sheep sorrel, ulei, mamane, bur clover, white clover, cranesbill, aalii, tarweed, ohia, evening primrose, ohelo, pukiawe, selfheal, common plantain, kukaenene, mountain pilo, catchfly, kupaoa, pamakani, hairy cat's-ear, Canadian horseweed, *Tetramalopium humile*, silversword.

KOOLAU GAP. Amaumau, rush, mauulaili, tree orchid, sandalwood, pawale, Hawaiian buttercup, trailing *akala*, native strawberry, nohoanu, apeape, highbush ohelo, trailing plantain, lobelia, wormwood, trailing kookoolau, greensword.

PALIKU. Amaumau, ekaha, akaha akolea, twayblade, alaawainui, hoawa, akala, kawau, olapa, kolea, manono, lobelia, naenae.

KAUPO GAP. Rush, mauulaili, sandalwood, hoawa, puaainaka, poha, common potato (naturalized), Jerusalem cherry, tree plantain, lobelia, naupaka, *Dubautia* sp., *Railliardia scabra*.

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Figure 1—SEdge, *Gahnia*

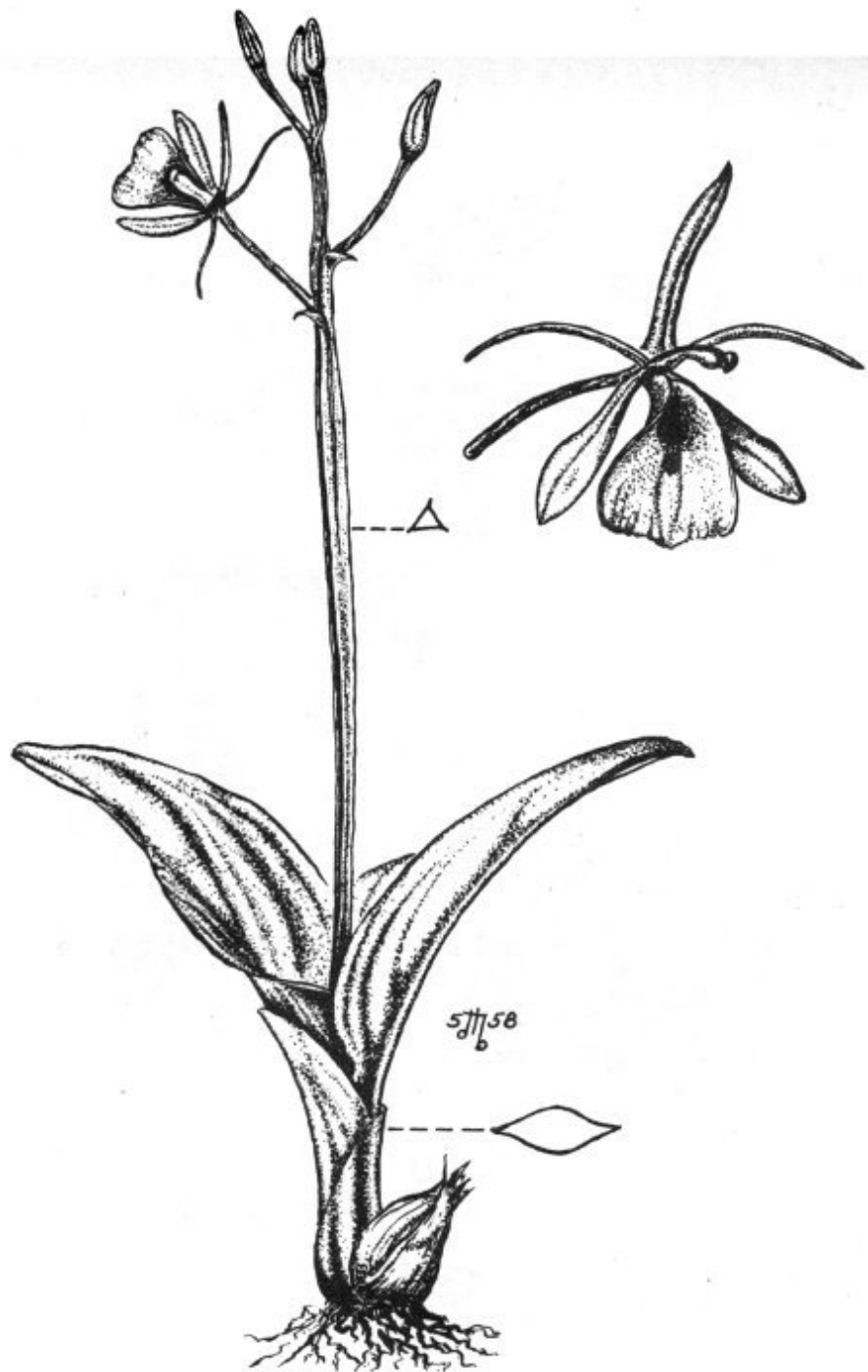


Figure 2—HAWAIIAN ORCHID, *Liparis*



Figure 3—SANDALWOOD, *Santalum haleakalae*

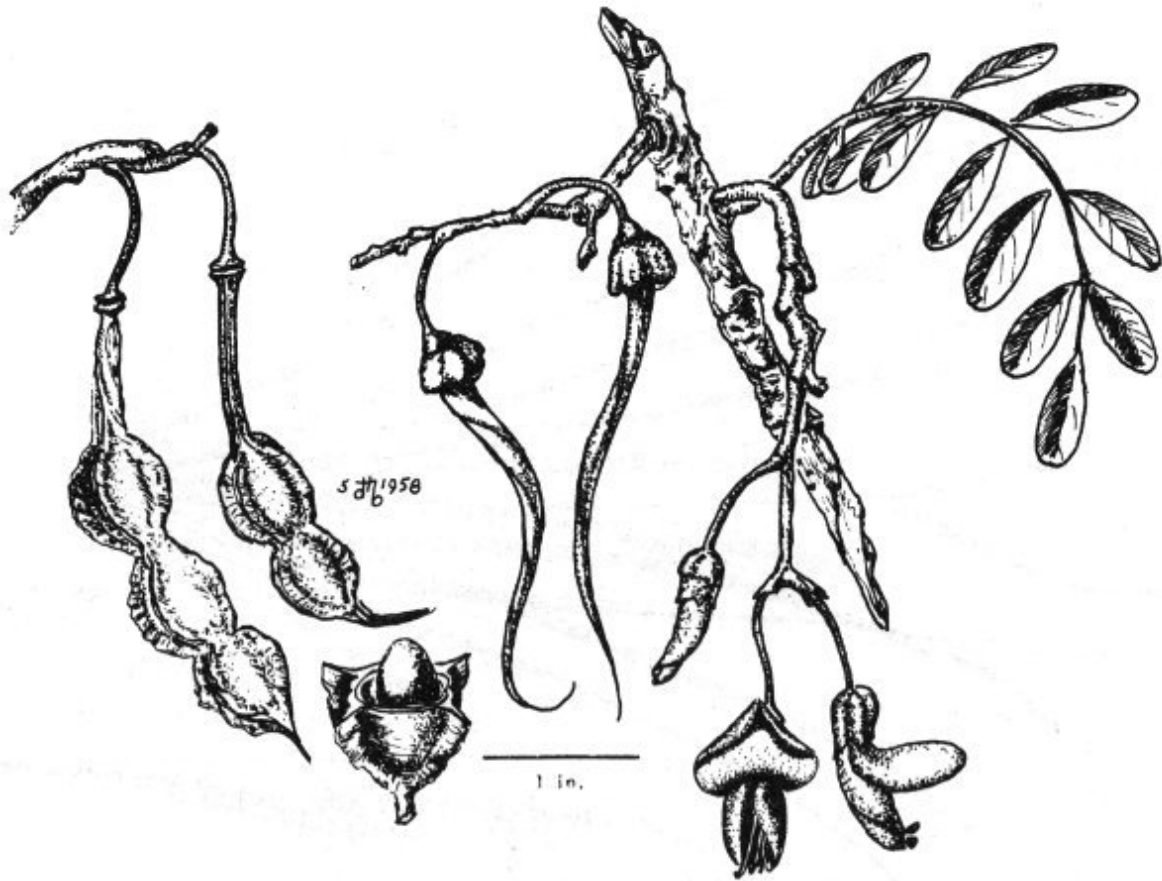


Figure 4—MAMANE, *Sophora chrysophylla*

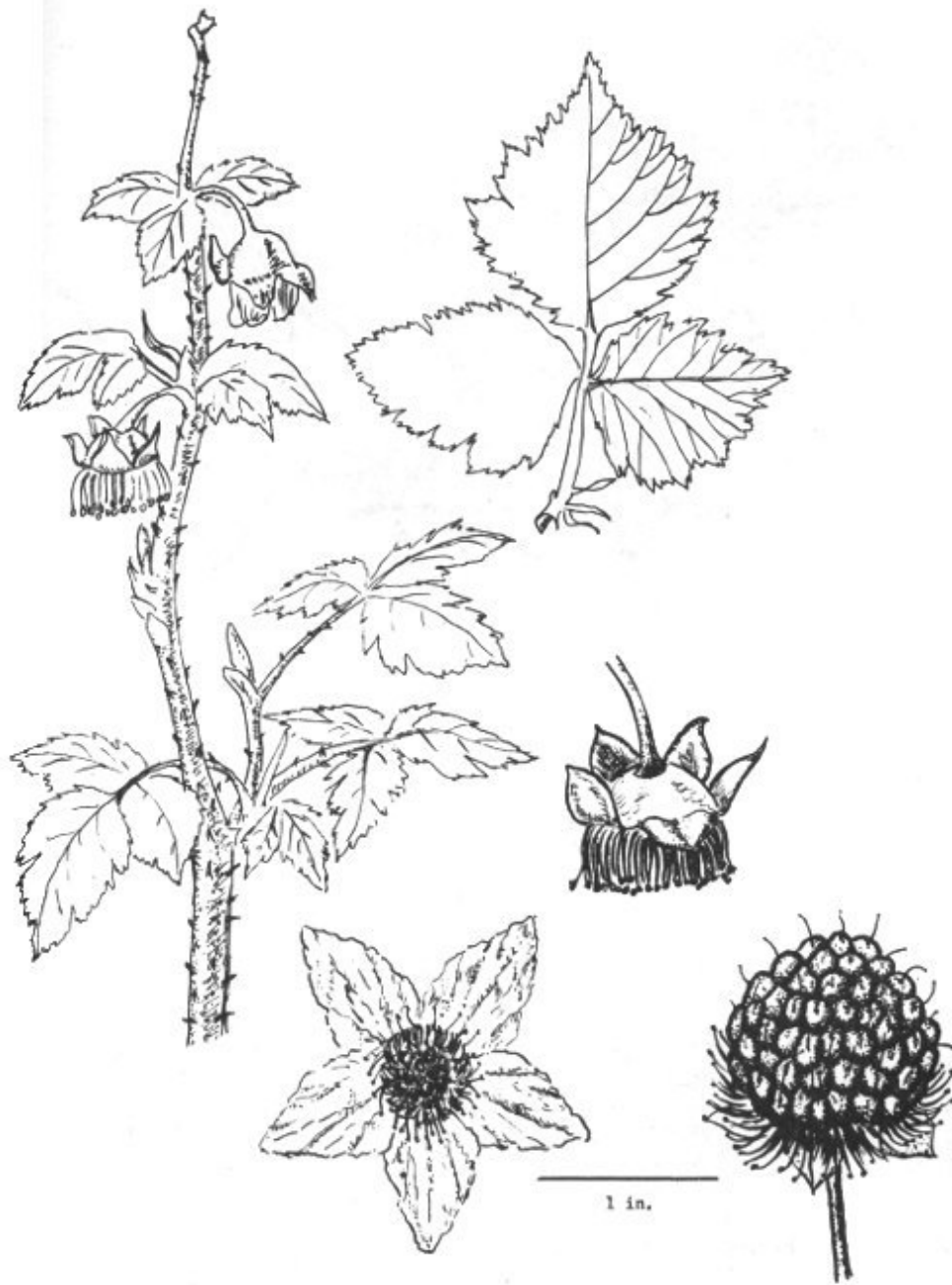


Figure 5—HAWAIIAN RASPBERRY, *Rubus hawaiiensis*



Figure 6—HINAHINA, *Geranium tridens*

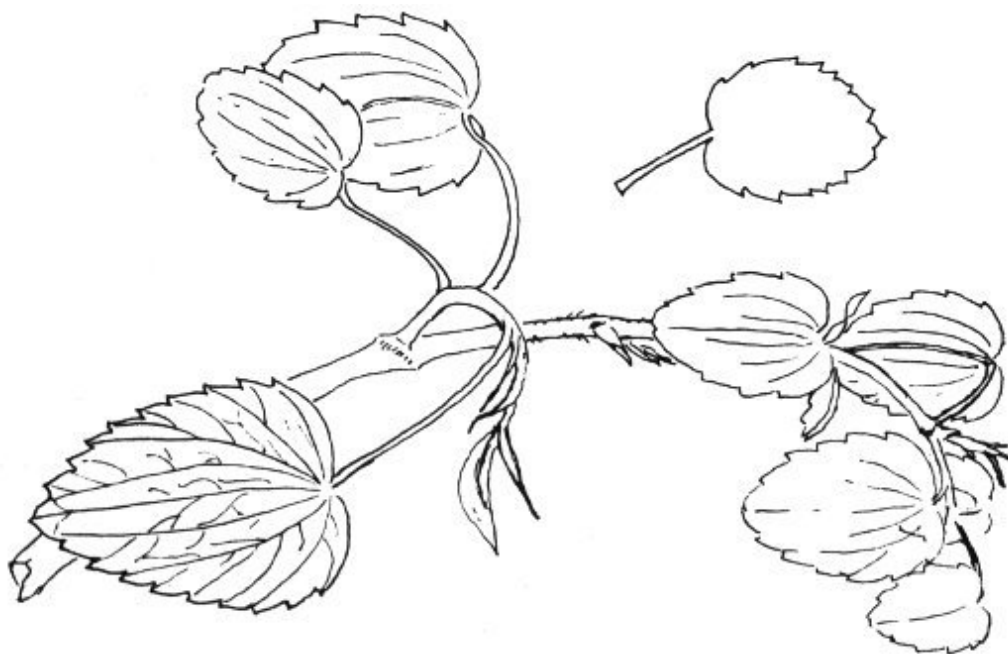


Figure 7—NOHOANU, *Geranium (Neurophyllodes) arboreum*



Figure 8—AALII, Seed capsules, leaf detail



Figure 9—OHIA LEHUA, Twig with flower beginning to open



Figure 10—OLAPA

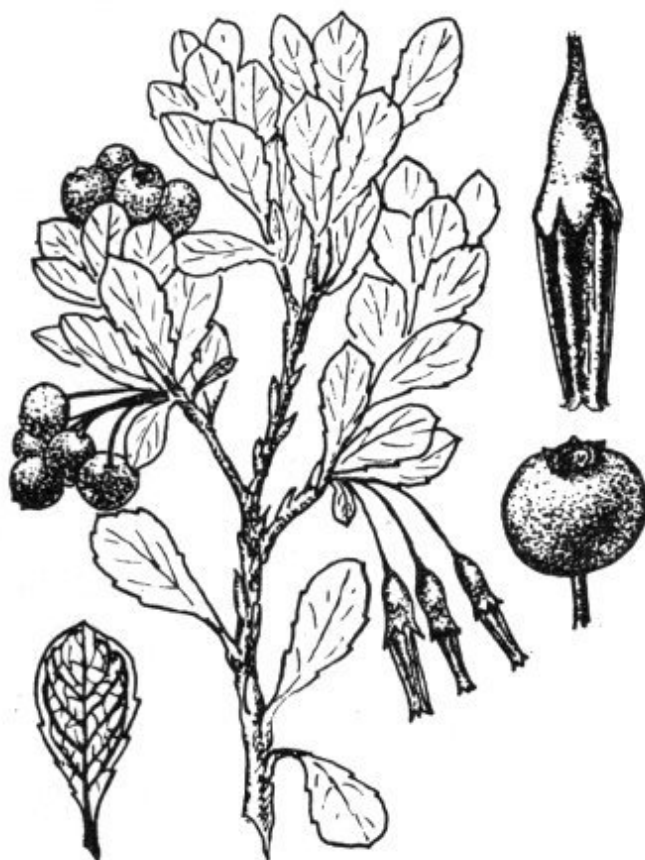


Figure 11—OHELO

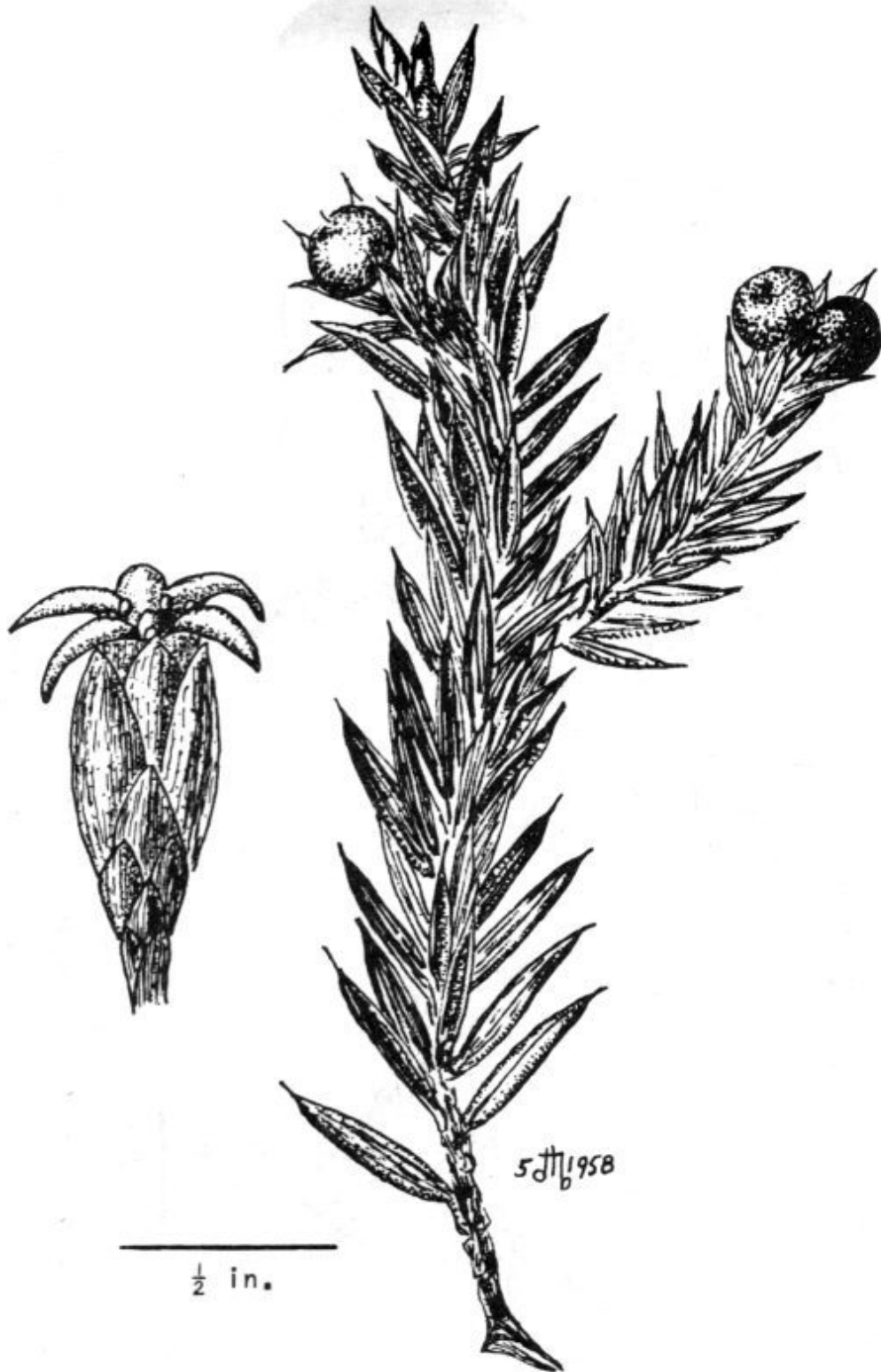


Figure 12—PUKIAWE, Twig, flower magnified



Figure 13—KOLEA, showing fruit

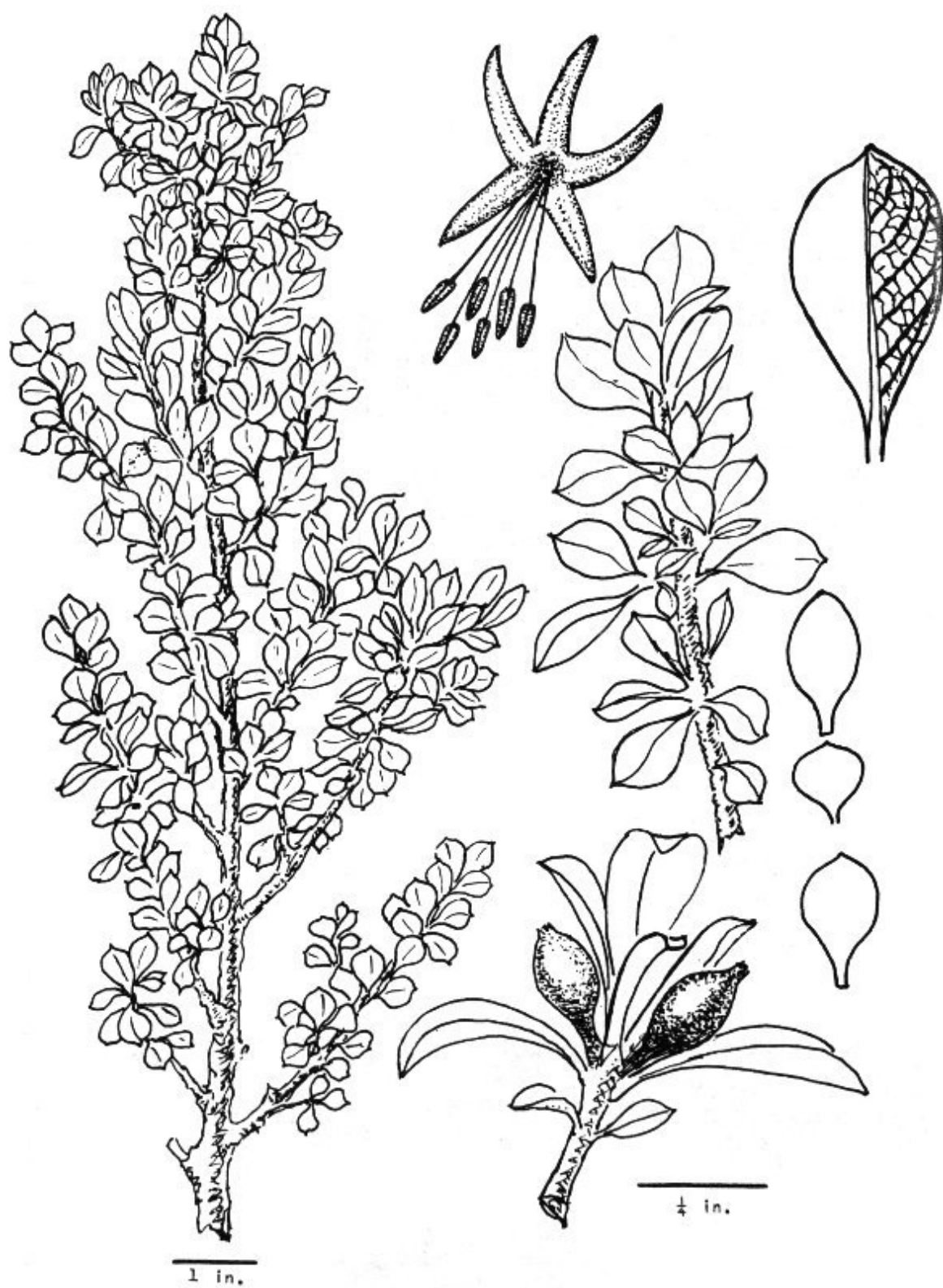


Figure 14—MOUNTAIN PILO. Twig, fruits, leaf shapes, magnified flower and leaf showing venation

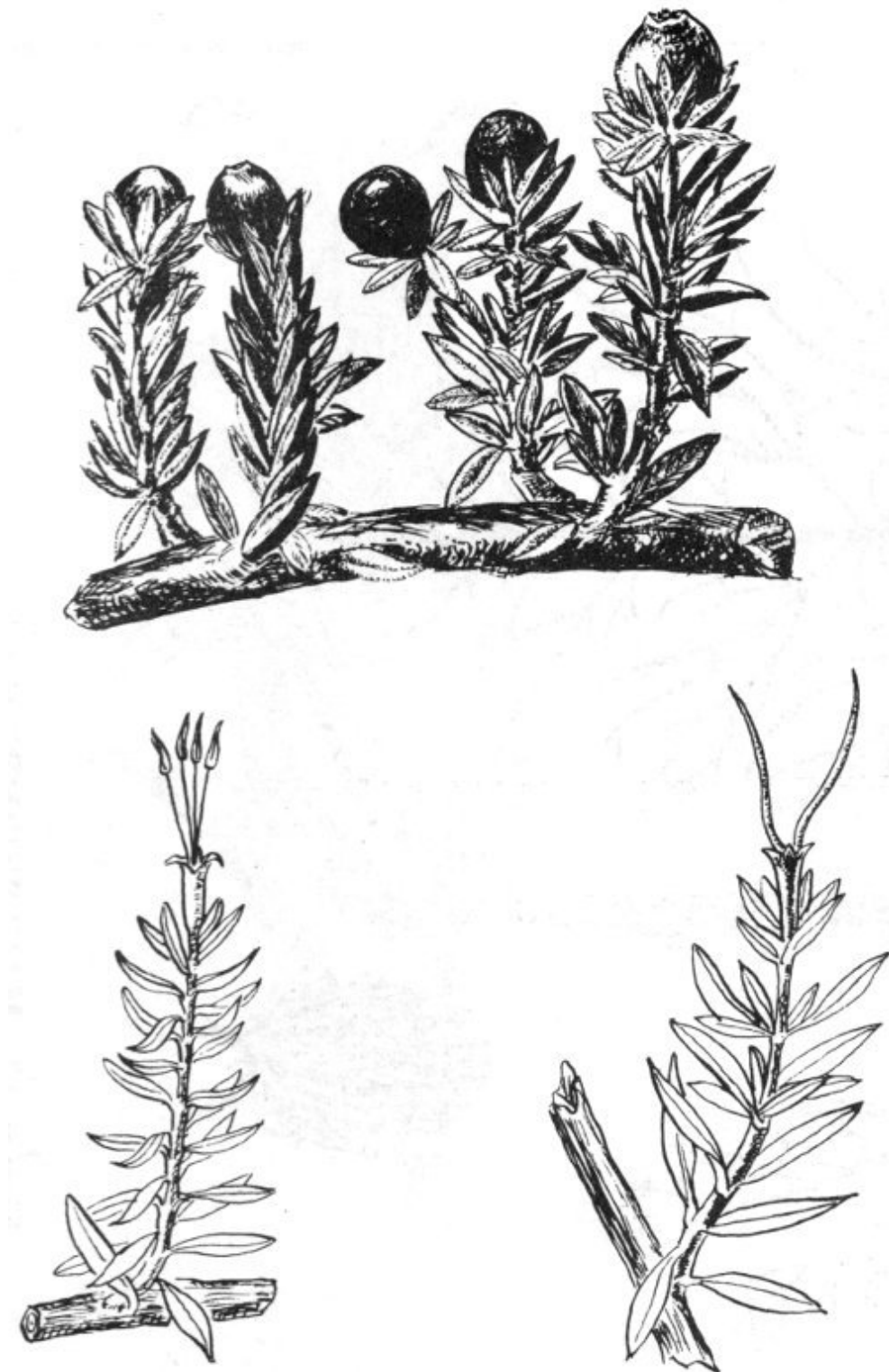


Figure 15—KUKAENENE, fruiting twig, male and female flowers



Figure 16—CATCHFLY, *Silene struthioloides*

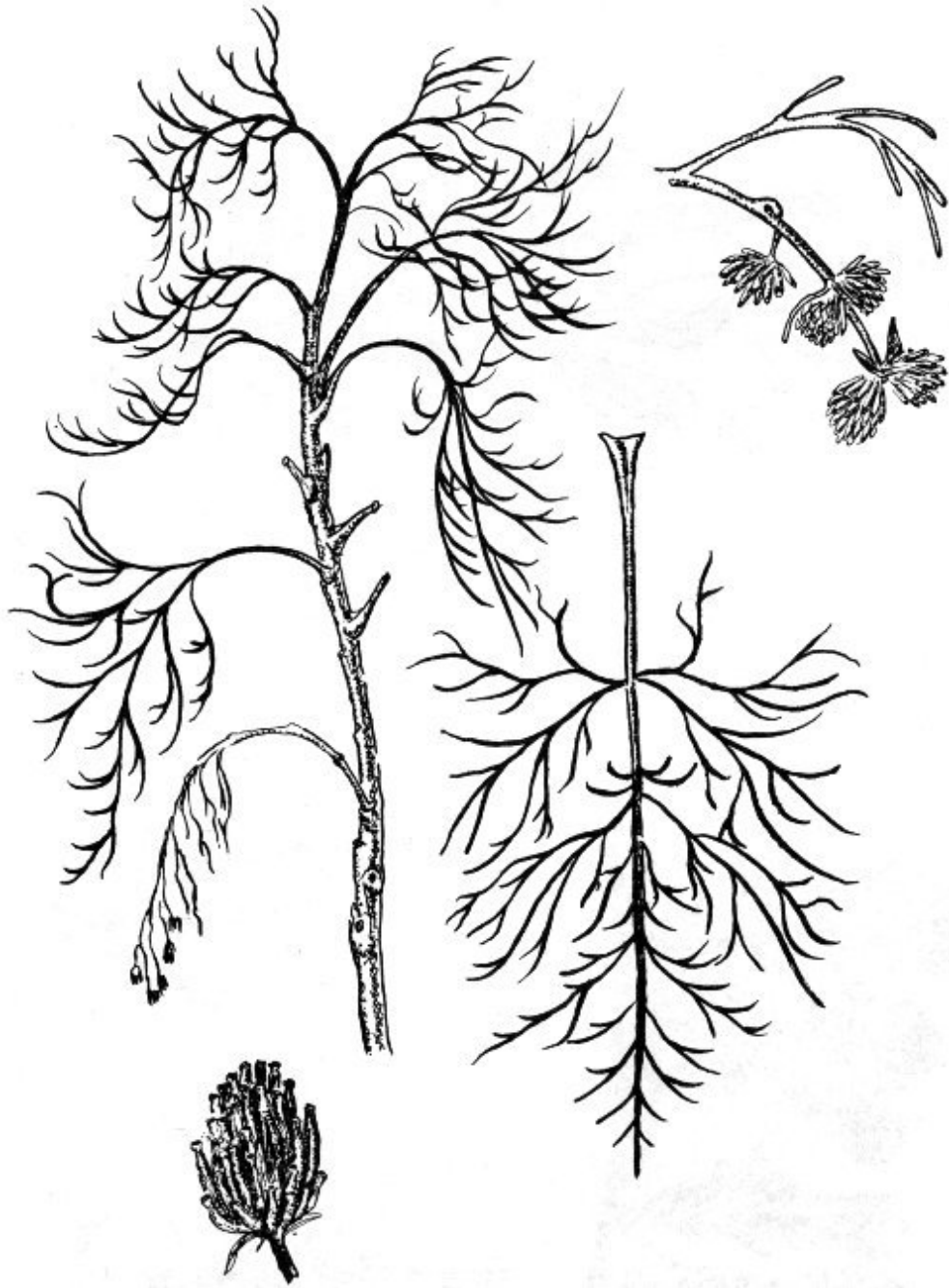


Figure 17—MAUI WORMWOOD, leaves, magnified flowers



Figure 18—NAENAE, *Dubautia plantaginea*



Figure 19—KUPAOA, *Raillardia menziesii*

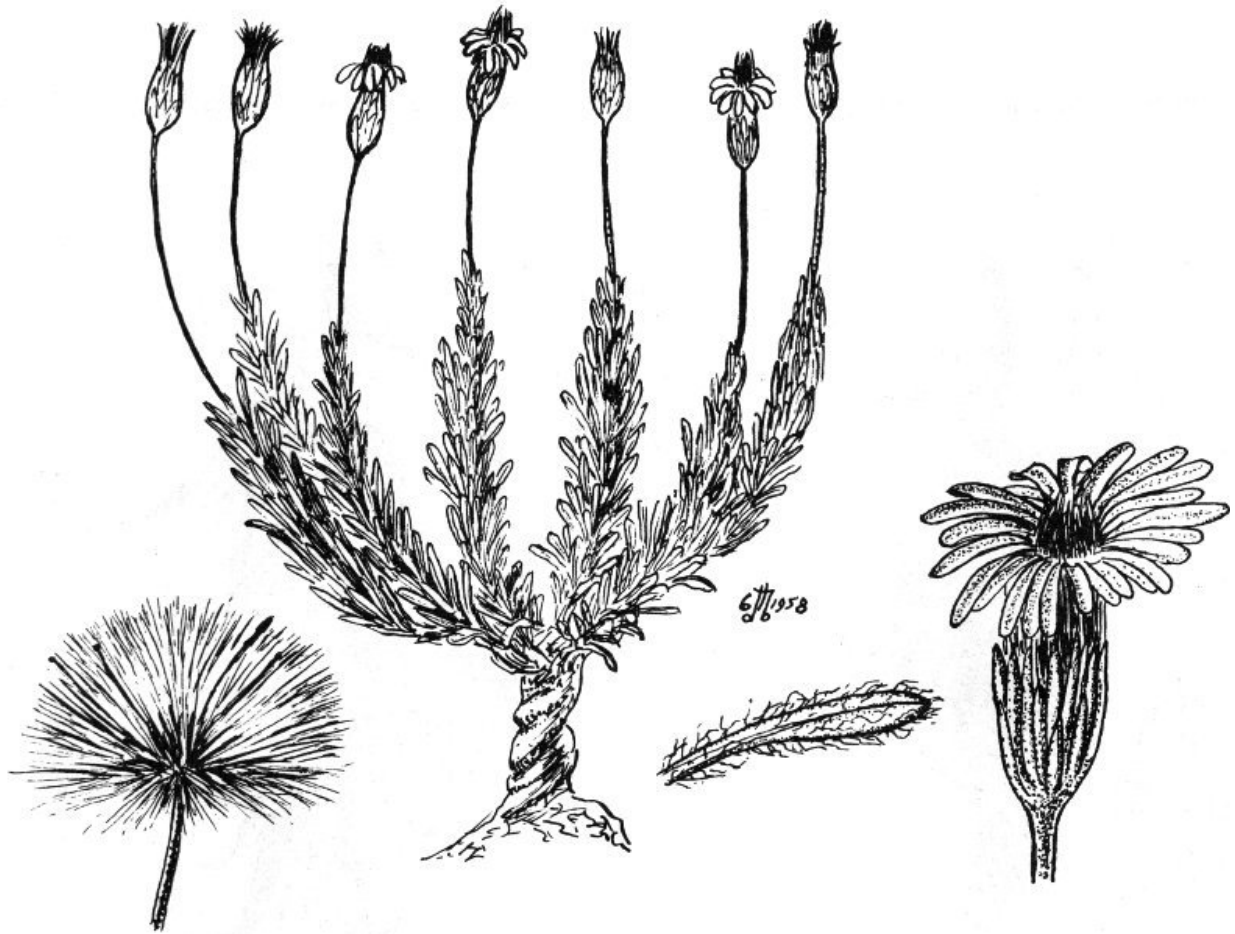


Figure 20—TETRAMALOPIMUM

THE BIRDS AND MAMMALS

It is a general rule that, as among plants, numbers and species of birds decrease as one goes up a mountain. It is true also that native Hawaiian birds are to be found mostly in upland forests. Of the members of the endemic nectar-sipping family, DREPANIDAE, *'apapane* and *'i'iwi* may be seen up to 8,500 feet. They are associated with ohia which give them their food. *'Amakihi* are fond of the mamane blossoms, so it is not surprising to find them widespread although not numerous in the park. Far rarer is the little Maui creeper, *'alauahio*, which finds insects and spiders for food in the bark of trunks and branches of trees and shrubs. It follows the latter up to the Alpine Zone.

Except for migrants and sea birds, other natives nowadays are absent. Occasionally an Hawaiian owl, *pueo*, has been seen flying over grasslands near the park entrance and at Paliku. Tropic birds, *koa'e*, occasionally soar around the cliffs inside the crater. The dark-rumped petrel, *'ua'u*, a vanishing species, is heard at night back of Holua and Kapalaoa cabins, and at Leleiwi Overlook. This pigeon-sized bird of the ocean nests in burrows 4-6 feet deep, at the bases of cliffs in the crater. Both young and adults were formerly hunted, often with nets, because they were highly prized as food. Bird catching was an important occupation of Hawaiians; at times one stumbles upon the remains of shelter caves and campsites used by them in a lonely mountain vastness. Except the period from early May to mid-August while it is away on its migration to Alaska, the golden plover, *kolea*, is a most conspicuous bird inside and outside the crater.

Introduced birds are everywhere. The Japanese white-eye or *mejiro* can be found in vegetation up to the Alpine Zone. It is especially common at Paliku, the most rewarding bird area in the park. The Pekin nightingale or Japanese hill robin, *sochi-cho*, goes all the way to the summit. I observed a pair at the structure on Red Hill, and picked up a dead one above the elevation of the Observatory. They sing in the trees around Paliku. Linnets and skylarks are often seen or heard anywhere up to the Alpine Zone. Commoner at low

elevations, a mockingbird appears infrequently above Park Headquarters, and mynahs visit the inn grounds in summer. They also appear at Paliku. English sparrows may linger around the horse barns, as expected.

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The most conspicuous and commonest Haleakala birds are introduced game-birds. Ringnecked pheasants are flushed or heard up to 8,500 feet, while Chukar partridges are numerous from 7,500 to 10,000 feet. Both of these are abundant in the crater. The Chukar, only a newcomer here, is already definitely at home. California quail scurry from the road or trail at elevations up to 7,500 feet. In 1958, 27 Erchel's francolin were released just below the park line near the inn, but success of their establishment is still a question.

The bat was apparently the only mammal living in Hawaii at the time when Polynesian navigators first visited here. They carried the pig, dog, and tramp rat along with them on their voyages. After Cook, many mammals were brought in, some of which readily reverted to a wild state. These include sheep, goats, cattle, horses, burros, dogs, cats, and mongooses.

The feral goat is the problem child of the Haleakala area. Often a beautiful creature, with long, black, flowing hair, it soon exterminates silversword, mamane, and desirable vegetation wherever it is left uncontrolled. Upon being forceably separated from its mother, the terrified kid quickly becomes tame and attaches itself as closely to the human associate as it did to its natural parent. Occasionally a stray dog that has reverted to the wild is seen in the park area. It is only a reminder of the days of the past century when packs roamed the side of the mountain and harassed visitors with their ferocity. Joining their members were wild bullocks that also constituted a threat. These were inevitably mentioned in all journals of early visitors. In the wet forested areas below Koolau Gap, on the fringe of upper Kipahulu Valley, and along the rim at Waianapanapa, pigs flourish in bliss that is broken only occasionally by a local hunter who seeks them just outside the boundary of the park.

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THE INSECT LIFE

Hawaii National Park is rich in insect life although, as is typical of insular areas, insects are much more sparse than in continental regions. Most of our insects are endemic species, i.e., are found only in these islands; many are very limited in distribution. For the most part the endemic insects are associated with the native plants. Most of our insects are comparatively small and inconspicuous; nevertheless, a great many are strange and unusual so that our fauna is particularly interesting to the scientist. We have none of the larger, showy butterflies and other insects which the visitor often expects to find here. No noxious or harmful species are present in areas used by park visitors, but the large blowflies which breed in goat carcasses in Haleakala Crater sometimes become a nuisance, due to their presence in large numbers. These are mainland species which have been accidentally introduced into Hawaii and which are now restricted to the highland areas where the climate is temperate. With the exception of lice and domestic flies, which were brought here by the Hawaiians themselves, Hawaii was free of pestiferous insects before the arrival of the Europeans.

One of the best known endemic Hawaiian insects is the butterfly named after King Kamehameha, *Vanessa tameamea*, a highly colored relative of the painted lady, the tortoise shells, and the red admiral butterflies of the mainland. This species is highly prized by amateur butterfly collectors since it is found no place else in the world. Its colors are orange, brown, and black. The female has small white spots in the apical portions of the front wings; these spots are rosy colored in the male. The caterpillars, green or purplish, feed on the leaves of the *mamaki*. Adults are found in forests throughout the islands. They are attracted to the native hydrangea, *kanawao*, to the introduced thimbleberry, or to the sap exuding from *koa* or *naio*. This butterfly is a strong flier, and ranges from the seacoast to the top of Haleakala. The female of *Hodegia apetala*, an endemic genus of only

one described species, is an Hawaiian moth unable to fly. The male

is unknown. This jumping insect, related to the bollworm, lives in bunch grass near the summit. The narrow, pointed, reduced, ashen-brown wings are ½ inch long; the abdomen, about as long as the wings, is brownish gray.

As one walks over open patches at higher elevations a brown moth, 1½ inches wide, rises readily before one. This is *Agrotis aulacias* that possesses [23] great powers of flight. The genus of 27 species is related to army worms, cut-worms, and a host of agricultural pests. *Fletcherana insularis* is a geometrid moth (inch worm, measuring worm, or looper) found in late spring high above the forest belt near the park entrance. The insect is an inch wide, its color is white, speckled with black. The genus of 5 species is confined to Hawaii.

Nesophrosyne haleakala is a mottled, gray-brown leafhopper, ⅛ inch long; the head and front part of the thorax are yellowish. In the park it is found in *pilo* and *ohelo* at 8,500 or 9,000 feet. The genus is endemic and contains over 60 forms distributed throughout the islands.

An undesirable pest has publicized its presence high above the park entrance by leaving its name on two caves which early visitors found convenient for shelter. Big Flea and Little Flea Caves often appear in accounts of early trips, but never without mention of the annoyance that was caused by their permanent occupants. Of the 7 different kinds of fleas recorded for Hawaii, only the so-called cat flea (*Ctenocephalides felis*) annoys people to any extent. Several species of fleas infest the Hawaiian rat and were presumably brought in when this rodent arrived with the earliest Polynesian immigrants.

Because the beautiful silversword has no very close relatives, which means that its ancestors arrived in Hawaii in earliest time, any species of insects associated solely with it as a host plant draws particular interest. Moreover, since the survival of silverswords is precarious in this day of rapid change, all agents that threaten must be carefully scrutinized. Silverswords are [24] specific hosts to a half dozen or more endemic species of insects. [87]

Among Lepidoptera, the family of moths and butterflies, the larvae of a pyralid moth, *Rhynchephestia rhabdotus*, feeds in the flower-heads, destroying the seeds. Caterpillars of a noctuid, *Euxoa epicremma*, have been collected beneath plants. The caterpillars of a tineid moth have been found

among dead leaves. Of the **DIPTERA**, great numbers of yellow maggots of a gray fly, *Tephritis cratericola*, feed on the seeds and prevent their development. The larvae of one of the Coleoptera, a beetle described as *Aescheithmysus terryi*, feeds apparently only on old and dead stems. A relative, *A. swezeyi*, similarly feeds on *Railliardia*. Great members of a leaf-hopper, *Iburnia argyroxiphii*, suck the sap of Haleakala silverswords. It is preyed upon by a wasp, *Polynema* sp., so tiny that the naked eye can hardly see it. The wasp lays its egg in the egg of the hopper. The larval wasp, upon hatching, feeds on the contents of the hopper egg, pupates within it, and emerges from it.

The Paliku area is very rich in insect life; many hundreds of species have been recorded from here and many are apparently very restricted in distribution and are known only from this locality. Species of pomace-flies (**DROSOPHILIDAE**) and the long-legged flies (**DOLICHOPODIDAE**) are especially prevalent at Paliku. Two species of the native genus *Idiomyia* are the largest members of **DROSOPHILIDAE** found any place in the world; they measure approximately 7 mm. in length. Two species of flightless Neuroptera (*Pseudopsectra cookeorum* and *lobipennis*) occur on the vegetation (*Dubautia*, *Metrosideros*, and *Cyanea*) in Haleakala; these represent one of the strangest entomological curiosities of the world.

Over five thousand species of insects have been recorded from the Hawaiian Islands and it is probable that several thousand more species remain to be described. In the past our knowledge of the insects was based largely upon

the monumental study, "Fauna Hawaiiensis,"^[25] published by a group of institutions including the British Museum (Natural History) and the Bernice P. Bishop Museum. A new faunistic study is now under way titled

"Insects of Hawaii."^[26] This is an extremely valuable reference work and is indispensable to anyone studying Hawaiian insects. It gives a complete review of the insects from the lowest orders through the butterflies and larger moths. Two volumes on the **DIPTERA** are now in press. Volume one gives a most comprehensive account of the mode of dispersal of our endemic fauna and flora to these islands. It is estimated that the 3,722 known endemic insects developed from approximately 250 ancestral species.

HAWAIIAN WORDS AND PLACE NAMES

THE HAMZA. During the centuries, changes occur in dialects. Hawaiian is one of the dialects of the Polynesian language. A striking change in Hawaiian is the dropping of the letter *k* that once appeared in some words. But instead of complete discard of the sound, its former existence is revealed as a little catch in the throat, called a glottal stop. This is represented in print by a mark (') called *hamza*. The omission of this mark can spell a word of entirely different meaning. Yet, without explanation, the hamza can be more bewildering than helpful to those unfamiliar with Hawaiian. Its use in this book therefore has been limited to this section and to the section on [plants](#), because it is needed if correct pronunciation is attempted. And who fails to want to try Hawaiian on his tongue!

These brief, literal interpretations are not mentioned in the text. Principal authorities are K. E. Emory and M. K. Pukui.

AHU—Pile, cairn, altar, shrine. Often a small or large pile of stones erected as a trailmark or landmark. *Ahu* were sometimes put up by passing parties as monuments or evidence that they had been there, even as mountain climbers do to this day. To be assured of a safe journey, an *ahu* of three stones was made as tribute to the god of the locality.

ALII—*Ali'i*, chief, chiefess, sovereign, ruler. One of the upper class.

HALALII—*Hala-li'i*, fun-making, from the name of a traditional chief of Ni'ihau.

HALEAKALA—*Hale-a-ka-la* is usually interpreted as “house of the sun,” a simple translation of the Hawaiian name given us for the mountain. It assumes vague reference to the Maui legend. An old explanation converted the translation to “house built by the sun.” Rev. A. O. Forbes ^[27] says that the name is a corruption of *Alehe-ka-la*, “snarer of the sun,” in reference to

Maui's deeds. Still another version would have it *Ahale-ka-la*, to be interpreted as "rays of the sun."

HALEMAUU—*Hale mau'u* is grass hut; *hale ma'u'u* signifies damp house. 90

HANA—Bay or valley, when used in a place name.

HANAKAUHI—*Hana-ka-uhi*, the mist-maker; yam valley.

HAOLE—A foreigner, one of foreign extraction; today, usually in reference to a Caucasian; adj., foreign.

HEIAU—Pre-christian place of worship, hence, usually translated temple. This was often a stone platform or an earth terrace.

HOLUA—A course used for the ancient royal sport of sliding down steep slopes; also the sled itself.

HONOKAHUA—*Hono-ka-hua*, joined foundation.

IAO—*'I-ao*, high; into the clouds.

KAHULUI—*Ka-hului*, a sea for drag-net fishing.

KALAHAKU—The proclamation of the Lord.

KALAPAWILI—*Ka-lapa-wili*, winding or twisting ridge.

KALUAAWA—*Ka-lua-'awa*, the 'awa pit. 'Awa or *kawa* is the well-known traditional drink of Polynesia made from *Piper methysticum*.

KALUAIKI—*Ka-lua-iki*, small pit or crater.

KALUANUI—*Ka-lua-nui*, large pit or crater.

KALUA O KA OO—*Ka lua o ka 'o'o*, the pit of the 'o'o. The 'o'o, now extinct, was an endemic, black, nectar-sipping bird. It had tufts of yellow

feathers under each wing and at the base of the tail, which were used in featherwork.

KALUA O UMI—Umi's Cave.

KAMOALII—*Ka moa li'i*, the little chicken.

KA MOA O PELE—Pele's chickens or chicken coop.

KANAKA—(*Haw. pl. kanaka; Eng. pl. kanakas*) human being, person, man.

KAPA—Tapa, a cloth made from the bark of *mamaki* or *wauke*; formerly, clothes of any kind; bedclothes.

KAPALAOA—*Ka palaoa*, the *palaoa* was a highly-prized pendant of whale-tooth ivory, a symbol of royalty over the theft of which wars were waged. 91

KAUPO—*Kau po*, to land at night. A variety of banana is given this name, probably after the place name.

KAWILINAU—*Ka wili nau*, literally, the twist of pain. This is the Hawaiian place name for Bottomless Pit.

KEAHUOKAHOLO—*Ke ahu o ka holo*, a heap resulting from a landslide.

KEANAE—*Ke 'anae*, the large mullet.

KEONEHEEHEE—*Ke one he'ehe'e*, the sliding sands.

KIHEI—Shoulder covering; a rectangular fine mat or tapa used as a mantle.

KIPAHULU—Worn-out soil.

KOLEKOLE—Bright red, blood red.

KOOLAU—*Ko'olau*, windward side.

KUIKI—*Ku iki*, a moment's stop, a short halt.

KUMUILIAHI—*Kumu 'iliahi*, sandalwood trunk.

LAUULU—*Lau 'ulu*, leaf of breadfruit tree.

LELEIWI—Carved figure on the bowsprit of a canoe or ship.

LILINOI—Goddess of Haleakala.

MAKAWAO—*Maka wao*, forest region.

MAUI—Contrary to popular belief and despite the similar spelling, the island does not bear the name of the demigod. The name of the island is pronounced mow-ee. This almost rhymes with an enthusiastic “WOWIE!” especially as a Virginian might pronounce it. The *au* must be treated in the fashion for Hawaiian diphthongs, that is, the crisp vowels are more loosely connected than in the English, and a slight accent is imposed on the leading one. The demigod’s name has three syllables, with accent on the second, i.e., the *u*. Thus, Ma-u’-i.

MAUNA—Mountain.

MAUNA HINA—Gray mountain.

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MOI—*Mo’i* (A 19th Century word), King, queen, sovereign.

NAMANA O KE AKUA—*Na mana o ke akua*, the miraculous power, *mana*, of Deity, gods, or spirits.

NA PIKO HANA—The hiding place for navel cords.

NIANIAU—*Ni’ani’au*, the sword fern.

OILI PUU—*O’ili pu’u*, hill appearing, hill shot out.

PALIKU—*Pali ku*, upright cliff.

POHAKU PALAHA—Broad stone; wide stone.

PUKALANI—*Puka lani*, heavenly entrance, chief’s doorway.

PUU—*Pu'u*, hill. This is contracted in some names to *pu*.

PUU HELE—*Pu'u hele*, moving hill.

PUU KAUAUA—*Pu'u ka uaua*, the stubborn hill; the tough hill.

PUU KUMU—*Pu'u kumu*, stump hill.

PUU MAILE—*Maile* is a fragrant vine, *Alyxia oliviformis*, used and loved like the laurel of Europe.

PUU MANEONEO—*Pu'u mane'one'o*, itching hill. *Maneoneo* means *barren*.

PUU NAUE—*Pu'u naue*, trembling hill; loose or insecure hill.

PUU NOLE—Grumbling hill.

WAIALE—*Wai'ale*, rippling water.

WAI ANAPANAPA—*Wai 'anapanapa*, sparkling water.

WAIHOI—*Wai ho'i*, water that returned.

WAIKAU—*Wai kau*, water on a high place.

WAIKEKEEHIA—*Keke'ehia* means to twist and wind like a rivulet or stream, hence, the name is interpreted as crooked waters.

WAILUKU—*Wai luku*, water of destruction.

HUI O AHINAHINA

The exclusive Society of the Silversword (Hui o Ahinahina) invites you to become a member if you have visited the summit or the crater of Haleakala. Only one class of membership. Life: one dollar! By joining you can help scientific study, interpretation, and display exhibits of the Park. Exchange your dollar at the Park or at the office of Silversword Inn for a silvery, engraved certificate, suitable for framing. With pride you can hand it down to your grandchildren, a souvenir of Haleakala.

ADDITIONAL HELP

The following publications contain information on the two sections of Hawaii National Park. They may be purchased at either Park Administration Office, at Silversword Inn, Haleakala, and at the Volcano House at Kilauea.

Volcanoes of Hawaii National Park. (pamphlet) by Gordon A. MacDonald and Douglas H. Hubbard.

Hawaii Natural History Association, 1951. 50¢

Ferns of Hawaii National Park. (pamphlet) by Douglas H. Hubbard.

Hawaii Natural History Association, 1952. 50¢

Trailside Plants of Hawaii National Park. (pamphlet) by Douglas H. Hubbard and Vernon R. Bender, Jr. Hawaii Natural History Association, 1960.

Plants of Hawaii National Park. By Otto Degener, 1945. \$4.00

Illustrative of Plants and Customs of the South Seas.

The Land of Pele. (pamphlet) by Nash Castro.

A historical sketch of Hawaii National Park.
Hilo Tribune Herald, Ltd., 1953. 85¢

Cartons, cigarette butts, and other trash do not add to your enjoyment of the park. You can easily dispose of these so that they cannot become obnoxious to you and those who come after you. With the number of visitors increasing each year, you can help protect and preserve the natural scene by placing all paper and other refuse in

containers provided for this purpose and by refraining from picking or breaking flowers, plants, and natural specimens. *Won't you do your share?* **THANK YOU.**

MILEAGES

Roads

Park Boundary to:	Miles
Silversword Inn	1.0
Hosmer Grove	1.5
Park Headquarters	2.0
Halemauu Trail	5.6
Leleiwi Overlook	8.4
Kalahaku Overlook	10.1
Observatory	12.1
Red Hill (Summit)	12.8
CAA Station	13.8

Halemauu Trail

Silversword Inn to:	Miles
Halemauu Trail Junction	2.9
Holua Visitor Cabin	5.9
Haleakala Park Road to:	
Holua Visitor Cabin	3.9
Holua Visitor Cabin to:	
Silversword Loop	.9
Bottomless Pit	2.3
Bubble Cave	3.3
Kapalaoa Visitor Cabin	3.8
Paliku Visitor Cabin	6.3
Kaupo Village	14.1

Sliding Sands Trail

Observatory to:	Miles
------------------------	--------------

Holua Visitor Cabin	7.4
Kapalaoa Visitor Cabin	5.8
Bubble Cave	6.5
Paliku Visitor Cabin	9.8
Kaupo Village	17.5

Miscellaneous Mileages:

Holua Visitor Cabin to Kapalaoa Visitor Cabin	3.8 Miles
Kapalaoa Visitor Cabin to Paliku Visitor Cabin	4.0
Paliku Visitor Cabin to Holua Visitor Cabin (Direct Route via Halemauu Trail)	6.3
White Hill	0.2
Crater Overlook Trail	350 Feet

A FEW STATISTICS

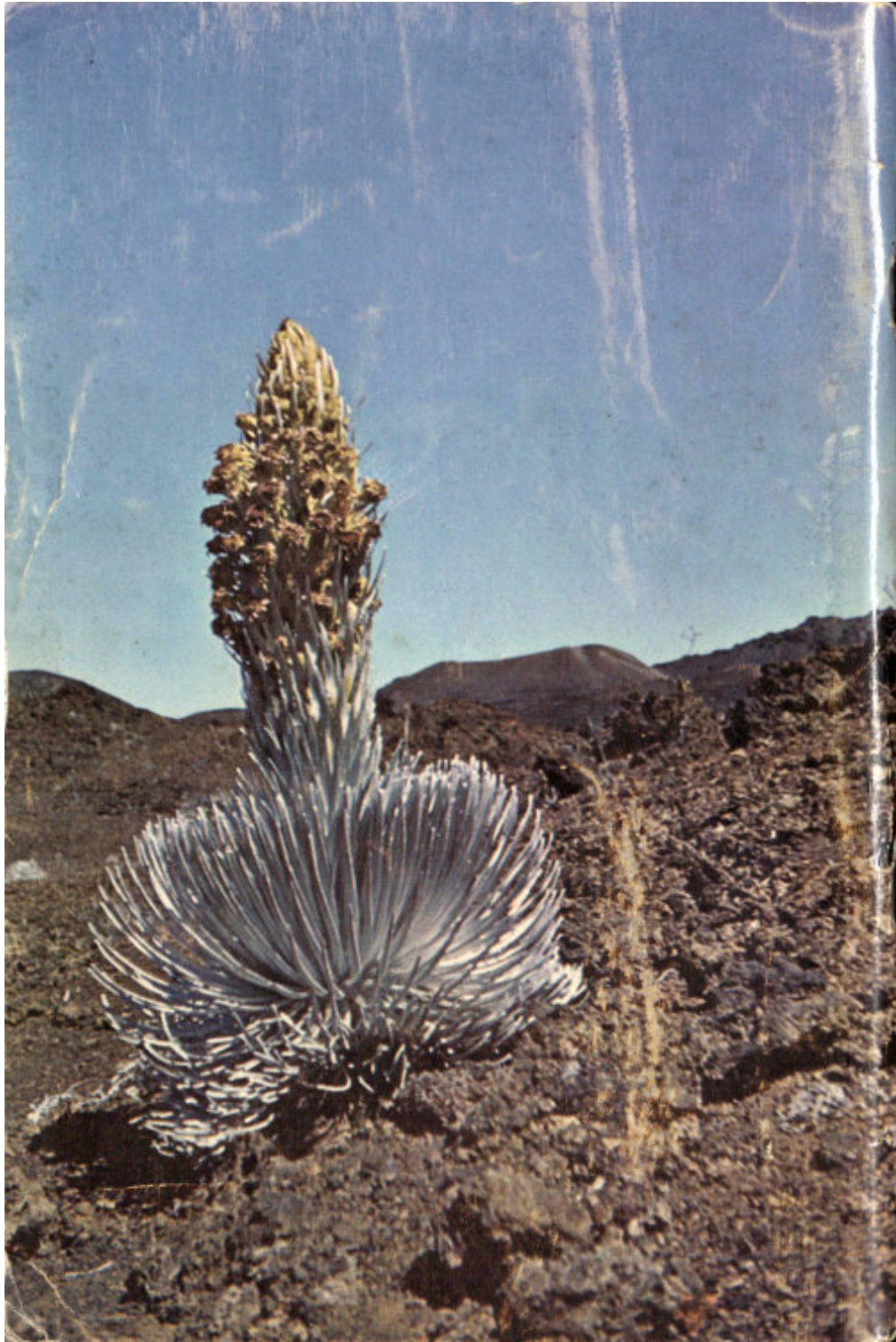
Area of the Crater	19 Square miles
Circumference of the Crater	20 Miles
Extreme length of the Crater	7½ Miles
Extreme width of the Crater	2½ Miles
Elevation of the summit of Red Hill	10,005 Feet
Elevation of Puu o Maui	8,133 Feet
Elevation of Park Boundary, Kaupo Trail	3,847 Feet
Elevation of Park Boundary, Koolau Gap	6,450 Feet
Area of the Haleakala Section, Hawaii National Park	26.7 Square miles

Footnotes

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Silversword in bloom

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